5.1 Introduction

In **Chapter 4**, we considered discrete random variable—that is, random variables whose set of possible values is either finite or countably infinite. However, there also exist random variables whose set of possible values is uncountable. Two examples are the time that a train arrives at a specified stop and the lifetime of a transistor. Let X be such a random variable. We say that X is a *continuous* † random variable if there exists a nonnegative function f, defined for all real $x \in (-\infty, \infty)$, having the property that for any set B of real numbers, ‡

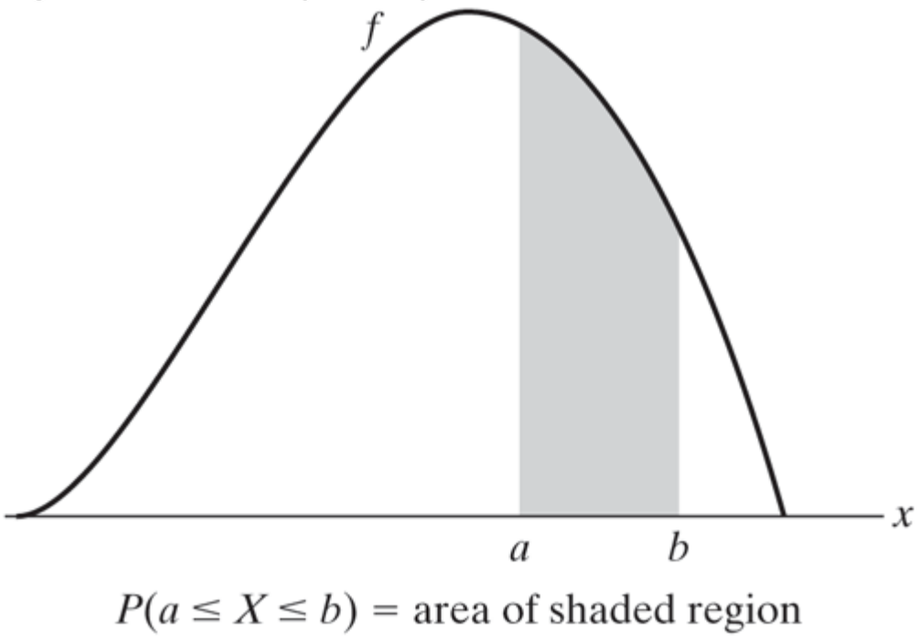
† Sometimes called *absolutely continuous*.

$$P\{X \in B\} = \int_B f(x)dx \tag{1.1}$$

‡ Actually, for technical reasons, **Equation (1.1)** is true only for the *measurable* sets B, which, fortunately, include all sets of practical interest.

The function f is called the *probability density function* of the random variable X. (See **Figure 5.1**.)

Figure 5.1 Probability density function f.



In words, **Equation (1.1)** states that the probability that X will be in B may be obtained by integrating the probability density function over the set B. Since X must assume some value, f must satisfy

$$1 = P\{X \in (-\infty, \infty)\} = \int_{-\infty}^{\infty} f(x)\,dx$$

All probability statements about X can be answered in terms of f. For instance, from **Equation (1.1)**, letting B=[a,b], we obtain

$$P\{a \leq X \leq b\} = \int_a^b f(x)\,dx \tag{1.2}$$

If we let a=b in **Equation (1.2)**, we get

$$P\{X=a\}=\int_a^a f(x)\,dx=0$$

In words, this equation states that the probability that a continuous random variable will assume any fixed value is zero. Hence, for a continuous random variable,

$$P\{X<a\}=P\{X\le a\}=F(a)=\int_{-\infty}^{a} f(x)\,dx$$

Example 1a

Suppose that X is a continuous random variable whose probability density function is given by

$$f(x)=\begin{cases}C(4x-2x^2) & 0<x<2\\ 0 & \text{otherwise}\end{cases}$$

a. What is the value of C?
b. Find $P\{X>1\}$.

Solution

a. Since f is a probability density function, we must have $\int_{-\infty}^{\infty} f(x)\,dx=1$, implying that
$C\int_0^2(4x-2x^2)\,dx=1$

or
$C[2x^2-\frac{2x^3}{3}]\Big|_{x=0}^{x=2}=1$

or
$C=\frac{3}{8}$

Hence,
b. $P\{X>1\}=\int_1^{\infty} f(x)\,dx=\frac{3}{8}\int_1^2(4x-2x^2)dx=\frac{1}{2}$

Example 1b

The amount of time in hours that a computer functions before breaking down is a continuous random variable with probability density function given by

$$f(x)=\begin{cases}\lambda e^{-x/100} & x\ge 0\\ 0 & x<0\end{cases}$$

What is the probability that

a. a computer will function between 50 and 150 hours before breaking down?
b. it will function for fewer than 100 hours?

Solution

a. Since
$1=\int_{-\infty}^{\infty} f(x)\,dx=\lambda\int_0^{\infty} e^{-x/100}\,dx$

we obtain
$1=-\lambda(100)e^{-x/100}\Big|_0^{\infty}=100\lambda \quad \text{or} \quad \lambda=\frac{1}{100}$

Hence, the probability that a computer will function between 50 and 150 hours before breaking

down is given by
$P\{50<X<150\}=\int_{50}^{150}\frac{1}{100}e^{-x/100}\,dx=-e^{-x/100}\big|_{50}^{150}=e^{-1/2}-e^{-3/2}\approx.383$

b. Similarly,
$P\{X<100\}=\int_{0}^{100}\frac{1}{100}e^{-x/100}\,dx=-e^{-x/100}\big|_{0}^{100}=1-e^{-1}\approx.632$

In other words, approximately 63.2 percent of the time, a computer will fail before registering 100 hours of use.

Example 1c

The lifetime in hours of a certain kind of radio tube is a random variable having a probability density function given by

$f(x)=\begin{cases}0 & x\le 100\\ \frac{100}{x^2} & x>100\end{cases}$

What is the probability that exactly 2 of 5 such tubes in a radio set will have to be replaced within the first 150 hours of operation? Assume that the events Ei, i=1,2,3,4,5, that the *i*th such tube will have to be replaced within this time are independent.

Solution

From the statement of the problem, we have

$P(E_i)=\int_{0}^{150}f(x)\,dx=100\int_{100}^{150}x^{-2}\,dx=\frac{1}{3}$

Hence, from the independence of the events Ei, it follows that the desired probability is

$\binom{5}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^3=\frac{80}{243}$

The relationship between the cumulative distribution F and the probability density f is expressed by

$F(a)=P\{X\in(-\infty,a]\}=\int_{-\infty}^{a}f(x)\,dx$

Differentiating both sides of the preceding equation yields

$\frac{d}{da}F(a)=f(a)$

That is, the density is the derivative of the cumulative distribution function. A somewhat more intuitive interpretation of the density function may be obtained from **Equation (1.2)** as follows:

$P\{a-\frac{\varepsilon}{2}\le X\le a+\frac{\varepsilon}{2}\}=\int_{a-\varepsilon/2}^{a+\varepsilon/2}f(x)\,dx\approx\varepsilon f(a)$

when ε is small and when f(·) is continuous at x=a. In other words, the probability that X will be contained in an interval of length ε around the point a is approximately εf(a). From this result, we see that f(a) is a measure of how likely it is that the random variable will be near a.

Example 1d

If X is continuous with distribution function FX and density function fX, find the density function of Y=2X.

Solution

We will determine fY in two ways. The first way is to derive, and then differentiate, the distribution function of Y:

$$F_Y(a) = P\{Y \le a\} = P\{2X \le a\} = P\{X \le a/2\} = F_X(a/2)$$

Differentiation gives

$$f_Y(a) = \frac{1}{2} f_X(a/2)$$

Another way to determine f_Y is to note that

$$\varepsilon f_Y(a) \approx P\{a - \frac{\varepsilon}{2} \le Y \le a + \frac{\varepsilon}{2}\} = P\{a - \frac{\varepsilon}{2} \le 2X \le a + \frac{\varepsilon}{2}\} = P\{\frac{a}{2} - \frac{\varepsilon}{4} \le X \le \frac{a}{2} + \frac{\varepsilon}{4}\} \approx \frac{\varepsilon}{2} f_X(a/2)$$

Dividing through by ε gives the same result as before.

5.2 Expectation and Variance of Continuous Random Variables

In **Chapter 4**, we defined the expected value of a discrete random variable X by

$$E[X]=\sum_x xP\{X=x\}$$

If X is a continuous random variable having probability density function f(x), then, because

$$f(x)\,dx \approx P\{x \le X \le x+dx\} \quad \text{for } dx \text{ small}$$

it is easy to see that the analogous definition is to define the expected value of X by

$$E[X]=\int_{-\infty}^{\infty} xf(x)\,dx$$

Example 2a

Find E[X] when the density function of X is

$$f(x)=\begin{cases}2x & \text{if } 0 \le x \le 1\\ 0 & \text{otherwise}\end{cases}$$

Solution

$$E[X]=\int xf(x)\,dx=\int_0^1 2x^2\,dx=\frac{2}{3}$$

Example 2b

The density function of X is given by

$$f(x)=\begin{cases}1 & \text{if } 0 \le x \le 1\\ 0 & \text{otherwise}\end{cases}$$

Find E[eX].

Solution

Let Y=eX. We start by determining FY, the cumulative distribution function of Y. Now, for 1≤x≤e,

$$F_Y(x)=P\{Y \le x\}=P\{eX \le x\}=P\{X \le \log(x)\}=\int_0^{\log(x)} f(y)dy=\log(x)$$

By differentiating FY(x), we can conclude that the probability density function of Y is given by

$$f_Y(x)=\frac{1}{x} \qquad 1 \le x \le e$$

Hence,

$$E[eX]=E[Y]=\int_{-\infty}^{\infty} xf_Y(x)\,dx=\int_1^e 1\,dx=e-1$$

Although the method employed in **Example 2b** to compute the expected value of a function of X is always applicable, there is, as in the discrete case, an alternative way of proceeding. The following is a direct analog of **Proposition 4.1** of **Chapter 4**.

Proposition 2.1

If X is a continuous random variable with probability density function f(x), then, for any real-valued

function g,

$$E[g(X)]=\int_{-\infty}^{\infty} g(x)f(x)\,dx$$

An application of **Proposition 2.1** to **Example 2b** yields

$$E[eX]=\int_0^1 ex\,dx \qquad \text{since } f(x)=1,\quad 0<x<1=e-1$$

which is in accord with the result obtained in that example.

The proof of **Proposition 2.1** is more involved than that of its discrete random variable analog. We will present such a proof under the provision that the random variable g(X) is nonnegative. (The general proof, which follows the argument in the case we present, is indicated in **Theoretical Exercises 5.2** and **5.3**.) We will need the following lemma, which is of independent interest.

Lemma 2.1

For a nonnegative random variable Y,

$$E[Y]=\int_0^{\infty} P\{Y>y\}\,dy$$

Proof We present a proof when Y is a continuous random variable with probability density function fY. We have

$$\int_0^{\infty} P\{Y>y\}\,dy=\int_0^{\infty}\int_y^{\infty} fY(x)\,dx\,dy$$

where we have used the fact that $P\{Y>y\}=\int_y^{\infty} fY(x)\,dx$. Interchanging the order of integration in the preceding equation yields

$$\int_0^{\infty} P\{Y>y\}\,dy=\int_0^{\infty}\left(\int_0^x dy\right)fY(x)\,dx=\int_0^{\infty} xfY(x)\,dx=E[Y]$$

Proof of Proposition 2.1 From **Lemma 2.1**, for any function g for which $g(x)\geq 0$,

$$E[g(X)]=\int_0^{\infty} P\{g(X)>y\}\,dy=\int_0^{\infty}\int_{x:g(x)>y} f(x)\,dx\,dy=\int_{x:g(x)>0}\int_0^{g(x)} dy\,f(x)\,dx=\int_{x:g(x)>0} g(x)f(x)\,dx$$

which completes the proof.

Example 2c

A stick of length 1 is split at a point U having density function f(u)=1, 0<u<1. Determine the expected length of the piece that contains the point p, $0\leq p\leq 1$.

Solution

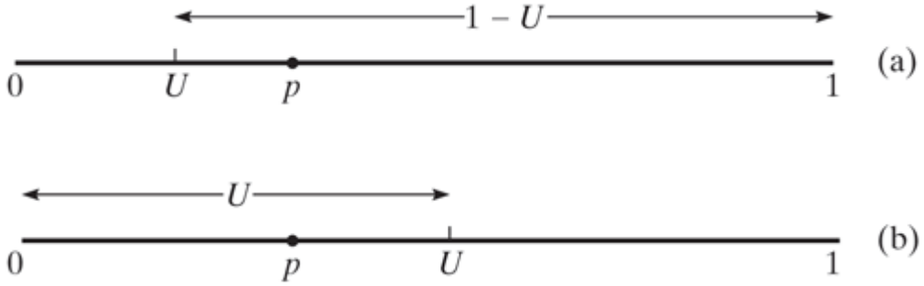
Let Lp(U) denote the length of the substick that contains the point p, and note that

$$Lp(U)=\begin{cases}1-U & U<p\\ U & U>p\end{cases}$$

(See **Figure 5.2**.) Hence, from **Proposition 2.1**,

$$E[Lp(U)]=\int_0^1 Lp(u)du=\int_0^p(1-u)du+\int_p^1 udu=12-(1-p)22+12-p22=12+p(1-p)$$

Figure 5.2 Substick containing point p: (a) U<p; (b) U>p.



Since p(1−p) is maximized when p=12, it is interesting to note that the expected length of the substick containing the point p is maximized when p is the midpoint of the original stick.

Example 2d

Suppose that if you are s minutes early for an appointment, then you incur the cost *cs*, and if you are s minutes late, then you incur the cost *ks*. Suppose also that the travel time from where you presently are to the location of your appointment is a continuous random variable having probability density function f. Determine the time at which you should depart if you want to minimize your expected cost.

Solution

Let X denote the travel time. If you leave t minutes before your appointment, then your cost—call it Ct(X)—is given by

$$Ct(X)=\begin{cases}c(t-X) & \text{if } X\le t\\ k(X-t) & \text{if } X\ge t\end{cases}$$

Therefore,

$$E[Ct(X)]=\int_0^\infty Ct(x)f(x)\,dx=\int_0^t c(t-x)f(x)\,dx+\int_t^\infty k(x-t)f(x)\,dx=ct\int_0^t f(x)\,dx-c\int_0^t xf(x)\,dx+k\int_t^\infty xf(x)\,dx-kt\int_t^\infty f(x)\,dx$$

The value of t that minimizes E[Ct(X)] can now be obtained by calculus. Differentiation yields

$$\frac{d}{dt}E[Ct(X)]=ct\,f(t)+cF(t)-ct\,f(t)-kt\,f(t)+kt\,f(t)-k[1-F(t)]=(k+c)F(t)-k$$

Equating the rightmost side to zero shows that the minimal expected cost is obtained when you leave t* minutes before your appointment, where t* satisfies

$$F(t^*)=\frac{k}{k+c}$$

As in **Chapter 4**, we can use **Proposition 2.1** to show the following.

Corollary 2.1

If a and b are constants, then

$$E[aX+b]=aE[X]+b$$

The proof of **Corollary 2.1** for a continuous random variable X is the same as the one given for a discrete random variable. The only modification is that the sum is replaced by an integral and the probability mass function by a probability density function.

The variance of a continuous random variable is defined exactly as it is for a discrete random variable, namely, if X is a random variable with expected value μ, then the variance of X is defined (for any type

of random variable) by

$$\text{Var}(X)=E[(X-\mu)2]$$

The alternative formula,

$$\text{Var}(X)=E[X2]-(E[X])2$$

is established in a manner similar to its counterpart in the discrete case.

Example 2e

Find Var(X) for X as given in **Example 2a**.

Solution

We first compute E[X2].

$$E[X2]=\int_{-\infty}^{\infty} x2f(x)\,dx=\int_{0}^{1} 2x3\,dx=12$$

Hence, since E[X]=23, we obtain

$$\text{Var}(X)=12-(23)2=118$$

It can be shown that, for constants a and b,

$$\text{Var}(aX+b)=a2\text{Var}(X)$$

The proof mimics the one given for discrete random variables.

There are several important classes of continuous random variables that appear frequently in applications of probability; the next few sections are devoted to a study of some of them.

5.3 The Uniform Random Variable

A random variable is said to be *uniformly* distributed over the interval (0, 1) if its probability density function is given by

$$f(x)=\begin{cases}1 & 0<x<1\\ 0 & \text{otherwise}\end{cases} \tag{3.1}$$

Note that **Equation (3.1)** is a density function, since $f(x)\geq 0$ and $\int_{-\infty}^{\infty} f(x)\,dx=\int_0^1 dx=1$. Because $f(x)>0$ only when $x\in(0,1)$, it follows that X must assume a value in interval (0, 1). Also, since f(x) is constant for $x\in(0,1)$, X is just as likely to be near any value in (0, 1) as it is to be near any other value. To verify this statement, note that for any $0<a<b<1$,

$$P\{a\leq X\leq b\}=\int_a^b f(x)\,dx=b-a$$

In other words, the probability that X is in any particular subinterval of (0, 1) equals the length of that subinterval.

In general, we say that X is a uniform random variable on the interval (α, β) if the probability density function of X is given by

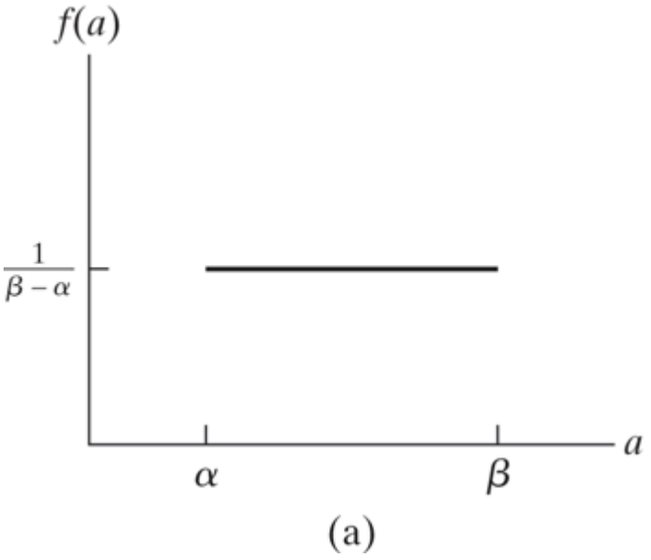
$$f(x)=\begin{cases}\frac{1}{\beta-\alpha} & \text{if } \alpha<x<\beta\\ 0 & \text{otherwise}\end{cases} \tag{3.2}$$

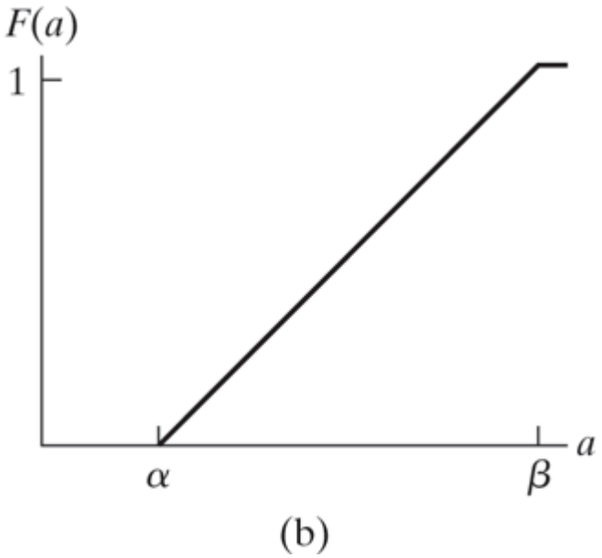
Since $F(a)=\int_{-\infty}^{a} f(x)\,dx$, it follows from **Equation (3.2)** that the distribution function of a uniform random variable on the interval (α, β) is given by

$$F(a)=\begin{cases}0 & a\leq\alpha\\ \frac{a-\alpha}{\beta-\alpha} & \alpha<a<\beta\\ 1 & a\geq\beta\end{cases}$$

Figure 5.3 presents a graph of f(a) and F(a).

Figure 5.3 Graph of (a) f(a) and (b) F(a) for a uniform (α, β) random variable.





(b)

Example 3a

Let X be uniformly distributed over (α, β). Find (a) E[X] and (b) Var(X).

Solution

a. $E[X]=\int_{-\infty}^{\infty} xf(x)\,dx=\int_{\alpha}^{\beta} x\beta-\alpha dx=\beta^2-\alpha^2 2(\beta-\alpha)=\beta+\alpha 2$

   In words, the expected value of a random variable that is uniformly distributed over some interval is equal to the midpoint of that interval.
b. To find Var(X), we first calculate E[X2].
   $E[X2]=\int_{\alpha}^{\beta} 1\beta-\alpha x2\,dx=\beta^3-\alpha^3 3(\beta-\alpha)=\beta^2+\alpha\beta+\alpha^2 3$

   Hence,
   $Var(X)=\beta^2+\alpha\beta+\alpha^2 3-(\alpha+\beta)^2 4=(\beta-\alpha)^2 12$

   Therefore, the variance of a random variable that is uniformly distributed over some interval is the square of the length of that interval divided by 12.

Example 3b

If X is uniformly distributed over (0, 10), calculate the probability that (a) X<3, (b) X>6, and (c) 3<X<8.

Solution

a. $P\{X<3\}=\int_0^3 110\,dx=310$
b. $P\{X>6\}=\int_6^{10} 110\,dx=410$
c. $P\{3<X<8\}=\int_3^8 110\,dx=12$

Example 3c

Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. That is, they arrive at 7, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits

a. less than 5 minutes for a bus;
b. more than 10 minutes for a bus.

Solution

Let X denote the number of minutes past 7 that the passenger arrives at the stop. Since X is a uniform random variable over the interval (0, 30), it follows that the passenger will have to wait less than 5 minutes if (and only if) he arrives between 7:10 and 7:15 or between 7:25 and 7:30. Hence, the desired probability for part (a) is

$$P\{10<X<15\}+P\{25<X<30\}=\int_{10}^{15}\frac{1}{30}dx+\int_{25}^{30}\frac{1}{30}dx=\frac{1}{3}$$

Similarly, he would have to wait more than 10 minutes if he arrives between 7 and 7:05 or between 7:15 and 7:20, so the probability for part (b) is

$$P\{0<X<5\}+P\{15<X<20\}=\frac{1}{3}$$

The next example was first considered by the French mathematician Joseph L. F. Bertrand in 1889 and is often referred to as *Bertrand's paradox*. It represents our initial introduction to a subject commonly referred to as *geometrical probability*.

Example 3d

Consider a random chord of a circle. What is the probability that the length of the chord will be greater than the side of the equilateral triangle inscribed in that circle?

Solution

As stated, the problem is incapable of solution because it is not clear what is meant by a random chord. To give meaning to this phrase, we shall reformulate the problem in two distinct ways.

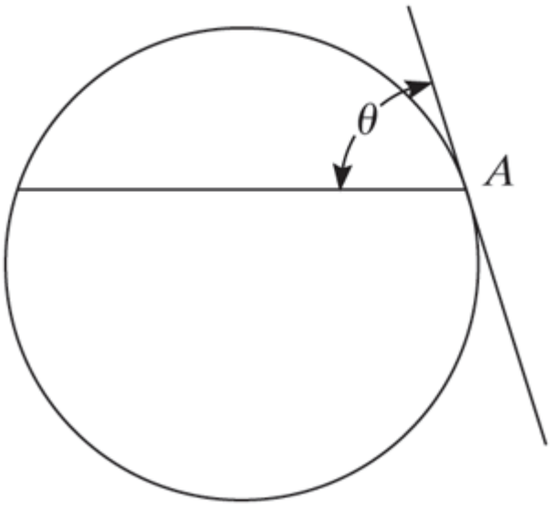
The first formulation is as follows: The position of the chord can be determined by its distance from the center of the circle. This distance can vary between 0 and r, the radius of the circle. Now, the length of the chord will be greater than the side of the equilateral triangle inscribed in the circle if the distance from the chord to the center of the circle is less than r/2. Hence, by assuming that a random chord is a chord whose distance D from the center of the circle is uniformly distributed between 0 and r, we see that the probability that the length of the chord is greater than the side of an inscribed equilateral triangle is

$$P\{D<\frac{r}{2}\}=\frac{r/2}{r}=\frac{1}{2}$$

For our second formulation of the problem, consider an arbitrary chord of the circle; through one end of the chord, draw a tangent. The angle θ between the chord and the tangent, which can vary from 0◦ to 180◦, determines the position of the chord. (See **Figure 5.4**.) Furthermore, the length of the chord will be greater than the side of the inscribed equilateral triangle if the angle θ is between 60◦ and 120◦. Hence, assuming that a random chord is a chord whose angle θ is uniformly distributed between 0◦ and 180◦, we see that the desired answer in this formulation is

$$P\{60<\theta<120\}=\frac{120-60}{180}=\frac{1}{3}$$

Figure 5.4



Note that random experiments could be performed in such a way that 12 or 13 would be the correct probability. For instance, if a circular disk of radius r is thrown on a table ruled with parallel lines a distance 2r apart, then one and only one of these lines would cross the disk and form a chord. All distances from this chord to the center of the disk would be equally likely, so that the desired probability that the chord's length will be greater than the side of an inscribed equilateral triangle is 12. In contrast, if the experiment consisted of rotating a needle freely about a point A on the edge (see **Figure 5.4**) of the circle, the desired answer would be 13.

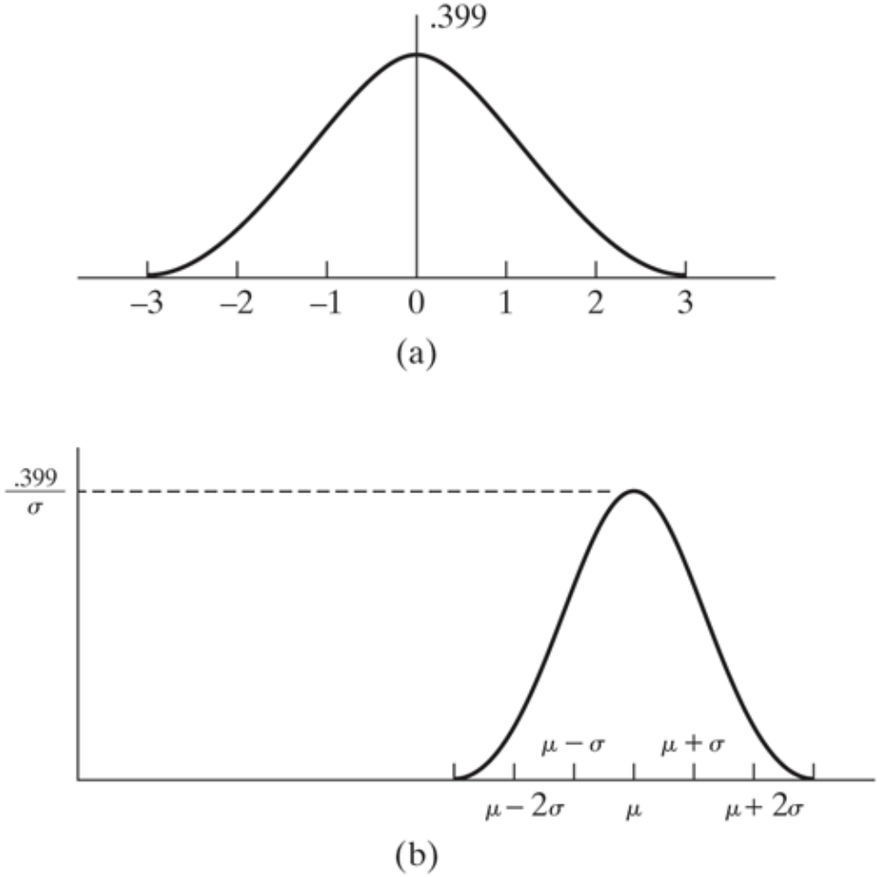
5.4 Normal Random Variables

We say that X is a normal random variable, or simply that X is normally distributed, with parameters μ and σ2 if the density of X is given by

$$f(x)=\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2} \qquad -\infty<x<\infty$$

This density function is a bell-shaped curve that is symmetric about μ. (See **Figure 5.5**.)

Figure 5.5 Normal density function: (a) μ=0, σ=1; (b) arbitrary μ, σ2.



The normal distribution was introduced by the French mathematician Abraham DeMoivre in 1733, who used it to approximate probabilities associated with binomial random variables when the binomial parameter n is large. This result was later extended by Laplace and others and is now encompassed in a probability theorem known as the central limit theorem, which is discussed in **Chapter 8**. The *central limit theorem*, one of the two most important results in probability theory,† gives a theoretical base to the often noted empirical observation that, in practice, many random phenomena obey, at least approximately, a normal probability distribution. Some examples of random phenomena obeying this behavior are the height of a man or woman, the velocity in any direction of a molecule in gas, and the error made in measuring a physical quantity.

† The other is the strong law of large numbers.

To prove that f(x) is indeed a probability density function, we need to show that

$$\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty}e^{-(x-\mu)^2/2\sigma^2}\,dx=1$$

Making the substitution $y=(x-\mu)/\sigma$, we see that

$$\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{-(x-\mu)^2/2\sigma^2}\,dx=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-y^2/2}\,dy$$

Hence, we must show that

$$\int_{-\infty}^{\infty} e^{-y^2/2}\,dy=\sqrt{2\pi}$$

Toward this end, let $I=\int_{-\infty}^{\infty} e^{-y^2/2}\,dy$. Then

$$I^2=\int_{-\infty}^{\infty} e^{-y^2/2}\,dy\int_{-\infty}^{\infty} e^{-x^2/2}\,dx=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(y^2+x^2)/2}\,dy\,dx$$

We now evaluate the double integral by means of a change of variables to polar coordinates. (That is, let $x=r\cos\theta$, $y=r\sin\theta$, and $dy\,dx=r\,d\theta\,dr$.) Thus,

$$I^2=\int_0^{\infty}\int_0^{2\pi} e^{-r^2/2}r\,d\theta\,dr=2\pi\int_0^{\infty} re^{-r^2/2}\,dr=-2\pi e^{-r^2/2}\Big|_0^{\infty}=2\pi$$

Hence, $I=\sqrt{2\pi}$, and the result is proved.

An important fact about normal random variables is that if X is normally distributed with parameters μ and σ^2, then $Y=aX+b$ is normally distributed with parameters $a\mu+b$ and $a^2\sigma^2$. To prove this statement, suppose that $a>0$. (The proof when $a<0$ is similar.) Let F_Y denote the cumulative distribution function of Y. Then

$$F_Y(x)=P\{Y\le x\}=P\{aX+b\le x\}=P\{X\le x-ba\}=F_X(x-ba)$$

where F_X is the cumulative distribution function of X. By differentiation, the density function of Y is then

$$f_Y(x)=\frac{1}{a}f_X(x-ba)=\frac{1}{\sqrt{2\pi}a\sigma}\exp\{-(x-ba-\mu)^2/2\sigma^2\}=\frac{1}{\sqrt{2\pi}a\sigma}\exp\{-(x-b-a\mu)^2/2(a\sigma)^2\}$$

which shows that Y is normal with parameters $a\mu+b$ and $a^2\sigma^2$.

An important implication of the preceding result is that if X is normally distributed with parameters μ and σ^2, then $Z=(X-\mu)/\sigma$ is normally distributed with parameters 0 and 1. Such a random variable is said to be a *standard*, or a *unit*, normal random variable.

We now show that the parameters μ and σ^2 of a normal random variable represent, respectively, its expected value and variance.

Example 4a

Find E[X] and Var(X) when X is a normal random variable with parameters μ and σ^2.

Solution

Let us start by finding the mean and variance of the standard normal random variable $Z=(X-\mu)/\sigma$. We have

$$E[Z]=\int_{-\infty}^{\infty} xf_Z(x)\,dx=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} xe^{-x^2/2}\,dx=-\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\Big|_{-\infty}^{\infty}=0$$

Thus,

$$\mathrm{Var}(Z)=E[Z^2]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} x^2e^{-x^2/2}\,dx$$

Integration by parts (with $u=x$ and $dv=xe^{-x^2/2}$) now gives

$$\mathrm{Var}(Z)=\frac{1}{\sqrt{2\pi}}\left(-xe^{-x^2/2}\Big|_{-\infty}^{\infty}+\int_{-\infty}^{\infty} e^{-x^2/2}\,dx\right)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}\,dx=1$$

Because $X=\mu+\sigma Z$, the preceding yields the results

$$E[X]=\mu+\sigma E[Z]=\mu$$

and

$$Var(X)=\sigma^2 Var(Z)=\sigma^2$$

It is customary to denote the cumulative distribution function of a standard normal random variable by $\Phi(x)$. That is,

$$\Phi(x)=\frac{1}{2\pi}\int_{-\infty}^{x} e^{-y^2/2}\,dy$$

The values of $\Phi(x)$ for nonnegative x are given in **Table 5.1**. For negative values of x, $\Phi(x)$ can be obtained from the relationship

$$\Phi(-x)=1-\Phi(x) \qquad -\infty<x<\infty \tag{4.1}$$

Table 5.1 Area Φ(x) Under the Standard Normal Curve to the Left of X.

X	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

The proof of **Equation (4.1)**, which follows from the symmetry of the standard normal density, is left as an exercise. This equation states that if Z is a standard normal random variable, then

$$P\{Z \le -x\} = P\{Z > x\} \qquad -\infty < x < \infty$$

Since $Z = (X - \mu)/\sigma$ is a standard normal random variable whenever X is normally distributed with parameters μ and σ^2, it follows that the distribution function of X can be expressed as

$$F_X(a) = P\{X \le a\} = P\left(\frac{X - \mu}{\sigma} \le \frac{a - \mu}{\sigma}\right) = \Phi\left(\frac{a - \mu}{\sigma}\right)$$

Example 4b

If X is a normal random variable with parameters $\mu = 3$ and $\sigma^2 = 9$, find (a) $P\{2 < X < 5\}$; (b) $P\{X > 0\}$; (c)

$P\{|X-3|>6\}$.

Solution

a. $P\{2<X<5\}=P\{2-33<X-33<5-33\}=P\{-13<Z<23\}=\Phi(23)-\Phi(-13)=\Phi(23)-[1-\Phi(13)]\approx.3779$

b. $P\{X>0\}=P\{X-33>0-33\}=P\{Z>-1\}=1-\Phi(-1)=\Phi(1)\approx.8413$

c. $P\{|X-3|>6\}=P\{X>9\}+P\{X<-3\}=P\{X-33>9-33\}+P\{X-33<-3-33\}=P\{Z>2\}+P\{Z<-2\}=1-\Phi(2)+\Phi(-2)=2[1-\Phi(2)]\approx.0456$

Example 4c

An examination is frequently regarded as being good (in the sense of determining a valid grade spread for those taking it) if the test scores of those taking the examination can be approximated by a normal density function. (In other words, a graph of the frequency of grade scores should have approximately the bell-shaped form of the normal density.) The instructor often uses the test scores to estimate the normal parameters μ and $\sigma 2$ and then assigns the letter grade A to those whose test score is greater than $\mu+\sigma$, B to those whose score is between μ and $\mu+\sigma$, C to those whose score is between $\mu-\sigma$ and μ, D to those whose score is between $\mu-2\sigma$ and $\mu-\sigma$, and F to those getting a score below $\mu-2\sigma$. (This strategy is sometimes referred to as grading “on the curve.”) Since

$P\{X>\mu+\sigma\} =P\{ \frac{X-\mu}{\sigma} >1 \}=1-\Phi(1)\approx .1587$ $P\{\mu<X<\mu+\sigma\} =P\{ 0< \frac{X-\mu}{\sigma} <1 \}=\Phi(1)-\Phi(0)\approx .3413$ $P\{\mu-\sigma<X<\mu\} =P\{ -1< \frac{X-\mu}{\sigma} <0 \} =\Phi(0)-\Phi(-1)\approx .3413$ $P\{\mu-2\sigma<X<\mu-\sigma\} =P\{ -2< \frac{X-\mu}{\sigma} <-1 \} =\Phi(2)-\Phi(1)\approx .1359$ $P\{X<\mu-2\sigma\} =P\{ \frac{X-\mu}{\sigma} <-2 \}=\Phi(-2)\approx .0228$

it follows that approximately 16 percent of the class will receive an A grade on the examination, 34 percent a B grade, 34 percent a C grade, and 14 percent a D grade; 2 percent will fail.

Example 4d

An expert witness in a paternity suit testifies that the length (in days) of human gestation is approximately normally distributed with parameters $\mu=270$ and $\sigma 2=100$. The defendant in the suit is able to prove that he was out of the country during a period that began 290 days before the birth of the child and ended 240 days before the birth. If the defendant was, in fact, the father of the child, what is the probability that the mother could have had the very long or very short gestation indicated by the testimony?

Solution

Let X denote the length of the gestation, and assume that the defendant is the father. Then the probability that the birth could occur within the indicated period is

$P\{X>290 \text{ or } X<240\}=P\{X>290\}+P\{X<240\}=P\{\frac{X-270}{10}>2\}+P\{\frac{X-270}{10}<-3\}=1-\Phi(2)+1-\Phi(3)\approx.0241$

Example 4e

Suppose that a binary message either 0 or 1 must be transmitted by wire from location A to location B. However, the data sent over the wire are subject to a channel noise disturbance, so, to reduce the possibility of error, the value 2 is sent over the wire when the message is 1 and the value -2 is sent when the message is 0. If $x, x=\pm 2$, is the value sent at location A, then R, the value received at location B, is given by $R=x+N$, where N is the channel noise disturbance. When the message is received at location B, the receiver decodes it according to the following rule:

If $R\geq.5$, then 1 is concluded.

If $R<.5$, then 0 is concluded.

Because the channel noise is often normally distributed, we will determine the error probabilities when N is a

standard normal random variable.

Two types of errors can occur: One is that the message 1 can be incorrectly determined to be 0, and the other is that 0 can be incorrectly determined to be 1. The first type of error will occur if the message is 1 and $2+N<.5$, whereas the second will occur if the message is 0 and $-2+N\geq.5$. Hence,

$$P\{\text{error}|\text{message is 1}\}=P\{N<-1.5\}=1-\Phi(1.5)\approx.0668$$

and

$$P\{\text{error}|\text{message is 0}\}=P\{N\geq 2.5\}=1-\Phi(2.5)\approx.0062$$

Example 4f

Value at Risk (VAR) has become a key concept in financial calculations. The VAR of an investment is defined as that value v such that there is only a 1 percent chance that the loss from the investment will be greater than v. If X, the gain from an investment, is a normal random variable with mean μ and variance σ^2, then because the loss is equal to the negative of the gain, the VAR of such an investment is that value v such that

$$.01=P\{-X>v\}$$

Using that $-X$ is normal with mean $-\mu$ and variance σ^2, we see that

$$.01=P\{-X+\mu\sigma>v+\mu\sigma\}=1-\Phi(v+\mu\sigma)$$

Because, as indicated by **Table 5.1**, $\Phi(2.33)=.99$, we see that

$$v+\mu\sigma=2.33$$

That is,

$$v=\text{VAR}=2.33\sigma-\mu$$

Consequently, among a set of investments all of whose gains are normally distributed, the investment having the smallest VAR is the one having the largest value of $\mu-2.33\sigma$.

5.4.1 The Normal Approximation to the Binomial Distribution

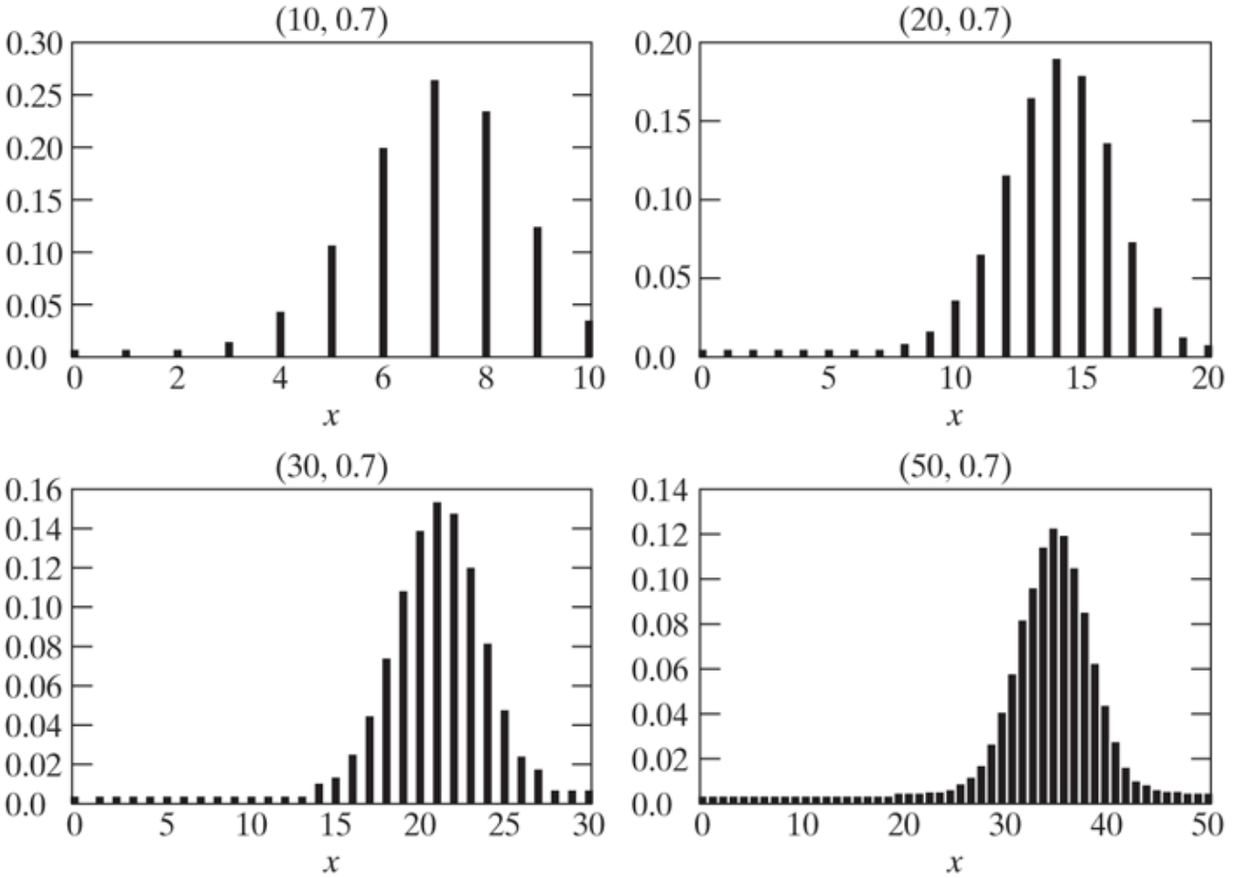
An important result in probability theory known as the DeMoivre–Laplace limit theorem states that when n is large, a binomial random variable with parameters n and p will have approximately the same distribution as a normal random variable with the same mean and variance as the binomial. This result was proved originally for the special case of $p=\frac{1}{2}$ by DeMoivre in 1733 and was then extended to general p by Laplace in 1812. It formally states that if we "standardize" the binomial by first subtracting its mean *np* and then dividing the result by its standard deviation $\sqrt{np(1-p)}$, then the distribution function of this standardized random variable (which has mean 0 and variance 1) will converge to the standard normal distribution function as $n\to\infty$.

> The DeMoivre–Laplace Limit Theorem
>
> If S_n denotes the number of successes that occur when n independent trials, each resulting in a success with probability p, are performed, then, for any $a<b$,
>
> $$P\left\{a\leq\frac{S_n-np}{\sqrt{np(1-p)}}\leq b\right\}\to\Phi(b)-\Phi(a)$$
>
> as $n\to\infty$.

Because the preceding theorem is only a special case of the central limit theorem, which is presented in **Chapter 8**, we shall not present a proof.

Note that we now have two possible approximations to binomial probabilities: the Poisson approximation, which is good when n is large and p is small, and the normal approximation, which can be shown to be quite good when np(1−p) is large. (See **Figure 5.6**.) [The normal approximation will, in general, be quite good for values of n satisfying np(1−p)≥10.]

Figure 5.6 The probability mass function of a binomial (*n, p*) random variable becomes more and more "normal" as *n* becomes larger and larger.



Example 4g

Let X be the number of times that a fair coin that is flipped 40 times lands on heads. Find the probability that X=20. Use the normal approximation and then compare it with the exact solution.

Solution

To employ the normal approximation, note that because the binomial is a discrete integer-valued random variable, whereas the normal is a continuous random variable, it is best to write P{X=i} as P{i−1/2<X<i+1/2} before applying the normal approximation (this is called the *continuity correction*). Doing so gives

$$P\{X=20\}=P\{19.5<X<20.5\}=P\{19.5-2010<X-2010<20.5-2010\}\approx P\{-.16<X-2010<.16\}\approx\Phi(.16)-\Phi(-.16)\approx.1272$$

The exact result is

$$P\{X=20\}=(4020)(12)40\approx.1254$$

Example 4h

The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year

students attend this college.

Solution

If X denotes the number of students who attend, then X is a binomial random variable with parameters n=450 and p=.3. Using the continuity correction, we see that the normal approximation yields

$$P\{X \geq 150.5\} = P\{X - (450)(.3) \over 450(.3)(.7) \geq 150.5 - (450)(.3) \over 450(.3)(.7)\} \approx 1 - \Phi(1.59) \approx .0559$$

Hence, less than 6 percent of the time do more than 150 of the first 450 accepted actually attend. (What independence assumptions have we made?)

Example 4i

To determine the effectiveness of a certain diet in reducing the amount of cholesterol in the bloodstream, 100 people are put on the diet. After they have been on the diet for a sufficient length of time, their cholesterol count will be taken. The nutritionist running this experiment has decided to endorse the diet if at least 65 percent of the people have a lower cholesterol count after going on the diet. What is the probability that the nutritionist endorses the new diet if, in fact, it has no effect on the cholesterol level?

Solution

Let us assume that if the diet has no effect on the cholesterol count, then, strictly by chance, each person's count will be lower than it was before the diet with probability 12. Hence, if X is the number of people whose count is lowered, then the probability that the nutritionist will endorse the diet when it actually has no effect on the cholesterol count is

$$\sum_{i=65}^{100} \binom{100}{i} \left(\frac{1}{2}\right)^{100} = P\{|X| \geq 64.5\} = P\left\{ \frac{X - (100)\left(\frac{1}{2}\right)}{100\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)} \geq 2.9 \right\} \approx 1 - \Phi(2.9) \approx .0019$$

Example 4j

Fifty-two percent of the residents of New York City are in favor of outlawing cigarette smoking on university campuses. Approximate the probability that more than 50 percent of a random sample of n people from New York are in favor of this prohibition when

a. n=11
b. n=101
c. n=1001

How large would n have to be to make this probability exceed .95?

Solution

Let N denote the number of residents of New York City. To answer the preceding question, we must first understand that a *random sample* of size n is a sample such that the n people were chosen in such a manner that each of the $\binom{N}{n}$ subsets of n people had the same chance of being the chosen subset. Consequently, Sn, the number of people in the sample who are in favor of the smoking prohibition, is a hypergeometric random variable. That is, Sn has the same distribution as the number of white balls obtained when n balls are chosen from an urn of N balls, of which .52N are white. But because N and .52N are both large in comparison with the sample size n, it follows from the binomial approximation to the hypergeometric (see **Section 4.8.3**) that the distribution of Sn is closely approximated by a binomial distribution with parameters n and p=.52. The normal approximation to the binomial distribution then shows that

$$P\{Sn > .5n\} = P\left\{ \frac{Sn - .52n}{n(.52)(.48)} > \frac{.5n - .52n}{n(.52)(.48)} \right\} = P\left\{ \frac{Sn - .52n}{n(.52)(.48)} > \frac{.04n}{} \right\} \approx \Phi(.04n)$$

Thus,

$$P\{Sn > .5n\} \approx \begin{cases} \Phi(.1328) = .5528, & \text{if } n = 11 \\ \Phi(.4020) = .6562, & \text{if } n = 101 \\ \Phi(1.2665) = .8973, & \text{if } n = 1001 \end{cases}$$

In order for this probability to be at least .95, we would need $\Phi(.04n)>.95$. Because $\Phi(x)$ is an increasing function and $\Phi(1.645)=.95$, this means that

$.04n>1.645$

or

$n\geq 1691.266$

That is, the sample size would have to be at least 1692.

Historical notes concerning the normal distribution

The normal distribution was introduced by the French mathematician Abraham DeMoivre in 1733. DeMoivre, who used this distribution to approximate probabilities connected with coin tossing, called it the exponential bell-shaped curve. Its usefulness, however, became truly apparent only in 1809, when the famous German mathematician Karl Friedrich Gauss used it as an integral part of his approach to predicting the location of astronomical entities. As a result, it became common after this time to call it the *Gaussian distribution*.

During the mid- to late 19th century, however, most statisticians started to believe that the majority of data sets would have histograms conforming to the Gaussian bell-shaped form. Indeed, it came to be accepted that it was "normal" for any well-behaved data set to follow this curve. As a result, following the lead of the British statistician Karl Pearson, people began referring to the Gaussian curve by calling it simply the *normal* curve. (A partial explanation as to why so many data sets conform to the normal curve is provided by the central limit theorem, which is presented in **Chapter 8**.)

Abraham DeMoivre (1667–1754)

Today there is no shortage of statistical consultants, many of whom ply their trade in the most elegant of settings. However, the first of their breed worked, in the early years of the 18th century, out of a dark, grubby betting shop in Long Acres, London, known as Slaughter's Coffee House. He was Abraham DeMoivre, a Protestant refugee from Catholic France, and, for a price, he would compute the probability of gambling bets in all types of games of chance.

Although DeMoivre, the discoverer of the normal curve, made his living at the coffee shop, he was a mathematician of recognized abilities. Indeed, he was a member of the Royal Society and was reported to be an intimate of Isaac Newton.

Listen to Karl Pearson imagining DeMoivre at work at Slaughter's Coffee House: "*I picture DeMoivre working at a dirty table in the coffee house with a broken-down gambler beside him and Isaac Newton walking through the crowd to his corner to fetch out his friend. It would make a great picture for an inspired artist*."

Karl Friedrich Gauss

Karl Friedrich Gauss (1777–1855), one of the earliest users of the normal curve, was one of the greatest mathematicians of all time. Listen to the words of the well-known mathematical historian E. T. Bell, as expressed in his 1954 book *Men of Mathematics*: In a chapter entitled "The Prince of Mathematicians," he writes, "*Archimedes, Newton, and Gauss; these three are in a class by themselves among the great mathematicians, and it is not for ordinary mortals to attempt to rank them in order of merit. All three started tidal waves in both pure and applied mathematics. Archimedes esteemed his pure mathematics more highly than its applications;*

Newton appears to have found the chief justification for his mathematical inventions in the scientific uses to which he put them; while Gauss declared it was all one to him whether he worked on the pure or on the applied side."

5.5 Exponential Random Variables

A continuous random variable whose probability density function is given, for some $\lambda>0$, by

$$f(x)=\begin{cases}\lambda e^{-\lambda x} & \text{if } x\geq 0\\ 0 & \text{if } x<0\end{cases}$$

is said to be an *exponential* random variable (or, more simply, is said to be exponentially distributed) with parameter λ. The cumulative distribution function $F(a)$ of an exponential random variable is given by

$$F(a) = P\{X\leq a\} = \int_0^a \lambda e^{-\lambda x}\, dx = -e^{-\lambda x}\,|_0^a = 1-e^{-\lambda a} \qquad a\geq 0$$

Note that $F(\infty)=\int_0^\infty \lambda e^{-\lambda x}\, dx=1$, as, of course, it must. The parameter λ will now be shown to equal the reciprocal of the expected value.

Example 5a

Let X be an exponential random variable with parameter λ. Calculate (a) $E[X]$ and (b) $Var(X)$.

Solution

(a) Since the density function is given by

$$f(x)=\begin{cases}\lambda e^{-\lambda x} & x\geq 0\\ 0 & x<0\end{cases}$$

we obtain, for $n>0$,

$$E[X^n]=\int_0^\infty x^n \lambda e^{-\lambda x}\, dx$$

Integrating by parts (with $\lambda e^{-\lambda x}=dv$ and $u=x^n$) yields

$$E[X^n] = -x^n e^{-\lambda x}\,|_0^\infty + \int_0^\infty e^{-\lambda x} n x^{n-1}\, dx = 0 + \frac{n}{\lambda}\int_0^\infty \lambda e^{-\lambda x} x^{n-1}\, dx = \frac{n}{\lambda} E[X^{n-1}]$$

Letting $n=1$ and then $n=2$ gives

$$E[X]= \frac{1}{\lambda} E[X^2]= \frac{2}{\lambda} E[X]= \frac{2}{\lambda^2}$$

(b) Hence,

$$Var(X)= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Thus, the mean of the exponential is the reciprocal of its parameter λ, and the variance is the mean squared.

In practice, the exponential distribution often arises as the distribution of the amount of time until some specific event occurs. For instance, the amount of time (starting from now) until an earthquake occurs, or until a new war breaks out, or until a telephone call you receive turns out to be a wrong number are all random variables that tend in practice to have exponential distributions. (For a theoretical explanation of this phenomenon, see **Section 4.7**.)

Example 5b

Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda=\frac{1}{10}$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait

a. more than 10 minutes;
b. between 10 and 20 minutes.

Solution

Let X denote the length of the call made by the person in the booth. Then the desired probabilities are

a. $P\{X>10\} = 1-F(10) = e^{-1} \approx .368$

b. $P\{10<X<20\} = F(20)-F(10) = e^{-1} - e^{-2} \approx .233$

We say that a nonnegative random variable X is *memoryless* if

$$P\{X>s+t \mid X>t\}=P\{X>s\} \quad \text{for all } s, t \geq 0 \tag{5.1}$$

If we think of X as being the lifetime of some instrument, **Equation (5.1)** states that the probability that the instrument survives for at least s+t hours, given that it has survived t hours, is the same as the initial probability that it survives for at least s hours. In other words, if the instrument is alive at age t, the distribution of the remaining amount of time that it survives is the same as the original lifetime distribution. (That is, it is as if the instrument does not "remember" that it has already been in use for a time t.)

Equation (5.1) is equivalent to

$$P\{X>s+t,X>t\}P\{X>t\}=P\{X>s\}$$

or

$$P\{X>s+t\}=P\{X>s\}P\{X>t\} \tag{5.2}$$

Since **Equation (5.2)** is satisfied when X is exponentially distributed (for $e^{-\lambda(s+t)}=e^{-\lambda s}e^{-\lambda t}$), it follows that exponentially distributed random variables are memoryless.

Example 5c

Consider a post office that is staffed by two clerks. Suppose that when Mr. Smith enters the system, he discovers that Ms. Jones is being served by one of the clerks and Mr. Brown by the other. Suppose also that Mr. Smith is told that his service will begin as soon as either Ms. Jones or Mr. Brown leaves. If the amount of time that a clerk spends with a customer is exponentially distributed with parameter λ, what is the probability that of the three customers, Mr. Smith is the last to leave the post office?

Solution

The answer is obtained by reasoning as follows: Consider the time at which Mr. Smith first finds a free clerk. At this point, either Ms. Jones or Mr. Brown would have just left, and the other one would still be in service. However, because the exponential is memoryless, it follows that the additional amount of time that this other person (either Ms. Jones or Mr. Brown) would still have to spend in the post office is exponentially distributed with parameter λ. That is, it is the same as if service for that person were just starting at this point. Hence, by symmetry, the probability that the remaining person finishes before Smith leaves must equal 12.

It turns out that not only is the exponential distribution memoryless, but it is also the unique distribution possessing this property. To see this, suppose that X is memoryless and let $\bar{F}(x)=P\{X>x\}$. Then, by **Equation (5.2)**,

$$\bar{F}(s+t)=\bar{F}(s)\bar{F}(t)$$

That is, $\bar{F}(\cdot)$ satisfies the functional equation

$$g(s+t)=g(s)g(t)$$

However, it turns out that the only right continuous solution of this functional equation is[†]

†One can prove **Equation (5.3)** as follows: If $g(s+t)=g(s)g(t)$, then

$$g(2n)=g(1n+1n)=g2(1n)$$

and repeating this yields $g(m/n)=gm(1/n)$. Also,

$$g(1)=g(1n+1n+\cdots+1n)=gn(1n) \quad \text{or} \quad g(1n)=(g(1))1/n$$

Hence, $g(m/n)=(g(1))m/n$, which, since g is right continuous, implies that $g(x)=(g(1))x$. Because $g(1)=(g(12))2\geq 0$, we obtain $g(x)=e-\lambda x$, where $\lambda=-\log(g(1))$.

$$g(x)=e-\lambda x \tag{5.3}$$

and, since a distribution function is always right continuous, we must have

$$\bar{F}(x)=e-\lambda x \quad \text{or} \quad F(x)=P\{X\leq x\}=1-e-\lambda x$$

which shows that X is exponentially distributed.

Example 5d

Suppose that the number of miles that a car can run before its battery wears out is exponentially distributed with an average value of 10,000 miles. If a person desires to take a 5000-mile trip, what is the probability that he or she will be able to complete the trip without having to replace the car battery? What can be said when the distribution is not exponential?

Solution

It follows by the memoryless property of the exponential distribution that the remaining lifetime (in thousands of miles) of the battery is exponential with parameter $\lambda=110$. Hence, the desired probability is

$$P\{\text{remaining lifetime} > 5\}=1-F(5)=e-5\lambda=e-1/2\approx .607$$

However, if the lifetime distribution *F* is not exponential, then the relevant probability is

$$P\{\text{lifetime}>t+5|\text{lifetime}>t\}=1-F(t+5)1-F(t)$$

where t is the number of miles that the battery had been in use prior to the start of the trip. Therefore, if the distribution is not exponential, additional information is needed (namely, the value of t) before the desired probability can be calculated.

A variation of the exponential distribution is the distribution of a random variable that is equally likely to be either positive or negative and whose absolute value is exponentially distributed with parameter $\lambda, \lambda \geq 0$. Such a random variable is said to have a *Laplace* distribution,‡ and its density is given by

‡ It also is sometimes called the double exponential random variable.

$$f(x)=\frac{1}{2}\lambda e^{-\lambda|x|} \quad -\infty<x<\infty$$

Its distribution function is given by

$$F(x) = \begin{cases} \frac{1}{2}\int_{-\infty}^{x} \lambda e^{\lambda y}\, dy & x<0 \\ \frac{1}{2}\int_{-\infty}^{0} \lambda e^{\lambda y}\, dy + \frac{1}{2}\int_{0}^{x} \lambda e^{-\lambda y}\, dy & x>0 \end{cases} = \begin{cases} \frac{1}{2} e^{\lambda x} & x<0 \\ 1-\frac{1}{2} e^{-\lambda x} & x>0 \end{cases}$$

Example 5e

Consider again **Example 4e**, which supposes that a binary message is to be transmitted from A to B, with the value 2 being sent when the message is 1 and −2 when it is 0. However, suppose now that rather than being a standard normal random variable, the channel noise N is a Laplacian random variable with parameter $\lambda=1$. Suppose again that if R is the value received at location B, then the message is decoded as follows:

If $R \geq .5$, then 1 is concluded.

If $R<.5$, then 0 is concluded.

In this case, where the noise is Laplacian with parameter $\lambda=1$, the two types of errors will have probabilities given by

$$P\{\text{error}|\text{message 1 is sent}\} = P\{N<-1.5\} = \frac{1}{2} e^{-1.5} \approx .1116$$
$$P\{\text{error}|\text{message 0 is sent}\} = P\{N \geq 2.5\} = \frac{1}{2} e^{-2.5} \approx .041$$

On comparing this with the results of **Example 4e**, we see that the error probabilities are higher when the noise is Laplacian with $\lambda=1$ than when it is a standard normal variable.

5.5.1 Hazard Rate Functions

Consider a positive continuous random variable X that we interpret as being the lifetime of some item. Let X have distribution function F and density f. The *hazard rate* (sometimes called the *failure rate*) function $\lambda(t)$ of F is defined by

$$\lambda(t)=\frac{f(t)}{\bar{F}(t)}, \quad \text{where } \bar{F} = 1 - F$$

To interpret $\lambda(t)$, suppose that the item has survived for a time t and we desire the probability that it will not survive for an additional time *dt*. That is, consider $P\{X \in (t,t+dt)|X>t\}$. Now,

$$P\{X \in (t,t+dt)|X>t\} = \frac{P\{X \in (t,t+dt), X>t\}}{P\{X>t\}} = \frac{P\{X \in (t,t+dt)\}}{P\{X>t\}} \approx \frac{f(t)}{\bar{F}(t)}\, dt$$

Thus, $\lambda(t)$ represents the conditional probability intensity that a t-unit-old item will fail.

Suppose now that the lifetime distribution is exponential. Then, by the memoryless property, it follows that the distribution of remaining life for a t-year-old item is the same as that for a new item. Hence, $\lambda(t)$ should be constant. In fact, this checks out, since

$$\lambda(t) = f(t)\,\bar{F}(t) = \lambda e^{-\lambda t} e^{-\lambda t} = \lambda$$

Thus, the failure rate function for the exponential distribution is constant. The parameter λ is often referred to as the *rate* of the distribution.

It turns out that the failure rate function $\lambda(s)$, $s \geq 0$, uniquely determines the distribution function F. To prove this, we integrate $\lambda(s)$ from 0 to t to obtain

$$\int_0^t \lambda(s)ds = \int_0^t \frac{f(s)}{1-F(s)}\,ds = -\log(1-F(s)) \Big|_0^t = -\log(1-F(t)) + \log(1-F(0)) = -\log(1-F(t))$$

where the second equality used that $f(s) = \frac{d}{ds}F(s)$ and the final equality used that $F(0) = 0$. Solving the preceding equation for $F(t)$ gives

$$F(t) = 1 - \exp\left\{-\int_0^t \lambda(s)\,ds\right\} \tag{5.4}$$

Hence, a distribution function of a positive continuous random variable can be specified by giving its hazard rate function. For instance, if a random variable has a linear hazard rate function—that is, if

$$\lambda(t) = a + bt$$

then its distribution function is given by

$$F(t) = 1 - e^{-at - bt^2/2}$$

and differentiation yields its density, namely,

$$f(t) = (a + bt)e^{-(at + bt^2/2)} \quad t \geq 0$$

When $a = 0$, the preceding equation is known as the *Rayleigh density function*.

Example 5f

One often hears that the death rate of a person who smokes is, at each age, twice that of a nonsmoker. What does this mean? Does it mean that a nonsmoker has twice the probability of surviving a given number of years as does a smoker of the same age?

Solution

If $\lambda_s(t)$ denotes the hazard rate of a smoker of age t and $\lambda_n(t)$ that of a nonsmoker of age t, then the statement at issue is equivalent to the statement that

$$\lambda_s(t) = 2\lambda_n(t)$$

The probability that an A-year-old nonsmoker will survive until age B, $A < B$, is

$$P\{\text{A-year-old nonsmoker reaches age B}\} = P\{\text{nonsmoker's lifetime} > B \mid \text{nonsmoker's lifetime} > A\} = \frac{\bar{F}_{non}(B)}{\bar{F}_{non}(A)} = \frac{\exp\left\{-\int_0^B \lambda_n(t)\,dt\right\}}{\exp\left\{-\int_0^A \lambda_n(t)\,dt\right\}} = \exp\left\{-\int_A^B \lambda_n(t)\,dt\right\} \tag{5.14}$$

whereas the corresponding probability for a smoker is, by the same reasoning,

$$P\{\text{A-year-old smoker reaches age B}\} = \exp\left\{-\int_A^B \lambda_s(t)\,dt\right\} = \exp\left\{-2\int_A^B \lambda_n(t)\,dt\right\} = \left[\exp\left\{-\int_A^B \lambda_n(t)\,dt\right\}\right]^2$$

In other words, for two people of the same age, one of whom is a smoker and the other a nonsmoker,

the probability that the smoker survives to any given age is the *square* (not one-half) of the corresponding probability for a nonsmoker. For instance, if $\lambda_n(t)=130$, $50 \le t \le 60$, then the probability that a 50-year-old nonsmoker reaches age 60 is $e^{-1/3} \approx .7165$, whereas the corresponding probability for a smoker is $e^{-2/3} \approx .5134$.

Equation (5.4) can be used to show that only exponential random variables are memoryless. For if a random variable has a memoryless distribution then the remaining life of a s year old must be the same for all s. That is, if X is memoryless, then $\lambda(s)=c$. But, by **Equation (5.4)**, this implies that the distribution function of X is $F(t)=1-e^{-ct}$, showing that X is exponential with rate c.

Summary

A random variable X is *continuous* if there is a nonnegative function f, called the *probability density function* of X, such that, for any set B,

$$P\{X \in B\} = \int_B f(x)\, dx$$

If X is continuous, then its distribution function F will be differentiable and

$$\frac{d}{dx}F(x) = f(x)$$

The expected value of a continuous random variable X is defined by

$$E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

A useful identity is that for any function g,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\, dx$$

As in the case of a discrete random variable, the variance of X is defined by

$$Var(X) = E[(X - E[X])^2]$$

A random variable X is said to be *uniform* over the interval (a, b) if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

Its expected value and variance are

$$E[X] = \frac{a+b}{2} \quad Var(X) = \frac{(b-a)^2}{12}$$

A random variable X is said to be *normal* with parameters μ and σ^2 if its probability density function is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2} \quad -\infty < x < \infty$$

It can be shown that

$$\mu = E[X] \quad \sigma^2 = Var(X)$$

If X is normal with mean μ and variance σ^2, then Z, defined by

$$Z = \frac{X - \mu}{\sigma}$$

is normal with mean 0 and variance 1. Such a random variable is said to be a *standard* normal random variable. Probabilities about X can be expressed in terms of probabilities about the standard normal variable Z, whose probability distribution function can be obtained either from **Table 5.1**, the normal calculator on StatCrunch, or a website.

When n is large, the probability distribution function of a binomial random variable with parameters n and p can be approximated by that of a normal random variable having mean *np* and variance $np(1-p)$.

A random variable whose probability density function is of the form

$$f(x)=\{\lambda e-\lambda x x\geq 0 \quad 0 \text{otherwise}$$

is said to be an *exponential* random variable with parameter λ. Its expected value and variance are, respectively,

$$E[X]=1\lambda \quad Var(X)=1\lambda 2$$

A key property possessed only by exponential random variables is that they are *memoryless*, in the sense that, for positive s and t,

$$P\{X>s+t|X>t\}=P\{X>s\}$$

If X represents the life of an item, then the memoryless property states that for any t, the remaining life of a t-year-old item has the same probability distribution as the life of a new item. Thus, one need not remember the age of an item to know its distribution of remaining life.

Let X be a nonnegative continuous random variable with distribution function F and density function f. The function

$$\lambda(t)=f(t)1-F(t) \quad t\geq 0$$

is called the *hazard rate*, or *failure rate*, function of F. If we interpret X as being the life of an item, then for small values of dt, λ(t) dt is approximately the probability that a t-unit-old item will fail within an additional time *dt*. If F is the exponential distribution with parameter λ, then

$$\lambda(t)=\lambda \quad t\geq 0$$

In addition, the exponential is the unique distribution having a constant failure rate.

A random variable is said to have a *gamma* distribution with parameters α and λ if its probability density function is equal to

$$f(x)=\lambda e-\lambda x(\lambda x)\alpha-1\Gamma(\alpha) \quad x\geq 0$$

and is 0 otherwise. The quantity Γ(α) is called the gamma function and is defined by

$$\Gamma(\alpha)=\int 0\infty e-xx\alpha-1 \, dx$$

The expected value and variance of a gamma random variable are, respectively,

$$E[X]=\alpha\lambda \quad Var(X)=\alpha\lambda 2$$

A random variable is said to have a *beta* distribution with parameters (a, b) if its probability density function is equal to

$$f(x)=1B(a,b)xa-1(1-x)b-1 \quad 0\leq x\leq 1$$

and is equal to 0 otherwise. The constant B(a, b) is given by

$$B(a,b)=\int 01xa-1(1-x)b-1 \, dx$$

The mean and variance of such a random variable are, respectively,

$E[X]=\frac{a}{a+b}$ $\quad Var(X)=\frac{ab}{(a+b)^2(a+b+1)}$

Problems

5.1. Let X be a random variable with probability density function
f(x)={c(1−x2)−1<x<10otherwise

a. What is the value of c?
b. What is the cumulative distribution function of X?

5.2. A system consisting of one original unit plus a spare can function for a random amount of time X. If the density of X is given (in units of months) by
f(x)={Cxe−x/2x>00x≤0

what is the probability that the system functions for at least 5 months?
5.3. Consider the function
f(x)={C(2x−x3)0<x<520otherwise

Could f be a probability density function? If so, determine *C*. Repeat if f(x) were given by
f(x)={C(2x−x2)0<x<520otherwise

5.4. The probability density function of X, the lifetime of a certain type of electronic device (measured in hours), is given by
f(x)={10x2x>100x≤10

a. Find P{X>20}.
b. What is the cumulative distribution function of X?
c. What is the probability that of 6 such types of devices, at least 3 will function for at least 15 hours? What assumptions are you making?

5.5. A filling station is supplied with gasoline once a week. If its weekly volume of sales in thousands of gallons is a random variable with probability density function
f(x)={5(1−x)40<x<10otherwise

what must the capacity of the tank be so that the probability of the supply being exhausted in a given week is .01?
5.6. Compute E[X] if X has a density function given by
a. f(x)={14xe−x/2x>00otherwise;
b. f(x)={c(1−x2)−1<x<10otherwise;
c. f(x)={5x2x>50x≤5.

5.7. The density function of X is given by
f(x)={a+bx20≤x≤10otherwise

If E[X]=35, find a and b.
5.8. The lifetime in hours of an electronic tube is a random variable having a probability density function given by
f(x)=xe−x x≥0

Compute the expected lifetime of such a tube.
5.9. Consider **Example 4b** of **Chapter 4**, but now suppose that the seasonal demand is a continuous random variable having probability density function f. Show that the optimal amount to stock is the value s^* that satisfies

$F(s^*)=bb+\ell$

where b is net profit per unit sale, ℓ is the net loss per unit unsold, and F is the cumulative distribution function of the seasonal demand.

5.10. Trains headed for destination A arrive at the train station at 15-minute intervals starting at 7 A.M., whereas trains headed for destination B arrive at 15-minute intervals starting at 7:05 A.M.

a. If a certain passenger arrives at the station at a time uniformly distributed between 7 and 8 A.M. and then gets on the first train that arrives, what proportion of time does he or she go to destination A?
b. What if the passenger arrives at a time uniformly distributed between 7:10 and 8:10 A.M.?

5.11. A point is chosen at random on a line segment of length L. Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than 14.

5.12. A bus travels between the two cities A and B, which are 100 miles apart. If the bus has a breakdown, the distance from the breakdown to city A has a uniform distribution over (0, 100). There is a bus service station in city A, in B, and in the center of the route between A and B. It is suggested that it would be more efficient to have the three stations located 25, 50, and 75 miles, respectively, from A. Do you agree? Why?

5.13. You arrive at a bus stop at 10 A.M., knowing that the bus will arrive at some time uniformly distributed between 10 and 10:30.

a. What is the probability that you will have to wait longer than 10 minutes?
b. If, at 10:15, the bus has not yet arrived, what is the probability that you will have to wait at least an additional 10 minutes?

5.14. Let X be a uniform (0, 1) random variable. Compute E[Xn] by using **Proposition 2.1**, and then check the result by using the definition of expectation.

5.15. If X is a normal random variable with parameters $\mu=10$ and $\sigma2=36$, compute

a. $P\{X>5\}$;
b. $P\{4<X<16\}$;
c. $P\{X<8\}$;
d. $P\{X<20\}$;
e. $P\{X>16\}$.

5.16. The annual rainfall (in inches) in a certain region is normally distributed with $\mu=40$ and $\sigma=4$. What is the probability that starting with this year, it will take more than 10 years before a year occurs having a rainfall of more than 50 inches? What assumptions are you making?

5.17. The salaries of physicians in a certain speciality are approximately normally distributed. If 25 percent of these physicians earn less than $180,000 and 25 percent earn more than $320,000, approximately what fraction earn

a. less than $200,000?
b. between $280,000 and $320,000?

5.18. Suppose that X is a normal random variable with mean 5. If $P\{X>9\}=.2$, approximately what is Var(X)?

5.19. Let X be a normal random variable with mean 12 and variance 4. Find the value of c such that $P\{X>c\}= .10$.

5.20. If 65 percent of the population of a large community is in favor of a proposed rise in school taxes, approximate the probability that a random sample of 100 people will contain

a. at least 50 who are in favor of the proposition;
b. between 60 and 70 inclusive who are in favor;
c. fewer than 75 in favor.

5.21. Suppose that the height, in inches, of a 25-year-old man is a normal random variable with

parameters $\mu=71$ and $\sigma^2=6.25$. What percentage of 25-year-old men are taller than 6 feet, 2 inches? What percentage of men in the 6-footer club are taller than 6 feet, 5 inches?

5.22. Every day Jo practices her tennis serve by continually serving until she has had a total of 50 successful serves. If each of her serves is, independently of previous ones, successful with probability .4, approximately what is the probability that she will need more than 100 serves to accomplish her goal?

Hint: Imagine even if Jo is successful that she continues to serve until she has served exactly 100 times. What must be true about her first 100 serves if she is to reach her goal?

5.23. One thousand independent rolls of a fair die will be made. Compute an approximation to the probability that the number 6 will appear between 150 and 200 times inclusively. If the number 6 appears exactly 200 times, find the probability that the number 5 will appear less than 150 times.

5.24. The lifetimes of interactive computer chips produced by a certain semiconductor manufacturer are normally distributed with parameters $\mu=1.4\times106$ hours and $\sigma=3\times105$ hours. What is the approximate probability that a batch of 100 chips will contain at least 20 whose lifetimes are less than 1.8×106?

5.25. Each item produced by a certain manufacturer is, independently, of acceptable quality with probability .95. Approximate the probability that at most 10 of the next 150 items produced are unacceptable.

5.26. Two types of coins are produced at a factory: a fair coin and a biased one that comes up heads 55 percent of the time. We have one of these coins but do not know whether it is a fair coin or a biased one. In order to ascertain which type of coin we have, we shall perform the following statistical test: We shall toss the coin 1000 times. If the coin lands on heads 525 or more times, then we shall conclude that it is a biased coin, whereas if it lands on heads fewer than 525 times, then we shall conclude that it is a fair coin. If the coin is actually fair, what is the probability that we shall reach a false conclusion? What would it be if the coin were biased?

5.27. In 10,000 independent tosses of a coin, the coin landed on heads 5800 times. Is it reasonable to assume that the coin is not fair? Explain.

5.28. Twelve percent of the population is left handed. Approximate the probability that there are at least 20 left-handers in a school of 200 students. State your assumptions.

5.29. A model for the movement of a stock supposes that if the present price of the stock is s, then after one period, it will be either *us* with probability p or *ds* with probability $1-p$. Assuming that successive movements are independent, approximate the probability that the stock's price will be up at least 30 percent after the next 1000 periods if $u=1.012$, $d=0.990$, and $p=.52$.

5.30. An image is partitioned into two regions, one white and the other black. A reading taken from a randomly chosen point in the white section will be normally distributed with $\mu=4$ and $\sigma^2=4$, whereas one taken from a randomly chosen point in the black region will have a normally distributed reading with parameters (6, 9). A point is randomly chosen on the image and has a reading of 5. If the fraction of the image that is black is α, for what value of α would the probability of making an error be the same, regardless of whether one concluded that the point was in the black region or in the white region?

5.31.

a. A fire station is to be located along a road of length $A, A<\infty$. If fires occur at points uniformly chosen on (0, A), where should the station be located so as to minimize the expected distance from the fire? That is, choose a so as to
minimize $E[|X-a|]$

when X is uniformly distributed over (0, A).

b. Now suppose that the road is of infinite length—stretching from point 0 outward to ∞. If the distance of a fire from point 0 is exponentially distributed with rate λ, where should the fire station now be located? That is, we want to minimize $E[|X-a|]$, where X is now exponential with rate λ.

5.32. The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda=12$. What is

a. What is the probability that a repair time exceeds 2 hours?
b. the conditional probability that a repair takes at least 10 hours, given that its duration exceeds 9 hours?

5.33. If U is uniformly distributed on (0, 1), find the distribution of $Y=-\log(U)$.
5.34. Jones figures that the total number of thousands of miles that a racing auto can be driven before it would need to be junked is an exponential random variable with parameter 120. Smith has a used car that he claims has been driven only 10,000 miles. If Jones purchases the car, what is the probability that she would get at least 20,000 additional miles out of it? Repeat under the assumption that the lifetime mileage of the car is not exponentially distributed, but rather is (in thousands of miles) uniformly distributed over (0, 40).
5.35. If X is an exponential random variable with parameter λ, and $c>0$, find the density function of cX. What kind of random variable is cX.
5.36. The lung cancer hazard rate $\lambda(t)$ of a t-year-old male smoker is such that
$\lambda(t)=.027+.00025(t-40)2 \qquad t\geq 40$

Assuming that a 40-year-old male smoker survives all other hazards, what is the probability that he survives to (a) age 50 and (b) age 60 without contracting lung cancer?
5.37. Suppose that the life distribution of an item has the hazard rate function $\lambda(t)=t3$, $t>0$. What is the probability that
a. the item survives to age 2?
b. the item's lifetime is between .4 and 1.4?
c. a 1-year-old item will survive to age 2?

5.38. If X is uniformly distributed over $(-1,1)$, find
(a) $P\{|X|>12\}$;
(b) the density function of the random variable $|X|$.
5.39. If Y is uniformly distributed over (0, 5), what is the probability that the roots of the equation $4x2+4xY+Y+2=0$ are both real?
5.40. If X is an exponential random variable with parameter $\lambda=1$, compute the probability density function of the random variable Y defined by $Y=\log X$.
5.41. If X is uniformly distributed over (a, b), find a and b if $E[X]=10$, $Var(X)=48$.
5.42. If X is uniformly distributed over (0, 1), find the density function of $Y=eX$.
5.43. Find the distribution of $R=A\sin\theta$, where A is a fixed constant and θ is uniformly distributed on $(-\pi/2,\pi/2)$. Such a random variable *R* arises in the theory of ballistics. If a projectile is fired from the origin at an angle α from the earth with a speed v, then the point R at which it returns to the earth can be expressed as $R=(v2/g)\sin2\alpha$, where g is the gravitational constant, equal to 980 centimeters per second squared.
5.44. Let Y be a lognormal random variable (see **Example 7e** for its definition) and let $c>0$ be a constant. Answer true or false to the following, and then give an explanation for your answer.
a. cY is lognormal;
b. c+Y is lognormal.

Theoretical Exercises

5.1. The speed of a molecule in a uniform gas at equilibrium is a random variable whose probability density function is given by
$f(x)=\{ax2e-bx2x\geq 00x<0$

where $b=m/2kT$ and k, T, and m denote, respectively, Boltzmann's constant, the absolute temperature of the gas, and the mass of the molecule. Evaluate a in terms of b.
5.2. Show that
$E[Y]=\int 0\infty P\{Y>y\}\,dy-\int 0\infty P\{Y<-y\}\,dy$

Hint: Show that
$\int 0\infty P\{Y<-y\}\,dy=-\int -\infty\ 0\ x\ f\ Y\ (x)\,dx \int 0\infty P\{Y>y\}\,dy=\int 0\infty x\ f\ Y\ (x)\,dx$

5.3. Show that if X has density function f, then
$E[g(X)]=\int -\infty\infty g(x)f(x)\,dx$

Hint: Using **Theoretical Exercise 5.2**, start with
$E[g(X)]=\int 0\infty P\{g(X)>y\}dy-\int 0\infty P\{g(X)<-y\}dy$

and then proceed as in the proof given in the text when $g(X)\geq 0$.
5.4. Prove **Corollary 2.1**.
5.5. Use the result that for a nonnegative random variable Y,
$E[Y]=\int 0\infty P\{Y>t\}dt$

to show that for a nonnegative random variable X,
$E[Xn]=\int 0\infty nxn-1P\{X>x\}dx$

Hint: Start with
$E[Xn]=\int 0\infty P\{Xn>t\}dt$

and make the change of variables $t=xn$.
5.6. Define a collection of events Ea, $0<a<1$, having the property that $P(Ea)=1$ for all a but $P(\cap aEa)=0$.
Hint: Let X be uniform over (0, 1) and define each Ea in terms of X.
5.7. The standard deviation of X, denoted *SD*(*X*), is given by
$SD(X)=Var(X)$

Find $SD(aX+b)$ if X has variance $\sigma 2$.
5.8. Let X be a random variable that takes on values between 0 and c. That is, $P\{0\leq X\leq c\}=1$. Show that
$Var(X)\leq c24$

Hint: One approach is to first argue that
$E[\ X\ 2\ ]\leq cE[X]$

and then use this inequality to show that
$Var(X)\leq c\ 2\ [\alpha(1-\alpha)]$ where $\alpha= E[X]\ c$

5.9. Show that Z is a standard normal random variable; then, for $x>0$,

a. $P\{Z>x\}=P\{Z<-x\}$;

b. $P\{|Z|>x\}=2P\{Z>x\}$;
c. $P\{|Z|<x\}=2P\{Z<x\}-1$.

5.10. Let $f(x)$ denote the probability density function of a normal random variable with mean μ and variance σ^2. Show that $\mu-\sigma$ and $\mu+\sigma$ are points of inflection of this function. That is, show that $f''(x)=0$ when $x=\mu-\sigma$ or $x=\mu+\sigma$.
5.11. Let Z be a standard normal random variable Z, and let g be a differentiable function with derivative g'.
a. Show that $E[g'(Z)]=E[Zg(Z)]$;
b. Show that $E[Z^{n+1}]=nE[Z^{n-1}]$.
c. Find $E[Z^4]$.

5.12. Use the identity of **Theoretical Exercises 5.5** to derive $E[X^2]$ when X is an exponential random variable with parameter λ.
5.13. The median of a continuous random variable having distribution function F is that value m such that $F(m)=\frac{1}{2}$. That is, a random variable is just as likely to be larger than its median as it is to be smaller. Find the median of X if X is
a. uniformly distributed over (a, b);
b. normal with parameters μ, σ^2;
c. exponential with rate λ.

5.14. The mode of a continuous random variable having density f is the value of x for which $f(x)$ attains its maximum. Compute the mode of X in cases (a), (b), and (c) of **Theoretical Exercises 5.13**.
5.15. If X is an exponential random variable with parameter λ, and $c>0$, show that cX is exponential with parameter λ/c.
5.16. Compute the hazard rate function of X when X is uniformly distributed over $(0, a)$.
5.17. If X has hazard rate function $\lambda_X(t)$, compute the hazard rate function of aX where a is a positive constant.
5.18. Verify that the gamma density function integrates to 1.
5.19. If X is an exponential random variable with mean $1/\lambda$, show that
$E[X^k]=k!\lambda^k \qquad k=1, 2, \ldots$

Hint: Make use of the gamma density function to evaluate $E[X^k]$.
5.20. Verify that
$\mathrm{Var}(X)=\alpha\lambda^2$

when X is a gamma random variable with parameters α and λ.
5.21. Show that $\Gamma(\frac{1}{2})=\sqrt{\pi}$.
Hint: $\Gamma(\frac{1}{2})=\int_0^\infty e^{-x}x^{-1/2}\,dx$. Make the change of variables $y=2x$ and then relate the resulting expression to the normal distribution.
5.22. Compute the hazard rate function of a gamma random variable with parameters (α,λ) and show it is increasing when $\alpha\geq 1$ and decreasing when $\alpha\leq 1$.
5.23. Compute the hazard rate function of a Weibull random variable and show it is increasing when $\beta\geq 1$ and decreasing when $\beta\leq 1$.
5.24. Show that a plot of $\log(\log(1-F(x))^{-1})$ against $\log x$ will be a straight line with slope β when $F(\cdot)$ is a Weibull distribution function. Show also that approximately 63.2 percent of all observations from such a distribution will be less than α. Assume that $\nu=0$.
5.25. Let
$Y=(X-\nu\alpha)^\beta$

Show that if X is a Weibull random variable with parameters ν, α, and β, then Y is an exponential random variable with parameter $\lambda=1$ and vice versa.

5.26. Let F be a continuous distribution function. If U is uniformly distributed on (0, 1), find the distribution function of Y=F−1(U), where F−1 is the inverse function of F. (That is, y=F−1(x) if F(y)=x.)
5.27. If X is uniformly distributed over (a, b), what random variable, having a linear relation with X, is uniformly distributed over (0, 1)?
5.28. Consider the beta distribution with parameters (a, b). Show that

a. when a>1 and b>1, the density is unimodal (that is, it has a unique mode) with mode equal to (a−1)/(a+b−2);
b. when a≤1, b≤1, and a+b<2, the density is either unimodal with mode at 0 or 1 or U-shaped with modes at both 0 and 1;
c. when a=1=b, all points in [0, 1] are modes.

5.29. Let X be a continuous random variable having cumulative distribution function *F*. Define the random variable Y by Y=F(X). Show that Y is uniformly distributed over (0, 1).
5.30. Let X have probability density fX. Find the probability density function of the random variable Y defined by Y=aX+b.
5.31. Find the probability density function of Y=eX when X is normally distributed with parameters μ and σ2. The random variable Y is said to have a *lognormal distribution* (since log Y has a normal distribution) with parameters μ and σ2.
5.32. Let X and Y be independent random variables that are both equally likely to be either 1, 2, ... , (10)N, where N is very large. Let D denote the greatest common divisor of X and Y, and let Qk=P{D=k}.

a. Give a heuristic argument that Qk=1k2Q1.
   Hint: Note that in order for D to equal k, k must divide both X and Y and also X/k, and Y/k must be relatively prime. (That is, X/k, and Y/k must have a greatest common divisor equal to 1.)
b. Use part (a) to show that
   Q 1 =P{X and Y are relatively prime} = 1 ∑ k=1 ∞ 1/ k 2

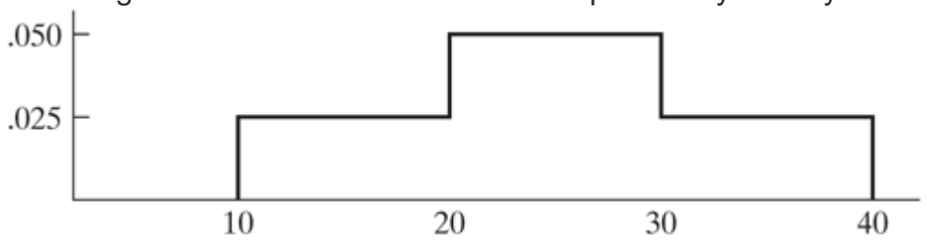
   It is a well-known identity that ∑1∞1/k2=π2/6, so Q 1 =6/ π 2 . (In number theory, this is known as the Legendre theorem.)
c. Now argue that
   Q1=∏i=1∞(Pi2−1Pi2)

   where Pi is the *i*th-smallest prime greater than 1.
   Hint: *X* and *Y* will be relatively prime if they have no common prime factors. Hence, from part (b), we see that
   ∏i=1∞(Pi2−1Pi2)=6π2

5.33. Prove **Theorem 7.1** when g(x) is a decreasing function.

Self-Test Problems and Exercises

5.1. The number of minutes of playing time of a certain high school basketball player in a randomly chosen game is a random variable whose probability density function is given in the following figure:



Find the probability that the player plays

a. more than 15 minutes;
b. between 20 and 35 minutes;
c. less than 30 minutes;
d. more than 36 minutes.

5.2. For some constant c, the random variable X has the probability density function
f(x)={cxn0<x<10otherwise

Find (a) c and (b) P{X>x}, 0<x<1.
5.3. For some constant c, the random variable X has the probability density function
f(x)={cx40<x<20otherwise

Find (a) E[X] and (b) Var(X).
5.4. The random variable X has the probability density function
f(x)={ax+bx20<x<10otherwise

If E[X]=.6, find (a) P{X<12} and (b) Var(X).
5.5. The random variable X is said to be a discrete uniform random variable on the integers 1,2, ..., n if
P{X=i}=1n i=1,2,..., n

For any nonnegative real number x, let Int(x) (sometimes written as [x]) be the largest integer that is less than or equal to x. Show that if U is a uniform random variable on (0, 1), then X=Int (nU)+1 is a discrete uniform random variable on 1, ..., n.
5.6. Your company must make a sealed bid for a construction project. If you succeed in winning the contract (by having the lowest bid), then you plan to pay another firm $100,000 to do the work. If you believe that the minimum bid (in thousands of dollars) of the other participating companies can be modeled as the value of a random variable that is uniformly distributed on (70, 140), how much should you bid to maximize your expected profit?
5.7. To be a winner in a certain game, you must be successful in three successive rounds. The game depends on the value of U, a uniform random variable on (0, 1). If U>.1, then you are successful in round 1; if U>.2, then you are successful in round 2; and if U>.3, then you are successful in round 3.

a. Find the probability that you are successful in round 1.
b. Find the conditional probability that you are successful in round 2 given that you were successful in round 1.
c. Find the conditional probability that you are successful in round 3 given that you were successful in rounds 1 and 2.
d. Find the probability that you are a winner.

5.8. A randomly chosen IQ test taker obtains a score that is approximately a normal random variable with mean 100 and standard deviation 15. What is the probability that the score of such a person is (a) more than 125; (b) between 90 and 110?
5.9. Suppose that the travel time from your home to your office is normally distributed with mean 40 minutes and standard deviation 7 minutes. If you want to be 95 percent certain that you will not be late for an office appointment at 1 P.M., what is the latest time that you should leave home?
5.10. The life of a certain type of automobile tire is normally distributed with mean 34,000 miles and standard deviation 4000 miles.

a. What is the probability that such a tire lasts more than 40,000 miles?
b. What is the probability that it lasts between 30,000 and 35,000 miles?
c. Given that it has survived 30,000 miles, what is the conditional probability that the tire survives another 10,000 miles?

5.11. The annual rainfall in Cleveland, Ohio, is approximately a normal random variable with mean 40.2 inches and standard deviation 8.4 inches. What is the probability that

a. next year's rainfall will exceed 44 inches?
b. the yearly rainfalls in exactly 3 of the next 7 years will exceed 44 inches?

Assume that if Ai is the event that the rainfall exceeds 44 inches in year i (from now), then the events Ai, i≥1, are independent.
5.12. The following table uses 1992 data concerning the percentages of male and female full-time workers whose annual salaries fall into different ranges:

Earnings range	Percentage of female	Percentage of males
≤9999	8.6	4.4
10,000–19,999	38.0	21.1
20,000–24,999	19.4	15.8
25,000–49,999	29.2	41.5
≥50,000	4.8	17.2

Suppose that random samples of 200 male and 200 female full-time workers are chosen. Approximate the probability that

a. at least 70 of the women earn $25,000 or more;
b. at most 60 percent of the men earn $25,000 or more;
c. at least three-fourths of the men and at least half the women earn $20,000 or more.

5.13. At a certain bank, the amount of time that a customer spends being served by a teller is an exponential random variable with mean 5 minutes. If there is a customer in service when you enter the bank, what is the probability that he or she will still be with the teller after an additional 4 minutes?
5.14. Suppose that the cumulative distribution function of the random variable X is given by
$F(x)=1-e-x2 \quad x>0$

Evaluate (a) $P\{X>2\}$; (b) $P\{1<X<3\}$; (c) the hazard rate function of F; (d) E[X]; (e) Var(X).
Hint: For parts (d) and (e), you might want to make use of the results of Theoretical Exercise 5.5.
5.15. The number of years that a washing machine functions is a random variable whose hazard rate function is given by
$\lambda(t)=\{.20<t<2.2+.3(t-2)2\leq t<51.1t>5$

a. What is the probability that the machine will still be working 6 years after being purchased?
b. If it is still working 6 years after being purchased, what is the conditional probability that it will fail within the next 2 years?

5.16. A standard Cauchy random variable has density function
$f(x)=1\pi(1+x2) \quad -\infty<x<\infty$

Show that if X is a standard Cauchy random variable, then 1/X is also a standard Cauchy random variable.

5.17. A roulette wheel has 38 slots, numbered 0, 00, and 1 through 36. If you bet 1 on a specified number, then you either win 35 if the roulette ball lands on that number or lose 1 if it does not. If you continually make such bets, approximate the probability that

a. you are winning after 34 bets;
b. you are winning after 1000 bets;
c. you are winning after 100,000 bets.

Assume that each roll of the roulette ball is equally likely to land on any of the 38 numbers.

5.18. There are two types of batteries in a bin. When in use, type i batteries last (in hours) an exponentially distributed time with rate λi, i=1, 2.. A battery that is randomly chosen from the bin will be a type i battery with probability $pi,\sum i=12pi=1$. If a randomly chosen battery is still operating after t hours of use, what is the probability that it will still be operating after an additional s hours?

5.19. Evidence concerning the guilt or innocence of a defendant in a criminal investigation can be summarized by the value of an exponential random variable X whose mean μ depends on whether the defendant is guilty. If innocent, $\mu=1$; if guilty, $\mu=2$. The deciding judge will rule the defendant guilty if X>c for some suitably chosen value of c.

a. If the judge wants to be 95 percent certain that an innocent man will not be convicted, what should be the value of *c*?
b. Using the value of c found in part (a), what is the probability that a guilty defendant will be convicted?

5.20. For any real number y, define y+ by
$y+=y,\text{if } y\geq 00,\text{if } y<0$

Let c be a constant.

a. Show that
$E[(Z-c)+]=12\pi e-c2/2-c(1-\Phi(c))$

when Z is a standard normal random variable.
b. Find E[(X−c)+] when X is normal with mean μ and variance $\sigma 2$.

5.21. With $\Phi(x)$ being the probability that a normal random variable with mean 0 and variance 1 is less than x, which of the following are true:

a. $\Phi(-x)=\Phi(x)$
b. $\Phi(x)+\Phi(-x)=1$
c. $\Phi(-x)=1/\Phi(x)$

5.22. Let U be a uniform (0, 1) random variable, and let a<b be constants.

a. Show that if b>0, then bU is uniformly distributed on (0, b), and if b<0, then bU is uniformly distributed on (b, 0).
b. Show that a+U is uniformly distributed on (a, 1+a).
c. What function of U is uniformly distributed on (a, b)?
d. Show that min(U, 1−U) is a uniform (0, 1/2) random variable.

e. Show that max(U, 1−U) is a uniform (1/2, 1) random variable.

5.23. Let
$$f(x)=\{13ex,\text{if } x<013\text{if } 0\le x<113e-(x-1)\text{if } x\ge 1$$

a. Show that f is a probability density function. (That is, show that $f(x)\ge 0$, and $\int_{-\infty}^{\infty} f(x)\,dx=1$.)
b. If X has density function f, find E[X].

5.24. Let
$$f(x)=\theta 21+\theta(1+x)e-\theta x, \quad x>0$$

where $\theta>0$.
a. Show that f(x) is a density function. That is, show that $f(x)\ge 0$, and that $\int_{0}^{\infty} f(x)\,dx=1$.
b. Find E[X]
c. Find Var(X).

Chapter 6 Jointly Distributed Random Variables

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6.1 Joint Distribution Functions

Thus far, we have concerned ourselves only with probability distributions for single random variables. However, we are often interested in probability statements concerning two or more random variables. In order to deal with such probabilities, we define, for any two random variables X and Y, the *joint cumulative probability distribution function* of X and Y by

$$F(a,b)=P\{X\le a,Y\le b\} \quad -\infty<a,b<\infty$$

All joint probability statements about X and Y can, in theory, be answered in terms of their joint distribution function. For instance,

$$P(a1<X\le a2,\ b1<Y\le b2)=F(a2,b2)+F(a1,b1)-F(a1,b2)-F(a2,b1) \tag{1.1}$$

whenever a1<a2, b1<b2. To verify **Equation (1.1)**, note that for a1<a2,

$$P(X\le a2,\ Y\le b)=P(X\le a1,\ Y\le b)+P(a1<X\le a2,\ Y\le b)$$

giving that

$$P(a1<X\le a2,\ Y\le b)=F(a2,b)-F(a1,b) \tag{1.2}$$

Also, because for b1<b2,

$$P(a1<X\le a2,\ Y\le b2)=P(a1<X\le a2,\ Y\le b1)+P(a1<X\le a2,\ b1<Y\le b2)$$

we have that when a1<a2, b1<b2

$$P(\ a\ 1<X\le a\ 2\ ,\ \ b\ 1<Y\le b\ 2\ )=P(\ a\ 1<X\le a\ 2\ ,\ Y\le b\ 2\ )-P(\ a\ 1<X\le a\ 2\ ,\ Y\le b\ 1\ )=F(\ a\ 2\ ,\ b\ 2\ )-F(\ a\ 1\ ,\ b\ 2\ )-F(\ a\ 2\ ,\ b\ 1\ )+F(\ a\ 1\ ,\ b\ 1\ )$$

where the final equality used **Equation (1.2)**.

When X and Y are discrete random variables, with X taking on one of the values xi, i≥1, and Y one of the values yj, j≥1, it is convenient to define the *joint probability mass function* of X and Y by

$$p(x,y)=P(X=x,Y=y)$$

Using that the event {X=x} is the union of the mutually exclusive events {X=x,Y=yj}, j≥1, it follows that the probability mass function of X can be obtained from the joint probability mass function by

$$p_X(x)=P(X=x)=P(\cup_j\{X=x,Y=y_j\})=\sum_j P(X=x,Y=y_j)=\sum_j p(x,y_j)$$

Similarly, the probability mass function of Y is obtained from

$$p_Y(y)=\sum_i p(x_i,y)$$

Example 1a

Suppose that 3 balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. If we let X and Y denote, respectively, the number of red and white balls chosen, then the joint probability mass function of X and Y,p(i,j)=P{X=i,Y=j}, is obtained by noting that X=i, Y=j if, of the 3 balls selected, i are red, j are white, and 3−i−j are blue. Because all subsets of size 3 are equally likely to be chosen, it

follows that

$$p(i,j) = \binom{3}{i}\binom{4}{j}\binom{5}{3-i-j}\Big/\binom{12}{3}$$

Consequently,

$$p(0,0) = \binom{5}{3}\Big/\binom{12}{3} = \frac{10}{220} \quad p(0,1) = \binom{4}{1}\binom{5}{2}\Big/\binom{12}{3} = \frac{40}{220} \quad p(0,2) = \binom{4}{2}\binom{5}{1}\Big/\binom{12}{3} = \frac{30}{220}$$
$$p(0,3) = \binom{4}{3}\Big/\binom{12}{3} = \frac{4}{220} \quad p(1,0) = \binom{3}{1}\binom{5}{2}\Big/\binom{12}{3} = \frac{30}{220} \quad p(1,1) = \binom{3}{1}\binom{4}{1}\binom{5}{1}\Big/\binom{12}{3} = \frac{60}{220}$$
$$p(1,2) = \binom{3}{1}\binom{4}{2}\Big/\binom{12}{3} = \frac{18}{220} \quad p(2,0) = \binom{3}{2}\binom{5}{1}\Big/\binom{12}{3} = \frac{15}{220} \quad p(2,1) = \binom{3}{2}\binom{4}{1}\Big/\binom{12}{3} = \frac{12}{220}$$
$$p(3,0) = \binom{3}{3}\Big/\binom{12}{3} = \frac{1}{220}$$

These probabilities can most easily be expressed in tabular form, as in **Table 6.1**. The reader should note that the probability mass function of X is obtained by computing the row sums, whereas the probability mass function of Y is obtained by computing the column sums. Because the individual probability mass functions of X and Y thus appear in the margin of such a table, they are often referred to as the *marginal probability mass functions* of X and Y, respectively.

Table 6.1 P{X=i,Y=j}.

i \ j	0	1	2	3	Row sum $= P\{X = i\}$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column sum $= P\{Y = j\}$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

Example 1b

Suppose that 15 percent of the families in a certain community have no children, 20 percent have 1 child, 35 percent have 2 children, and 30 percent have 3. Suppose further that in each family each child is equally likely (independently) to be a boy or a girl. If a family is chosen at random from this community, then B, the number of boys, and G, the number of girls, in this family will have the joint probability mass function shown in **Table 6.2**.

Table 6.2 P{B=i,G=j}.

i \ j	0	1	2	3	Row sum $= P\{B = i\}$
0	.15	.10	.0875	.0375	.3750
1	.10	.175	.1125	0	.3875
2	.0875	.1125	0	0	.2000
3	.0375	0	0	0	.0375
Column sum $= P\{G = j\}$	.3750	.3875	.2000	.0375	

The probabilities shown in **Table 6.2** are obtained as follows:

P{ B=0,G=0 } =P{ no children }=.15 P{ B=0,G=1 } =P{ 1 girl and total of 1 child } =P{ 1 child }P{ 1 girl | 1 child }=(.20)(1 2) P{ B=0,G=2 } =P{ 2 girls and total of 2 children } =P{ 2 children }P{ 2 girls | 2 children }=(.35) (1 2) 2

We leave the verification of the remaining probabilities in the table to the reader.

Example 1c

Consider independent trials where each trial is a success with probability p. Let Xr denote the number of trials until there have been r successes, and let Ys denote the number of trials until there have been s failures. Suppose we want to derive their joint probability mass function P(Xr=i,Ys=j). To do so, first consider the case i<j. In this case, write

P(Xr=i,Ys=j)=P(Xr=i)P(Ys=j|Xr=i)

Now, if there have been r successes after trial i then there have been i−r failures by that point. Hence, the conditional distribution of Ys, given that Xr=i, is the distribution of i plus the number of additional trials after trial i until there have been an additional s−i+r failures. Hence,

P(Xr=i,Ys=j)=P(Xr=i)P(Ys−i+r=j−i) , i<j

Because Xr is a negative binomial random variable with parameters (r,p) and Ys−i+r is a negative binomial random variable with parameters (s−i+r,1−p), the preceding yields

P(X r =i, Y s =j)=(i−1 r−1) p r (1−p) i−r (j−i−1 s−i+r−1) (1−p) s−i+r p j−s−r , i<j

We leave it as an exercise to determine the analogous expression when j<i.

We say that X and Y are *jointly continuous* if there exists a function f(x,y), defined for all real x and y, having the property that for every set C of pairs of real numbers (that is, C is a set in the two-dimensional plane),

$$P\{(X,Y)\in C\}= \iint_{(x,y)\in C} f(x,y)\,dx\,dy \qquad (1.3)$$

The function f(x,y) is called the *joint probability density function* of X and Y. If A and B are any sets of real numbers, then by defining C={(x,y):x∈A,y∈B}, we see from **Equation (1.3)** that

$$P\{ X\in A,Y\in B \}= \int B \int A f(x,y)\, dx\, dy \tag{1.4}$$

Because

$$F(a,b) =P\{ X\in(-\infty, a] ,Y\in(-\infty,b] \} = \int -\infty\ b \int -\infty\ a f(x,y)\, dx\, dy$$

it follows, upon differentiation, that

$$f(a,b)= \partial\ 2\ \partial a \partial b\ F(a,b)$$

wherever the partial derivatives are defined. Another interpretation of the joint density function, obtained from **Equation (1.4)**, is

$$P\{a<X<a+da,b<Y<b+db\} = \int b\ d+db \int a\ a+da f(x,y)\, dx\, dy \approx f(a,b)\, da\, db$$

when da and db are small and f(x,y) is continuous at a, b. Hence, f(a,b) is a measure of how likely it is that the random vector (X, Y) will be near (a, b).

If X and Y are jointly continuous, they are individually continuous, and their probability density functions can be obtained as follows:

$$P\{ X\in A \} =P\{ X\in A,Y\in(-\infty,\infty) \} = \int A \int -\infty\ \infty f(x,y)\, dy\, dx = \int A f\ X\ (x)\, dx$$

where

$$f\ X\ (x)= \int -\infty\ \infty f(x,y)\, dy$$

is thus the probability density function of X. Similarly, the probability density function of Y is given by

$$f\ Y\ (y)= \int -\infty\ \infty f(x,y)\, dx$$

Example 1d

The joint density function of X and Y is given by

$$f(x,y)=\{\ 2\ e\ -x\ e\ -2y \quad 0<x<\infty,\ 0<y<\infty\ 0 \quad \text{otherwise}$$

Compute (a) P{X>1,Y<1}, (b) P{X<Y}, and (c) P{X<a}.

Solution

a. $P(X>1, Y<1)= \int 0\ 1 \int 1\ \infty\ 2\ e\ -x\ e\ -2y\ dx\, dy$

   Now,
   $\int 1\ \infty\ e\ -x\, dx=-\ e\ -x\ |\ 1\ \infty = e\ -1$

   giving that
   $P(X>1, Y<1)= e\ -1 \int 0\ 1\ 2\ e\ -2y\, dy= e\ -1\ (1-\ e\ -2\ )$

b. $P\{X<Y\} = \iint (\ x,y\ ):x<y\ 2\ e\ -x\ e\ -2y\ dx\, dy = \int 0\ \infty \int 0\ y\ 2\ e\ -x\ e\ -2y\ dx\, dy = \int 0\ \infty\ 2\ e\ -2y\ (1-\ e\ -y\ )dy = \int 0\ \infty\ 2\ e\ -2y\, dy- \int 0\ \infty\ 2\ e\ -3y\, dy =1-\ 2\ 3 = 1\ 3$

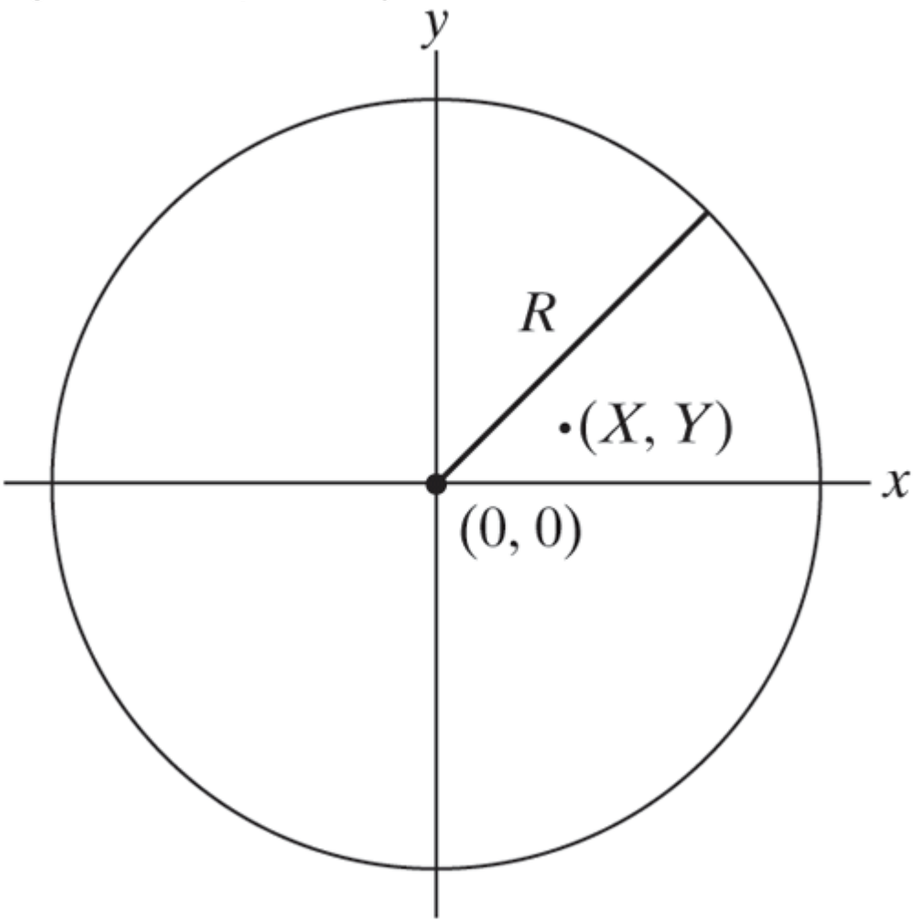
c. $P\{X<a\} = \int_0^a \int_0^\infty 2e^{-2y} e^{-x}\, dy\, dx = \int_0^a e^{-x}\, dx = 1 - e^{-a}$

Example 1e

Consider a circle of radius R, and suppose that a point within the circle is randomly chosen in such a manner that all regions within the circle of equal area are equally likely to contain the point. (In other words, the point is uniformly distributed within the circle.) If we let the center of the circle denote the origin and define X and Y to be the coordinates of the point chosen (**Figure 6.1**), then, since (X, Y) is equally likely to be near each point in the circle, it follows that the joint density function of X and Y is given by

$$f(x,y) = \begin{cases} c & \text{if } x^2+y^2 \le R^2 \\ 0 & \text{if } x^2+y^2 > R^2 \end{cases}$$

Figure 6.1 Joint probability distribution.



for some value of c.

a. Determine c.
b. Find the marginal density functions of X and Y.
c. Compute the probability that D, the distance from the origin of the point selected, is less than or equal to a.
d. Find E [D].

Solution

a. Because
$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dy\, dx = 1$

it follows that
c $\iint x^2 + y^2 \le R^2 \, dy\, dx = 1$

We can evaluate $\int \int x^2 + y^2 \le R^2 \, dy\, dx$ either by using polar coordinates or, more simply, by noting that it represents the area of the circle and is thus equal to πR^2. Hence,
$c = 1\pi R^2$

b. $f_X(x) = \int_{-\infty}^{\infty} f(x,y)\, dy = 1\pi R^2 \int x^2 + y^2 \le R^2 \, dy = 1\pi R^2 \int_{-a}^{a} dy$, where $a = R^2 - x^2$
$= 2\pi R^2 R^2 - x^2, \quad x^2 \le R^2$

and it equals 0 when $x^2 > R^2$. By symmetry, the marginal density of Y is given by
$f_Y(y) = 2\pi R^2 R^2 - y^2, \; y^2 \le R^2 = 0 \; y^2 > R^2$

c. The distribution function of $D = X^2 + Y^2$, the distance from the origin, is obtained as follows: For $0 \le a \le R$,
$F_D(a) = P\{X^2 + Y^2 \le a\} = P\{X^2 + Y^2 \le a^2\} = \iint x^2 + y^2 \le a^2 f(x,y)\, dy\, dx = 1\pi R^2 \iint x^2 + y^2 \le a^2 \, dy\, dx = \pi a^2 \pi R^2 = a^2 R^2$

where we have used the fact that $\int \int x^2 + y^2 \le a^2 \, dy\, dx$ is the area of a circle of radius *a* and thus is equal to $\pi\alpha^2$.

d. From part (c), the density function of D is
$f_D(a) = 2aR^2 \quad 0 \le a \le R$

Hence,
$E[D] = 2R^2 \int_0^R a^2 da = 2R^3$

Example 1f

The joint density of X and Y is given by

$f(x,y) = \{ e^{-(x+y)} \; 0 < x < \infty, \; 0 < y < \infty \; 0$ otherwise

Find the density function of the random variable X/Y.

Solution

We start by computing the distribution function of X/Y. For $a > 0$,

$F_{X/Y}(a) = P\{X Y \le a\} = \iint x/y \le a \; e^{-(x+y)} \, dx\, dy = \int_0^{\infty} \int_0^{ay} e^{-(x+y)} \, dx\, dy = \int_0^{\infty} (1 - e^{-ay}) e^{-y} dy = \{ -e^{-y} + e^{-(a+1)y}/(a+1) \} \mid_0^{\infty} = 1 - 1/(a+1)$

Differentiation shows that the density function of X/Y is given by $f_{X/Y}(a) = 1/(a+1)^2, 0 < a < \infty$.

We can also define joint probability distributions for n random variables in exactly the same manner as we did for n=2. For instance, the joint cumulative probability distribution function $F(a_1, a_2, \ldots, a_n)$ of the n random variables $X_1, X_2, \ldots, X_n$ is defined by

$F(a_1, a_2, \ldots, a_n) = P\{X_1 \le a_1, X_2 \le a_2, \ldots, X_n \le a_n\}$

Further, the n random variables are said to be *jointly continuous* if there exists a function $f(x_1, x_2, \ldots, x_n)$, called the *joint probability density function*, such that, for any set C in n-space,

$P\{(X_1, X_2, \ldots, X_n) \in C\} = \iint \cdots \int_{(x_1, \ldots, x_n) \in C} f(x_1, \ldots, x_n) dx_1\, dx_2 \cdots dx_n$

In particular, for any *n* sets of real numbers A1,A2,… ,An,

$$P\{X_1 \in A_1, X_2 \in A_2, \ldots, X_n \in A_n\} = \int_{A_n} \int_{A_{n-1}} \cdots \int_{A_1} f(x_1, \ldots, x_n)\, dx_1\, dx_2 \cdots dx_n$$

Example 1g The Multinomial Distribution

One of the most important joint distributions is the multinomial distribution, which arises when a sequence of n independent and identical experiments is performed. Suppose that each experiment can result in any one of r possible outcomes, with respective probabilities p1,p2,… ,pr,∑i=1rpi=1. If we let Xi denote the number of the n experiments that result in outcome number i, then

$$P\{X_1 = n_1, X_2 = n_2, \ldots, X_r = n_r\} = \frac{n!}{n_1!\, n_2! \cdots n_r!} p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r} \qquad (1.5)$$

whenever ∑i=1rni=n.

Equation (1.5) is verified by noting that any sequence of outcomes for the n experiments that leads to outcome i occurring ni times for i=1,2,… ,r will, by the assumed independence of experiments, have probability p 1 n 1 p 2 n2 … p r nr of occurring. Because there are n!/(n1!n2!… nr!) such sequences of outcomes (there are n!/n1! … nr! different permutations of n things of which n1 are alike, n2 are alike, … ,nr are alike), **Equation (1.5)** is established. The joint distribution whose joint probability mass function is specified by **Equation (1.5)** is called the *multinomial distribution*. Note that when r=2, the multinomial reduces to the binomial distribution.

Note also that any sum of a fixed set of the X i ' s will have a binomial distribution. That is, if N⊂{1,2,… ,r}, then ∑ i∈N X i will be a binomial random variable with parameters n and p= ∑ i∈N p i . This follows because ∑ i∈N X i represents the number of the n experiments whose outcome is in N, and each experiment will independently have such an outcome with probability ∑i∈Npi.

As an application of the multinomial distribution, suppose that a fair die is rolled 9 times. The probability that 1 appears three times, 2 and 3 twice each, 4 and 5 once each, and 6 not at all is

9!3!2!2!1!1!0!(16)3(16)2(16)2(16)1(16)1(16)0=9!3!2!2!(16)9

We can also use the multinomial distribution to analyze a variation of the classical birthday problem which asks for the probability that no 3 people in a group of size n have the same birthday when the birthdays of the n people are independent and each birthday is equally likely to be any of the 365 days of the year. Because this probability is 0 when n>730 (why is this), we will suppose that n≤730. To find the desired probability, note that there will be no set of 3 people having the same birthday if each of the 365 days of the year is the birthday of at most 2 persons. Now, this will be the case if for some i≤n/2 the event Ai occurs, where Ai is the event that the 365 days of the year can be partitioned into three groups of respective sizes i,n−2i, and 365−n+i such that every day in the first group is the birthday of exactly 2 of the n individuals, every day in the second group is the birthday of exactly 1 of the n individuals, and every day in the third group is the birthday of none of the n individuals. Now, because each day of the year is equally likely to be the birthday of an individual, it follows, for a given partition of the 365 days into three groups of respective sizes i, n−2i, and 365−n+i, that the probability each day in the first group is the birthday of exactly 2 of the n individuals, each day in the second group is the birthday of exactly 1 of the n individuals, and each day in the third group is the birthday of none of the n individuals is equal to the multinomial probability

n!(2!)i(1!)n−2i(0!)365−n+i(1365)n.

As the number of partitions of the 365 days of the year into 3 groups of respective sizes i,n−2i,365−n+i is 365! i!(n−2i)!(365−n+i)! , it follows that

$$P(A_i)=\frac{365!}{i!(n-2i)!(365-n+i)!}\frac{n!}{2^i(1365)^n}, \quad i\le n/2$$

As the events A_i, $i\le n/2$, are mutually exclusive we have that

$$P\{\text{no set of three with same birthday}\}=\sum_{i=0}^{[n/2]}\frac{365!}{i!(n-2i)!(365-n+i)!}\frac{n!}{2^i(1365)^n}$$

When $n=88$, the preceding gives

$$P\{\text{no set of three with same birthday}\}=\sum_{i=0}^{44}\frac{365!}{i!(88-2i)!(277+i)!}\frac{88!}{2^i(1365)^{88}}\approx .504$$

6.2 Independent Random Variables

The random variables X and Y are said to be *independent* if, for any two sets of real numbers A and B,

$$P\{X\in A, Y\in B\}=P\{X\in A\}P\{Y\in B\} \tag{2.1}$$

In other words, X and Y are independent if, for all A and B, the events $E_A=\{X\in A\}$ and $F_B=\{Y\in B\}$ are independent.

It can be shown by using the three axioms of probability that **Equation (2.1)** will follow if and only if, for all a, b

$$P\{X\le a, Y\le b\}=P\{X\le a\}P\{Y\le b\}$$

Hence, in terms of the joint distribution function F of X and Y, X and Y are independent if

$$F(a,b)=F_X(a)F_Y(b) \quad \text{for all } a,b$$

When X and Y are discrete random variables, the condition of independence (2.1) is equivalent to

$$p(x,y)=p_X(x)p_Y(y) \quad \text{for all } x,y \tag{2.2}$$

The equivalence follows because, if **Equation (2.1)** is satisfied, then we obtain **Equation (2.2)** by letting A and B be, respectively, the one-point sets A={x} and B={y}. Furthermore, if **Equation (2.2)** is valid, then for any sets A,B,

$$P\{X\in A, Y\in B\} = \sum_{y\in B}\sum_{x\in A} p(x,y) = \sum_{y\in B}\sum_{x\in A} p_X(x)\, p_Y(y) = \sum_{y\in B} p_Y(y) \sum_{x\in A} p_X(x) = P\{Y\in B\}P\{X\in A\}$$

and **Equation (2.1)** is established.

In the jointly continuous case, the condition of independence is equivalent to

$$f(x,y)=f_X(x)f_Y(y) \quad \text{for all } x,y$$

Thus, loosely speaking, X and Y are independent if knowing the value of one does not change the distribution of the other. Random variables that are not independent are said to be *dependent*.

Example 2a

Suppose that n+m independent trials having a common probability of success p are performed. If X is the number of successes in the first n trials, and Y is the number of successes in the final m trials, then X and Y are independent, since knowing the number of successes in the first n trials does not affect the distribution of the number of successes in the final m trials (by the assumption of independent trials). In fact, for integral x and y,

$$P\{X=x,Y=y\} = \binom{n}{x} p^x (1-p)^{n-x} \binom{m}{y} p^y (1-p)^{m-y} \qquad 0\le x\le n,\ 0\le y\le m = P\{X=x\}P\{Y=y\}$$

In contrast, X and Z will be dependent, where Z is the total number of successes in the n+m trials. (Why?)

Example 2b

Suppose that the number of people who enter a post office on a given day is a Poisson random variable with parameter λ. Show that if each person who enters the post office is a male with probability p and a female with probability 1−p, then the number of males and females entering the post office are independent Poisson random variables with respective parameters λp and λ(1−p).

Solution

Let X and Y denote, respectively, the number of males and females that enter the post office. We shall show the independence of X and Y by establishing **Equation (2.2)**. To obtain an expression for P{X=i,Y=j}, we condition on whether or not X+Y=i+j. This gives:

$$P\{ X=i,Y=j \} =P\{X=i,Y=j|X+Y=i+j\}P\{ X+Y=i+j \} \quad +P\{X=i,Y=j|X+Y\neq i+j\}P\{ X+Y\neq i+j \}$$

[Note that this equation is merely a special case of the formula P(E)=P(E|F)P(F)+P(E|Fc)P(Fc).]

Since P{X=i,Y=j|X+Y≠i+j} is clearly 0, we obtain

$$P\{X=i,Y=j\}=P\{X=i,Y=j|X+Y=i+j\}P\{X+Y=i+j\} \tag{2.3}$$

Now, because X+Y is the total number of people who enter the post office, it follows, by assumption, that

$$P\{X+Y=i+j\}=e-\lambda\lambda i+j(i+j)! \tag{2.4}$$

Furthermore, given that i+j people do enter the post office, since each person entering will be male with probability p, it follows that the probability that exactly i of them will be male (and thus j of them female) is just the binomial probability (i+j i) p i (1−p) j . That is,

$$P\{X=i,Y=j|X+Y=i+j\}=(i+j\ i\ ) p\ i\ (1-p)\ j \tag{2.5}$$

Substituting **Equations (2.4)** and (**2.5**) into **Equation (2.3)** yields

$$P\{ X=i,Y=j \} =(i+j\ i\ ) p\ i\ (1-p)\ j\ e\ -\lambda\ \lambda\ i+j\ (i+j)! = e\ -\lambda\ (\lambda p)\ i\ i!j!\ [\ \lambda(1-p\ )\ ]\ j = e\ -\lambda\ (\lambda p)\ i\ i!\ e\ -\lambda\ (1-p)\ [\ \lambda(1-p\ )\ ]\ j\ j! \tag{2.6}$$

Hence,

$$P\{X=i\}=e-\lambda p(\lambda p)ii!\sum je-\lambda(1-p)[\lambda(1-p)]jj!=e-\lambda p(\lambda p)ii! \tag{2.7}$$

and similarly,

$$P\{Y=j\}=e-\lambda(1-p)[\lambda(1-p)]jj! \tag{2.8}$$

Equations (2.6), **(2.7)**, and **(2.8)** establish the desired result.

Example 2c

A man and a woman decide to meet at a certain location. If each of them independently arrives at a time uniformly distributed between 12 noon and 1 P.M., find the probability that the first to arrive has to wait longer than 10 minutes.

Solution

If we let X and Y denote, respectively, the time past 12 that the man and the woman arrive, then X and Y are independent random variables, each of which is uniformly distributed over (0, 60). The desired

probability, P{X+10<Y}+P{Y+10<X}, which, by symmetry, equals 2P{X+10<Y}, is obtained as follows:

$$2P\{X+10<Y\} = 2\iint_{x+10<y} f(x,y)\,dx\,dy = 2\iint_{x+10<y} f_X(x) f_Y(y)\,dx\,dy = 2\int_{10}^{60}\int_{0}^{y-10}\left(\frac{1}{60}\right)^2 dx\,dy = 2\left(\frac{1}{60}\right)^2\int_{10}^{60}(y-10)\,dy = \frac{25}{36}$$

Our next example presents the oldest problem dealing with geometrical probabilities. It was first considered and solved by Buffon, a French naturalist of the eighteenth century, and is usually referred to as *Buffon's needle problem*.

Example 2d Buffon's Needle Problem

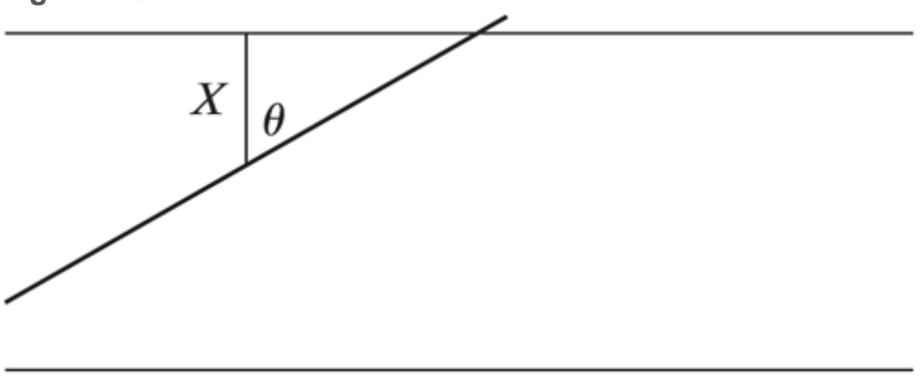
A table is ruled with equidistant parallel lines a distance D apart. A needle of length L, where L≤D, is randomly thrown on the table. What is the probability that the needle will intersect one of the lines (the other possibility being that the needle will be completely contained in the strip between two lines)?

Solution

Let us determine the position of the needle by specifying (1) the distance X from the middle point of the needle to the nearest parallel line and (2) the angle θ between the needle and the projected line of length X. (See **Figure 6.2**.) The needle will intersect a line if the hypotenuse of the right triangle in **Figure 6.2** is less than L/2—that is, if

$$X\cos\theta < \frac{L}{2} \quad \text{or} \quad X < \frac{L}{2}\cos\theta$$

Figure 6.2



As X varies between 0 and D/2 and θ between 0 and π/2, it is reasonable to assume that they are independent, uniformly distributed random variables over these respective ranges. Hence,

$$P\{X < \frac{L}{2}\cos\theta\} = \iint_{x<\frac{L}{2}\cos y} f_X(x) f_\theta(y)\,dx\,dy = \frac{4}{\pi D}\int_0^{\pi/2}\int_0^{\frac{L}{2}\cos y} dx\,dy = \frac{4}{\pi D}\int_0^{\pi/2}\frac{L}{2}\cos y\,dy = \frac{2L}{\pi D}$$

***Example 2e Characterization of the Normal Distribution**

Let X and Y denote the horizontal and vertical miss distances when a bullet is fired at a target, and assume that

1. X and Y are independent continuous random variables having differentiable density functions.
2. The joint density $f(x,y)=f_X(x)f_Y(y)$ of X and Y depends on (x, y) only through x^2+y^2.

Loosely put, assumption 2 states that the probability of the bullet landing on any point of the x–y plane depends only on the distance of the point from the target and not on its angle of orientation. An equivalent way of phrasing this assumption is to say that the joint density function is rotation invariant.

It is a rather interesting fact that assumptions 1 and 2 imply that X and Y are normally distributed random variables. To prove this, note first that the assumptions yield the relation

$$f(x,y)=f_X(x)f_Y(y)=g(x^2+y^2) \tag{2.9}$$

for some function g. Differentiating **Equation (2.9)** with respect to x yields

$$f'_X(x)f_Y(y)=2x\, g'(x^2+y^2) \tag{2.10}$$

Dividing **Equation (2.10)** by **Equation (2.9)** gives

$$f'_X(x)f_X(x)=2x\, g'(x^2+y^2)g(x^2+y^2)$$

or

$$f'_X(x)2x\, f_X(x)=g'(x^2+y^2)g(x^2+y^2) \tag{2.11}$$

Because the value of the left-hand side of **Equation (2.11)** depends only on x, whereas the value of the right-hand side depends on x^2+y^2, it follows that the left-hand side must be the same for all x. To see this, consider any x_1, x_2 and let y_1, y_2 be such that $x_1^2+y_1^2=x_2^2+y_2^2$. Then, from **Equation (2.11)**, we obtain

$$f'_X(x_1)2x_1\, f_X(x_1)=g'(x_1^2+y_1^2)g(x_1^2+y_1^2)=g'(x_2^2+y_2^2)g(x_2^2+y_2^2)=f'_X(x_2)2x_2\, f_X(x_2)$$

Hence,

$$f'_X(x)x\, f_X(x)=c \quad \text{or} \quad \frac{d}{dx}(\log f_X(x))=cx$$

which implies, upon integration of both sides, that

$$\log f_X(x)=a+cx^2/2 \quad \text{or} \quad f_X(x)=ke^{cx^2/2}$$

Since $\int_{-\infty}^{\infty} f_X(x)\, dx=1$, it follows that c is necessarily negative, and we may write $c=-1/\sigma^2$. Thus,

$$f_X(x)=ke^{-x^2/2\sigma^2}$$

That is, X is a normal random variable with parameters $\mu=0$ and σ^2. A similar argument can be applied to $f_Y(y)$ to show that

$$f_Y(y)=\frac{1}{\sqrt{2\pi}\,\sigma}e^{-y^2/2\sigma^2}$$

Furthermore, it follows from assumption 2 that $\sigma_2=\sigma^2$ and that X and Y are thus independent, identically distributed normal random variables with parameters $\mu=0$ and σ^2.

A necessary and sufficient condition for the random variables X and Y to be independent is for their joint probability density function (or joint probability mass function in the discrete case) f(x, y) to factor into two terms, one depending only on x and the other depending only on y.

Proposition 2.1

The continuous (discrete) random variables X and Y are independent if and only if their joint probability density (mass) function can be expressed as

$$f_{X,Y}(x,y)=h(x)\, g(y) \qquad -\infty<x<\infty, -\infty<y<\infty$$

Proof Let us give the proof in the continuous case. First, note that independence implies that the joint density is the product of the marginal densities of X and Y, so the preceding factorization will hold when the random variables are independent. Now, suppose that

$$f_{X,Y}(x,y)=h(x)\,g(y)$$

Then

$$1=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx\,dy=\int_{-\infty}^{\infty} h(x)\,dx\int_{-\infty}^{\infty} g(y)\,dy=C_1C_2$$

where $C_1=\int_{-\infty}^{\infty} h(x)\,dx$ and $C_2=\int_{-\infty}^{\infty} g(y)\,dy$. Also,

$$f_X(x)=\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy=C_2\,h(x)\qquad f_Y(y)=\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx=C_1\,g(y)$$

Since $C_1C_2=1$, it follows that

$$f_{X,Y}(x,y)=f_X(x)f_Y(y)$$

and the proof is complete.

Example 2f

If the joint density function of X and Y is

$$f(x,y)=6e^{-2x}e^{-3y}\qquad 0<x<\infty,\ 0<y<\infty$$

and is equal to 0 outside this region, are the random variables independent? What if the joint density function is

$$f(x,y)=24xy\qquad 0<x<1,\ 0<y<1,\ 0<x+y<1$$

and is equal to 0 otherwise?

Solution

In the first instance, the joint density function factors, and thus the random variables, are independent (with one being exponential with rate 2 and the other exponential with rate 3). In the second instance, because the region in which the joint density is nonzero cannot be expressed in the form $x\in A, y\in B$, the joint density does not factor, so the random variables are not independent. This can be seen clearly by letting

$$I(x,y)=\begin{cases}1 & \text{if } 0<x<1,\ 0<y<1,\ 0<x+y<1\\ 0 & \text{otherwise}\end{cases}$$

and writing

$$f(x,y)=24xy\,I(x,y)$$

which clearly does not factor into a part depending only on x and another depending only on y.

The concept of independence may, of course, be defined for more than two random variables. In general, the n random variables $X_1,X_2,\ldots,X_n$ are said to be independent if, for all sets of real numbers $A_1,A_2,\ldots,A_n$,

$$P\{X1\in A1, X2\in A2, \ldots, Xn\in An\} = \prod_{i=1}^{n} P\{Xi\in Ai\}$$

As before, it can be shown that this condition is equivalent to

$$P\{X_1 \le a_1, X_2 \le a_2, \ldots, X_n \le a_n\} = \prod_{i=1}^{n} P\{X_i \le a_i\} \quad \text{for all } a_1, a_2, \ldots, a_n$$

Finally, we say that an infinite collection of random variables is independent if every finite subcollection of them is independent.

Example 2g How can a computer choose a random subset?

Most computers are able to generate the value of, or *simulate*, a uniform (0, 1) random variable by means of a built-in subroutine that (to a high degree of approximation) produces such "random numbers." As a result, it is quite easy for a computer to simulate an indicator (that is, a Bernoulli) random variable. Suppose I is an indicator variable such that

$$P\{I=1\} = p = 1 - P\{I=0\}$$

The computer can simulate I by choosing a uniform (0, 1) random number U and then letting

$$I = \begin{cases} 1 & \text{if } U < p \\ 0 & \text{if } U \ge p \end{cases}$$

Suppose that we are interested in having the computer select k, $k \le n$, of the numbers $1, 2, \ldots, n$ in such a way that each of the $\binom{n}{k}$ subsets of size k is equally likely to be chosen. We now present a method that will enable the computer to solve this task. To generate such a subset, we will first simulate, in sequence, n indicator variables $I_1, I_2, \ldots, I_n$, of which exactly k will equal 1. Those i for which $I_i = 1$ will then constitute the desired subset.

To generate the random variables $I_1, \ldots, I_n$, start by simulating n independent uniform (0, 1) random variables $U_1, U_2, \ldots, U_n$. Now define

$$I_1 = \begin{cases} 1 & \text{if } U_1 < \frac{k}{n} \\ 0 & \text{otherwise} \end{cases}$$

and then, once $I_1, \ldots, I_i$ are determined, recursively set

$$I_{i+1} = \begin{cases} 1 & \text{if } U_{i+1} < \frac{k - (I_1 + \cdots + I_i)}{n-i} \\ 0 & \text{otherwise} \end{cases}$$

In words, at the (i+1)th stage, we set I_{i+1} equal to 1 (and thus put i+1 into the desired subset) with a probability equal to the remaining number of places in the subset (namely, $k - \sum_{j=1}^{i} I_j$), divided by the remaining number of possibilities (namely, n−i). Hence, the joint distribution of $I_1, I_2, \ldots, I_n$ is determined from

$$P\{I_1 = 1\} = \frac{k}{n} \quad P\{I_{i+1} = 1 \mid I_1, \ldots, I_i\} = \frac{k - \sum_{j=1}^{i} I_j}{n-i} \quad 1 < i < n$$

The proof that the preceding formula results in all subsets of size k being equally likely to be chosen is by induction on k+n. It is immediate when k+n=2 (that is, when k=1, n=1), so assume it to be true whenever $k+n \le l$. Now, suppose that k+n=l+1, and consider any subset of size k—say, $i_1 \le i_2 \le \cdots \le i_k$—and consider the following two cases.

Case 1: $i_1 = 1$

$$P\{I_1 = I_{i_2} = \cdots = I_{i_k} = 1, I_j = 0 \text{ otherwise}\} = P\{I_1 = 1\} P\{I_{i_2} = \cdots = I_{i_k} = 1, I_j = 0 \text{ otherwise} \mid I_1 = 1\}$$

Now given that $I_1 = 1$, the remaining elements of the subset are chosen as if a subset of size k−1 were to

be chosen from the n−1 elements 2,3,… ,n. Hence, by the induction hypothesis, the conditional probability that this will result in a given subset of size k−1 being selected is 1/(n−1k−1). Hence,

P{ I 1 = I i 2 =⋯= I i k =1, I j =0 otherwise} = k n 1 (n−1 k−1) = 1 (n k)

Case 2: i 1 ≠1

P{ I i1 = I i 2 =⋯= I i k =1, I j =0 otherwise} =P{ I i 1 =⋯= I i k =1, I j =0 otherwise| I 1 =0}P{ I 1 =0 } = 1 (n−1 k) (1− k n)= 1 (n k)

where the induction hypothesis was used to evaluate the preceding conditional probability.

Thus, in all cases, the probability that a given subset of size k will be the subset chosen is 1/(nk).

Remark The foregoing method for generating a random subset has a very low memory requirement. A faster algorithm that requires somewhat more memory is presented in **Section 10.1**. (The latter algorithm uses the last k elements of a random permutation of 1,2,… ,n.)

Example 2h

Let X, Y, Z be independent and uniformly distributed over (0, 1). Compute P{X≥YZ}.

Solution

Since

f X,Y,Z (x,y,z) = f X (x) f Y (y) f Z (z) =1, 0≤x≤1, 0≤y≤1, 0≤z≤1

we have

P{ X≥YZ } = ∫∫∫ x≥yz f X, Y, Z (x,y,z) dx dy dz = ∫ 0 1 ∫ 0 1 ∫ yz 1 dx dy dz = ∫ 0 1 ∫ 0 1 (1−yz) dy dz = ∫ 0 1 (1− z 2) dz = 3 4

Example 2i Probabilistic Interpretation of Half-Life

Let N(t) denote the number of nuclei contained in a radioactive mass of material at time t. The concept of half-life is often defined in a deterministic fashion by stating this it is an empirical fact that, for some value h, called the *half-life*,

N(t)=2−t/hN(0) t>0

[Note that N(h)=N(0)/2.] Since the preceding implies that, for any nonnegative s and t,

N(t+s)=2−(s+t)/hN(0)=2−t/hN(s)

it follows that no matter how much time s has already elapsed, in an additional time t, the number of existing nuclei will decrease by the factor 2−t/h.

Because the deterministic relationship just given results from observations of radioactive masses containing huge numbers of nuclei, it would seem that it might be consistent with a probabilistic interpretation. The clue to deriving the appropriate probability model for half-life resides in the empirical observation that the proportion of decay in any time interval depends neither on the total number of nuclei at the beginning of the interval nor on the location of this interval [since N(t+s)/N(s) depends neither on N(s) nor on s]. Thus, it appears that the individual nuclei act independently and with a memoryless life distribution. Consequently, since the unique life distribution that is memoryless is the

exponential distribution, and since exactly one-half of a given amount of mass decays every h time units, we propose the following probabilistic model for radioactive decay.

Probabilistic interpretation of the half-life *h*: The lifetimes of the individual nuclei are independent random variables having a life distribution that is exponential with median equal to h. That is, if L represents the lifetime of a given nucleus, then

$$P\{L<t\}=1-2^{-t/h}$$

(Because $P\{L<h\}=12$ and the preceding can be written as

$$P\{L<t\}=1-\exp\{-t\log 2h\}$$

it can be seen that L indeed has an exponential distribution with median h.)

Note that under the probabilistic interpretation of half-life just given, if one starts with N(0) nuclei at time 0, then N(t), the number of nuclei that remain at time t will have a binomial distribution with parameters n=N(0) and $p=2^{-t/h}$. Results of **Chapter 8** will show that this interpretation of half-life is consistent with the deterministic model when considering the proportion of a large number of nuclei that decay over a given time frame. However, the difference between the deterministic and probabilistic interpretation becomes apparent when one considers the actual number of decayed nuclei. We will now indicate this with regard to the question of whether protons decay.

There is some controversy over whether or not protons decay. Indeed, one theory predicts that protons should decay with a half-life of about h=1030 years. To check this prediction empirically, it has been suggested that one follow a large number of protons for, say, one or two years and determine whether any of them decay within that period. (Clearly, it would not be feasible to follow a mass of protons for 1030 years to see whether one-half of it decays.) Let us suppose that we are able to keep track of N(0)=1030 protons for c years. The number of decays predicted by the deterministic model would then be given by

$$N(0)-N(c) = h(1-2^{-c/h}) = 1-2^{-c/h}\ 1/h \approx \lim_{x\to 0} \frac{1-2^{-cx}}{x} \quad \text{since } 1\ h = 10^{-30} \approx 0 = \lim_{x\to 0} \frac{(c\,2^{-cx}\log 2)}{1} \text{ by L'Hôpital's rule } = c\log 2 \approx .6931c$$

For instance, the deterministic model predicts that in 2 years there should be 1.3863 decays, and it would thus appear to be a serious blow to the hypothesis that protons decay with a half-life of 1030 years if no decays are observed over those 2 years.

Let us now contrast the conclusions just drawn with those obtained from the probabilistic model. Again, let us consider the hypothesis that the half-life of protons is h=1030 years, and suppose that we follow h protons for c years. Since there is a huge number of independent protons, each of which will have a very small probability of decaying within this time period, it follows that the number of protons that decay will have (to a very strong approximation) a Poisson distribution with parameter equal to $h(1-2^{-c/h}) \approx c\log 2$. Thus,

$$P\{0 \text{ decays}\} = e^{-c\log 2} = e^{-\log(2^c)} = 1/2^c$$

and, in general,

$$P\{n \text{ decays}\} = 2^{-c}[c\log 2]^n/n! \qquad n \geq 0$$

Thus, we see that even though the average number of decays over 2 years is (as predicted by the deterministic model) 1.3863, there is 1 chance in 4 that there will not be any decays, thereby indicating that such a result in no way invalidates the original hypothesis of proton decay.

Remarks *Independence is a symmetric relation*. The random variables X and Y are independent if their joint density function (or mass function in the discrete case) is the product of their individual density (or mass) functions. Therefore, to say that X is independent of Y is equivalent to saying that Y is independent of X — or just that X and Y are independent. As a result, in considering whether X is independent of Y in situations where it is not at all intuitive that knowing the value of Y will not change the probabilities concerning X, it can be beneficial to interchange the roles of X and Y and ask instead whether Y is independent of X. The next example illustrates this point.

Example 2j

If the initial throw of the dice in the game of craps results in the sum of the dice equaling 4, then the player will continue to throw the dice until the sum is either 4 or 7. If this sum is 4, then the player wins, and if it is 7, then the player loses. Let N denote the number of throws needed until either 4 or 7 appears, and let X denote the value (either 4 or 7) of the final throw. Is N independent of X? That is, does knowing which of 4 or 7 occurs first affect the distribution of the number of throws needed until that number appears? Most people do not find the answer to this question to be intuitively obvious. However, suppose that we turn it around and ask whether X is independent of N. That is, does knowing how many throws it takes to obtain a sum of either 4 or 7 affect the probability that that sum is equal to 4? For instance, suppose we know that it takes n throws of the dice to obtain a sum of either 4 or 7. Does this affect the probability distribution of the final sum? Clearly not, since all that is important is that its value is either 4 or 7, and the fact that none of the first n−1 throws were either 4 or 7 does not change the probabilities for the nth throw. Thus, we can conclude that X is independent of N, or equivalently, that N is independent of X.

As another example, let X1,X2,… be a sequence of independent and identically distributed continuous random variables, and suppose that we observe these random variables in sequence. If Xn>Xi for each i=1,… ,n−1, then we say that Xn is a *record value*. That is, each random variable that is larger than all those preceding it is called a record value. Let An denote the event that Xn is a record value. Is An+1 independent of An? That is, does knowing that the nth random variable is the largest of the first n change the probability that the (n+1) random variable is the largest of the first n+1? While it is true that An+1 is independent of An, this may not be intuitively obvious. However, if we turn the question around and ask whether An is independent of An+1, then the result is more easily understood. For knowing that the (n+1) value is larger than X1,… ,Xn clearly gives us no information about the relative size of Xn among the first n random variables. Indeed, by symmetry, it is clear that each of these n random variables is equally likely to be the largest of this set, so P(An|An+1)=P(An)=1/n. Hence, we can conclude that An and An+1 are independent events.

Remark It follows from the identity

$$P\{X_1 \le a_1, \ldots, X_n \le a_n\} = P\{X_1 \le a_1\} P\{X_2 \le a_2 \mid X_1 \le a_1\} \cdots P\{X_n \le a_n \mid X_1 \le a_1, \ldots, X_{n-1} \le a_{n-1}\}$$

that the independence of X1,… ,Xn can be established sequentially. That is, we can show that these random variables are independent by showing that

$$X_2 \text{ is independent of } X_1 \quad X_3 \text{ is independent of } X_1, X_2 \quad X_4 \text{ is independent of } X_1, X_2, X_3 \vdots X_n \text{ is independent of } X_1, \ldots, X_{n-1}$$

6.3 Sums of Independent Random Variables

It is often important to be able to calculate the distribution of X+Y from the distributions of X and Y when X and Y are independent. Suppose that X and Y are independent, continuous random variables having probability density functions fX and fY. The cumulative distribution function of X+Y is obtained as follows:

$$F_{X+Y}(a) = P\{X+Y \le a\} = \iint_{x+y \le a} f_X(x) f_Y(y)\, dx\, dy = \int_{-\infty}^{\infty} \int_{-\infty}^{a-y} f_X(x) f_Y(y)\, dx\, dy = \int_{-\infty}^{\infty} \int_{-\infty}^{a-y} f_X(x) dx\, f_Y(y)\, dy = \int_{-\infty}^{\infty} F_X(a-y)\, f_Y(y)\, dy \quad (3.1)$$

The cumulative distribution function FX+Y is called the *convolution* of the distributions FX and FY (the cumulative distribution functions of X and Y, respectively).

By differentiating **Equation (3.1)**, we find that the probability density function fX+Y of X+Y is given by

$$f_{X+Y}(a) = \frac{d}{da} \int_{-\infty}^{\infty} F_X(a-y)\, f_Y(y) dy = \int_{-\infty}^{\infty} \frac{d}{da} F_X(a-y)\, f_Y(y)\, dy = \int_{-\infty}^{\infty} f_X(a-y)\, f_Y(y)\, dy \quad (3.2)$$

6.3.1 Identically Distributed Uniform Random Variables

It is not difficult to determine the density function of the sum of two independent uniform (0,1) random variables.

Example 3a Sum of two independent uniform random variables

If X and Y are independent random variables, both uniformly distributed on (0, 1), calculate the probability density of X+Y.

Solution

From **Equation (3.2)**, since

$$f_X(a) = f_Y(a) = \begin{cases} 1 & 0 < a < 1 \\ 0 & \text{otherwise} \end{cases}$$

we obtain

$$f_{X+Y}(a) = \int_0^1 f_X(a-y)\, dy$$

For $0 \le a \le 1$, this yields

$$f_{X+Y}(a) = \int_0^a dy = a$$

For $1 < a < 2$, we get

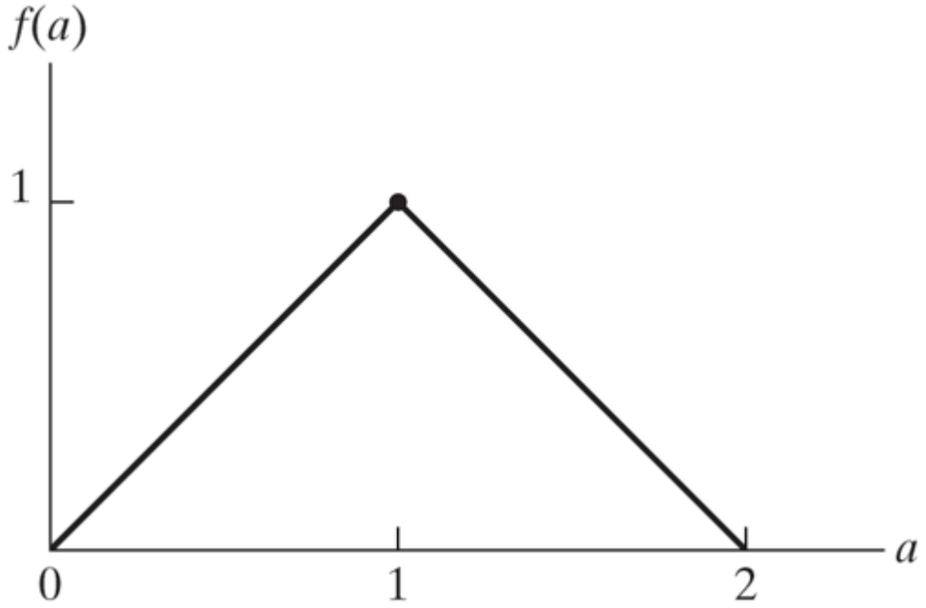
$$f_{X+Y}(a) = \int_{a-1}^1 dy = 2-a$$

Hence,

$$f_{X+Y}(a) = \begin{cases} a & 0 \le a \le 1 \\ 2-a & 1 < a < 2 \\ 0 & \text{otherwise} \end{cases}$$

Because of the shape of its density function (see **Figure 6.3**), the random variable X+Y is said to have a *triangular* distribution.

Figure 6.3 Triangular density function.



Now, suppose that X1,X2,… ,Xn are independent uniform (0,1) random variables, and let

Fn(x)=P{X1+… +Xn≤x}

Whereas a general formula for Fn(x) is messy, it has a particularly nice form when x≤1. Indeed, we now use mathematical induction to prove that

Fn(x)=xn/n! , 0≤x≤1

Because the proceeding equation is true for n=1, assume that

Fn−1(x)=xn−1/(n−1)! , 0≤x≤1

Now, writing

∑i=1nXi=∑i=1n−1Xi+Xn

and using the fact that the Xi are all nonnegative, we see from **Equation (3.1)** that, for 0≤x≤1,

F n (x) = ∫ 0 1 F n−1 (x−y) f X n (y)dy = 1 (n−1)! ∫ 0 x (x−y) n−1 dy by the induction hypothesis = 1 (n−1)! ∫ 0 x w n−1 dw (by w=x−y) = x n /n!

which completes the proof.

For an interesting application of the preceding formula, let us use it to determine the expected number of independent uniform (0,1) random variables that need to be summed to exceed 1. That is, with X1,X2,… being independent uniform (0,1) random variables, we want to determine E[N], where

N=min{n:X1+… +Xn>1}

Noting that N is greater than n>0 if and only if X 1 +…+ X n ≤1, we see that

P{N>n}=Fn(1)=1/n! , n>0

Because

P{N>0}=1=1/0!

we see that, for n>0,

P{ N=n }=P{N>n−1}−P{N>n}= 1 (n−1)! − 1 n! = n−1 n!

Therefore,

E[N] = ∑ n=1 ∞ n(n−1) n! = ∑ n=2 ∞ 1 (n−2)! =e

That is, the mean number of independent uniform (0,1) random variables that must be summed for the sum to exceed 1 is equal to e.

6.3.2 Gamma Random Variables

Recall that a gamma random variable has a density of the form

f(y)=λe−λy(λy)t−1Γ(t) 0<y<∞

An important property of this family of distributions is that for a fixed value of λ, it is closed under convolutions.

> ***Proposition 3.1***
>
> If X and Y are independent gamma random variables with respective parameters (s,λ) and (t,λ), then X+Y is a gamma random variable with parameters (s+t,λ).
>
> **Proof** Using **Equation (3.2)**, we obtain
>
> f X+Y (a) = 1 Γ(s)Γ(t) ∫ 0 a λ e −λ(a−y) [λ(a−y)] s−1 λ e −λy (λy) t−1 dy =K e −λa ∫ 0 a (a−y) s−1 y t−1 dy =K e −λa a s+t−1 ∫ 0 1 (1−x) s−1 x t−1 dx by letting x= y a =C e −λa a s+t−1
>
> where C is a constant that does not depend on a. But, as the preceding is a density function and thus must integrate to 1, the value of C is determined, and we have
>
> fX+Y(a)=λe−λa(λa)s+t−1Γ(s+t)
>
> Hence, the result is proved.

It is now a simple matter to establish, by using **Proposition 3.1** and induction, that if Xi,i=1,… ,n are independent gamma random variables with respective parameters (ti,λ),i=1,… ,n, then ∑i=1nXi is gamma with parameters (∑i=1nti,λ). We leave the proof of this statement as an exercise.

> ***Example 3b***
>
> Let X1,X2,… ,Xn be n independent exponential random variables, each having parameter λ. Then, since an exponential random variable with parameter λ is the same as a gamma random variable with parameters (1,λ), it follows from **Proposition 3.1** that X1+X2+⋯+Xn is a gamma random variable with parameters (n,λ).

If Z1,Z2,… ,Zn are independent standard normal random variables, then ∑ i=1 n Z i 2 is said to have the *chi-squared* (sometimes seen as χ2) distribution with n degrees of freedom. Let us compute the density function of Y. When n=1,Y=Z21, and from **Example 7b** of **Chapter 5**, we see that its probability density function is given by

$$f_{Z^2}(y) = \frac{1}{2\sqrt{y}}\left[f_Z(\sqrt{y}) + f_Z(-\sqrt{y})\right] = \frac{1}{2\sqrt{y}}\frac{2}{\sqrt{2\pi}}e^{-y/2} = \frac{1}{2}e^{-y/2}(y/2)^{1/2-1}\pi$$

But we recognize the preceding as the gamma distribution with parameters (12,12). [A by-product of this analysis is that $\Gamma(12)=\pi$.] But since each Z_i^2 is gamma (12,12), it follows from **Proposition 3.1** that the chi-squared distribution with n degrees of freedom is just the gamma distribution with parameters (n/2,12) and hence has a probability density function given by

$$f_Y(y) = \frac{1}{2}e^{-y/2}\left(\frac{y}{2}\right)^{n/2-1}\frac{1}{\Gamma\left(\frac{n}{2}\right)} \quad y>0 = \frac{e^{-y/2}y^{n/2-1}}{2^{n/2}\Gamma\left(\frac{n}{2}\right)} \quad y>0$$

When n is an even integer, $\Gamma(n/2)=[(n/2)-1]!$, whereas when n is odd, $\Gamma(n/2)$ can be obtained from iterating the relationship $\Gamma(t)=(t-1)\Gamma(t-1)$ and then using the previously obtained result that $\Gamma(12)=\sqrt{\pi}$. [For instance, $\Gamma(52)=32\Gamma(32)=3212\Gamma(12)=34\sqrt{\pi}$.]

In practice, the chi-squared distribution often arises as the distribution of the square of the error involved when one attempts to hit a target in n-dimensional space when the coordinate errors are taken to be independent standard normal random variables. It is also important in statistical analysis.

6.3.3 Normal Random Variables

We can also use **Equation (3.2)** to prove the following important result about normal random variables.

Proposition 3.2

If $X_i, i=1,\ldots,n$, are independent random variables that are normally distributed with respective parameters $\mu_i,\sigma_i^2, i=1,\ldots,n$, then $\sum_{i=1}^{n}X_i$ is normally distributed with parameters $\sum_{i=1}^{n}\mu_i$ and $\sum_{i=1}^{n}\sigma_i^2$.

Proof of Proposition 3.2: To begin, let X and Y be independent normal random variables with X having mean 0 and variance σ^2 and Y having mean 0 and variance 1. We will determine the density function of X+Y by utilizing **Equation (3.2)**. Now, with

$$c=\frac{1}{2\sigma^2}+\frac{1}{2}=\frac{1+\sigma^2}{2\sigma^2}$$

we have

$$f_X(a-y)f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\left\{-\frac{(a-y)^2}{2\sigma^2}\right\}\frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{y^2}{2}\right\} = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2\sigma^2}\right\}\exp\left\{-cy^2+\frac{ay}{\sigma^2}\right\} = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2\sigma^2}-c\left(y^2-\frac{2ay}{1+\sigma^2}\right)\right\}$$

where the preceding follows because $\frac{2c}{1+\sigma^2}=\frac{1}{\sigma^2}$. Now,

$$\frac{a^2}{2\sigma^2}+c\left(y^2-\frac{2ya}{1+\sigma^2}\right) = \frac{a^2}{2\sigma^2}+c\left(y-\frac{a}{1+\sigma^2}\right)^2-c\frac{a^2}{(1+\sigma^2)^2} = \frac{a^2}{2\sigma^2}+c\left(y-\frac{a}{1+\sigma^2}\right)^2-\frac{a^2}{2\sigma^2(1+\sigma^2)} = \frac{a^2}{2\sigma^2}\left(1-\frac{1}{1+\sigma^2}\right)+c\left(y-\frac{a}{1+\sigma^2}\right)^2 = \frac{a^2}{2(1+\sigma^2)}+c\left(y-\frac{a}{1+\sigma^2}\right)^2$$

Hence,

$$f_X(a-y)f_Y(y) = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2(1+\sigma^2)}\right\}\exp\left\{-c\left(y-\frac{a}{1+\sigma^2}\right)^2\right\}$$

From **Equation (3.2)**, we obtain that

$$f_{X+Y}(a) = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2(1+\sigma^2)}\right\}\int_{-\infty}^{\infty}\exp\left\{-c\left(y-\frac{a}{1+\sigma^2}\right)^2\right\}dy = \frac{1}{2\pi\sigma}\exp\left\{-\frac{a^2}{2(1+\sigma^2)}\right\}\int_{-\infty}^{\infty}\exp\{-cx^2\}\,dx = C\exp\left\{-\frac{a^2}{2(1+\sigma^2)}\right\}$$

where C does not depend on a. But this implies that X+Y is normal with mean 0 and variance 1+σ2.

Now, suppose that X1 and X2 are independent normal random variables with Xi having mean μi and variance σ i 2 , i=1,2. Then

X1+X2=σ2(X1−μ1σ2+X2−μ2σ2)+μ1+μ2

But since (X1−μ1)/σ2 is normal with mean 0 and variance σ12/σ22, and (X2−μ2)/σ2 is normal with mean 0 and variance 1, it follows from our previous result that (X1−μ1)/σ2+(X2−μ2)/σ2 is normal with mean 0 and variance 1+ σ 1 2 / σ 2 2 , implying that X1+X2 is normal with mean μ1+μ2 and variance σ 2 2 (1+ σ 1 2 / σ 2 2)= σ 1 2 + σ 2 2 .

Thus, **Proposition 3.2** is established when n=2. The general case now follows by induction. That is, assume that **Proposition 3.2** is true when there are n−1 random variables. Now consider the case of n, and write

∑i=1n Xi=∑i=1n−1Xi+Xn

By the induction hypothesis, ∑i=1n−1Xi is normal with mean ∑i=1n−1μi and variance ∑ i−1 n−1 σ i 2 . Therefore, by the result for n=2, ∑i=1nXi is normal with mean ∑i=1nμi and variance ∑ i−1 n σ i 2 .

Example 3c

A basketball team will play a 44-game season. Twenty-six of these games are against class A teams and 18 are against class B teams. Suppose that the team will win each game against a class A team with probability .4 and will win each game against a class B team with probability .7. Suppose also that the results of the different games are independent. Approximate the probability that

a. the team wins 25 games or more;
b. the team wins more games against class A teams than it does against class B teams.

Solution

a. Let XA and XB respectively denote the number of games the team wins against class A and against class B teams. Note that XA and XB are independent binomial random variables and E[X A]=26(.4)=10.4 Var(X A)=26(.4)(.6)=6.24 E[X B]=18(.7)=12.6 Var(X B)=18(.7)(.3)=3.78

   By the normal approximation to the binomial, XA and XB will have approximately the same distribution as would independent normal random variables with the preceding expected values and variances. Hence, by **Proposition 3.2**, XA+XB will have approximately a normal distribution with mean 23 and variance 10.02. Therefore, letting Z denote a standard normal random variable, we have
   P{ X A + X B ≥25 } =P{ X A + X B ≥24.5 } =P{ X A + X B −23 10.02 ≥ 24.5−23 10.02 } ≈P{ Z≥ 1.5 10.02 } ≈1−P{Z<.4739} ≈.3178

b. We note that XA−XB will have approximately a normal distribution with mean −2.2 and variance 10.02. Hence,
   P{ X A − X B ≥1 } =P{ X A − X B ≥.5 } =P{ X A − X B +2.2 10.02 ≥ .5+2.2 10.02 } ≈P{ Z≥ 2.7 10.02 } ≈1−P{Z<.8530} ≈.1968

Therefore, there is approximately a 31.78 percent chance that the team will win at least 25 games and approximately a 19.68 percent chance that it will win more games against class A teams than against class B teams.

The random variable Y is said to be a *lognormal* random variable with parameters μ and σ if log (Y) is a normal random variable with mean μ and variance σ2. That is, Y is lognormal if it can be expressed as

Y=eX

where X is a normal random variable.

Example 3d

Starting at some fixed time, let S(n) denote the price of a certain security at the end of n additional weeks, n≥1. A popular model for the evolution of these prices assumes that the price ratios S(n)/S(n−1),n≥1, are independent and identically distributed lognormal random variables. Assuming this model, with parameters μ=.0165,σ=.0730, what is the probability that

a. the price of the security increases over each of the next two weeks?
b. the price at the end of two weeks is higher than it is today?

Solution

Let Z be a standard normal random variable. To solve part (a), we use the fact that log(x) increases in x to conclude that x>1 if and only if log (x)> log (1)=0. As a result, we have

P{ S(1) S(0) >1} =P{log(S(1) S(0))>0} =P{Z> −.0165 .0730 } =P{Z<.2260} =.5894

In other words, the probability that the price is up after one week is .5894. Since the successive price ratios are independent, the probability that the price increases over each of the next two weeks is (.5894)2=.3474.

To solve part (b), we reason as follows:

P{ S(2) S(0) >1} =P{ S(2) S(1) S(1) S(0) >1} =P{log(S(2) S(1))+log(S(1) S(0))>0}

However, log (S(2)S(1))+log (S(1)S(0)), being the sum of two independent normal random variables with a common mean .0165 and a common standard deviation .0730, is a normal random variable with mean .0330 and variance 2(.0730)2. Consequently,

P{ S(2) S(0) >1} =P{Z> −.0330 .0730 2 } =P{Z<.31965} =.6254

6.3.4 Poisson and Binomial Random Variables

Rather than attempt to derive a general expression for the distribution of X+Y in the discrete case, we shall consider some examples.

Example 3e Sums of independent poisson random variables

If X and Y are independent Poisson random variables with respective parameters λ1 and λ2, compute the distribution of X+Y.

Solution

Because the event {X+Y=n} may be written as the union of the disjoint events {X=k,Y=n−k},0≤k≤n, we have

P{ X+Y=n } = ∑ k=0 n P{ X=k,Y=n−k } = ∑ k=0 n P{ X=k }P{ Y=n−k } = ∑ k=0 n e − λ 1 λ k 1 k! e − λ 2 λ n−k 2 (n−k)! = e −(λ 1 + λ 2) ∑ k=0 n λ k 1 λ n−k 2 k!(n−k)! = e −(λ 1 + λ 2) n! ∑ k=0 n n! k!(n−k)! λ k 1

λ n−k 2 = e −(λ 1 + λ 2) n! (λ 1 + λ 2) n

Thus, X+X has a Poisson distribution with parameter λ1+λ2.

Example 3f Sums of independent binomial random variables

Let X and Y be independent binomial random variables with respective parameters (n, p) and (m, p). Calculate the distribution of X+Y.

Solution

Recalling the interpretation of a binomial random variable, and without any computation at all, we can immediately conclude that X+Y is binomial with parameters (n+m,p). This follows because X represents the number of successes in n independent trials, each of which results in a success with probability p; similarly, Y represents the number of successes in m independent trials, each of which results in a success with probability p. Hence, given that X and Y are assumed independent, it follows that X+Y represents the number of successes in n+m independent trials when each trial has a probability p of resulting in a success. Therefore, X+Y is a binomial random variable with parameters (n+m,p). To check this conclusion analytically, note that

P{ X+Y=k } = ∑ i=0 n P{ X=i,Y=k−i } = ∑ i=0 n P{ X=i }P{ Y=k−i } = ∑ i=0 n (n i) p i q n−i (m k−i) p k−i q m−k+i

where q=1−p and where (rj)=0 when j<0. Thus,

P{X+Y=k}=pkqn+m−k∑i=0n(ni)(mk−i)

and the conclusion follows upon application of the combinatorial identity

(n+mk)=∑i=0n(ni)(mk−i)

6.4 Conditional Distributions: Discrete Case

Recall that for any two events E and F, the conditional probability of E given F is defined, provided that P(F)>0, by

P(E|F)=P(EF)P(F)

Hence, if X and Y are discrete random variables, it is natural to define the conditional probability mass function of X given that Y=y, by

p X|Y (x|y) =P{X=x|Y=y} = P{ X=x,Y=y } P{ Y=y } = p(x,y) p Y (y)

for all values of y such that pY(y)>0. Similarly, the conditional probability distribution function of X given that Y=y is defined, for all y such that pY(y)>0, by

F X|Y (x|y) =P{X≤x|Y=y} = ∑ a≤x p X|Y (a|y)

In other words, the definitions are exactly the same as in the unconditional case, except that everything is now conditional on the event that Y=y. If X is independent of Y, then the conditional mass function and the distribution function are the same as the respective unconditional ones. This follows because if X is independent of Y, then

p X|Y (x|y) =P{X=x|Y=y} = P{ X=x,Y=y } P{ Y=y } = P{ X=x }P{ Y=y } P{ Y=y } =P{ X=x }

Example 4a

Suppose that p(x,y), the joint probability mass function of X and Y, is given by

p(0,0)=.4 p(0,1)=.2 p(1,0)=.1 p(1,1)=.3

Calculate the conditional probability mass function of X given that Y=1.

Solution

We first note that

pY(1)=∑xp(x,1)=p(0,1)+p(1,1)=.5

Hence,

pX|Y(0|1)=p(0,1)pY(1)=25

and

pX|Y(1|1)=p(1,1)pY(1)=35

Example 4b

If X and Y are independent Poisson random variables with respective parameters λ1 and λ2, calculate the conditional distribution of X given that X+Y=n.

Solution

We calculate the conditional probability mass function of X given that X+Y=n as follows:

$$P\{X=k|X+Y=n\} = P\{ X=k,X+Y=n \} P\{ X+Y=n \} = P\{ X=k,Y=n-k \} P\{ X+Y=n \} = P\{ X=k \}P\{ Y=n-k \} P\{ X+Y=n \}$$

where the last equality follows from the assumed independence of X and Y. Recalling (**Example 3e**) that X+Y has a Poisson distribution with parameter λ1+λ2, we see that the preceding equals

$$P\{X=k|X+Y=n = e - \lambda 1 \lambda 1 k k! e - \lambda 2 \lambda 2 n-k (n-k)! [e -(\lambda 1 + \lambda 2) (\lambda 1 + \lambda 2) n n!] -1 = n! (n-k)! \lambda 1 k \lambda 2 n-k (\lambda 1 + \lambda 2) n =(n k) (\lambda 1 \lambda 1 + \lambda 2) k (\lambda 2 \lambda 1 + \lambda 2) n-k$$

In other words, the conditional distribution of X given that X+Y=n is the binomial distribution with parameters n and λ1/(λ1+λ2).

We can also talk about joint conditional distributions, as is indicated in the next two examples.

Example 4c

Consider the multinomial distribution with joint probability mass function

$$P\{ X i = n i ,i=1,\dots ,k\}= n! n 1 !\cdots n k ! p 1 n 1 \cdots p k n k , \quad n i \geq 0, \ \sum i=1 k n i =n$$

Such a mass function results when n independent trials are performed, with each trial resulting in outcome i with probability pi, ∑ i=1 k p i =1. The random variables Xi, i=1,… ,k, represent, respectively, the number of trials that result in outcome i, i=1,… ,k. Suppose we are given that nj of the trials resulted in outcome j, for j=r+1,… ,k, where Σ j=r+1 k n j =m≤n. Then, because each of the other n−m trials must have resulted in one of the outcomes 1,… ,r, it would seem that the conditional distribution of X1,… ,Xr is the multinomial distribution on n−m trials with respective trial outcome probabilities

$$P\{\text{outcome } i|\text{outcome is not any of } r+1,\dots ,k\}=piFr, i=1,\dots ,r$$

where F r = Σ i=1 k p i is the probability that a trial results in one of the outcomes 1,… ,r.

Solution

To verify this intuition, let n1,… ,nr, be such that ∑i=1rni=n−m. Then

$$P\{ X 1 = n 1 ,\dots , X r = n r | X r+1 = n r+1 ,\dots X k = n k \} = P\{ X 1 = n 1 , \dots , X k = n k \} P\{ X r+1 = n r+1 , \dots , X k = n k \} = n! n!\cdots n k ! p 1 n 1 \cdots p r n r p r+1 n r+1 \cdots p k n k n! (n-m)! n r+1 !\cdots n k ! F r n-m p r+1 n r+1 \cdots p k n k$$

where the probability in the denominator was obtained by regarding outcomes 1,… ,r as a single outcome having probability Fr, thus showing that the probability is a multinomial probability on n trials with outcome probabilities Fr,pr+1,… ,pk. Because ∑i=1rni=n−m, the preceding can be written as

$$P\{ X 1 = n 1 ,\dots , X r = n r | X r+1 = n r+1 ,\dots X k = n k \} = (n-m)! n 1 !\cdots n r ! (p 1 F r) n 1 \cdots (p r F r) n r$$

and our intuition is upheld.

Example 4d

Consider n independent trials, with each trial being a success with probability p. Given a total of k successes, show that all possible orderings of the k successes and n−k failures are equally likely.

Solution

We want to show that given a total of k successes, each of the (n k) possible orderings of k successes and n−k failures is equally likely. Let X denote the number of successes, and consider any ordering of k successes and n−k failures, say, o=(s,… ,s,f,… ,f). Then

P(o|X=k) = P(o,X=k) P(X=k) = P(o) P(X=k) = P k (1−p) n−k (n k) p k (1−p) n−k = 1 (n k)

6.5 Conditional Distributions: Continuous Case

If X and Y have a joint probability density function f(x,y), then the conditional probability density function of X given that Y=y is defined, for all values of y such that fY(y)>0, by

fX|Y(x|y)=f(x,y)fY(y)

To motivate this definition, multiply the left-hand side by dx and the right-hand side by (dx dy)/dy to obtain

f X|Y (x|y) dx = f(x,y) dx dy f Y (y) dy ≈ P{ x≤X≤x+dx,y≤Y≤y+dy } P{ y≤Y≤y+dy } =P{x≤X≤x+dx|y≤Y≤y+dy}

In other words, for small values of dx and dy,fX|Y(x|y)dx represents the conditional probability that X is between x and x+dx given that Y is between y and y+dy.

The use of conditional densities allows us to define conditional probabilities of events associated with one random variable when we are given the value of a second random variable. That is, if X and Y are jointly continuous, then, for any set A,

P{X∈A|Y=y}=∫AfX|Y(x|y) dx

In particular, by letting A=(−∞,a) we can define the conditional cumulative distribution function of *X* given that *Y* = *y* by

FX|Y(a|y)≡P{X≤a|Y=y}=∫−∞afX|Y(x|y) dx

The reader should note that by using the ideas presented in the preceding discussion, we have been able to give workable expressions for conditional probabilities, even though the event on which we are conditioning (namely, the event {Y=y}) has probability 0.

If X and Y are independent continuous random variables, the conditional density of X given that Y=y is just the unconditional density of X. This is so because, in the independent case,

fX|Y(x|y)=f(x,y)fY(y)=fX(x)fY(y)fY(y)=fX(x)

Example 5a

The joint density of X and Y is given by

f(x,y)={ 12 5 x(2−x−y) 0<x<1,0<y<1 0 otherwise

Compute the conditional density of X given that Y=y, where 0<y<1.

Solution

For 0<x<1,0<y<1, we have

f X|Y (x|y) = f(x,y) f Y (y) = f(x,y) ∫ −∞ ∞ f(x,y) dx = x(2−x−y) ∫ 0 1 x(2−x−y) dx = x(2−x−y) 2 3 −y/2 = 6x(2−x−y) 4−3y

Example 5b

Suppose that the joint density of X and Y is given by

f(x,y)={ e −x/y e −y y 0<x<∞,0<y<∞ 0 otherwise

Find P{X>1|Y=y}.

Solution

We first obtain the conditional density of X given that Y=y.

$$f_{X|Y}(x|y) = f(x,y)\, f_Y(y) = e^{-x/y} e^{-y} / y\, e^{-y} \int_0^\infty (1/y) e^{-x/y}\, dx = 1\, y\, e^{-x/y}$$

Hence,

$$P\{X>1|Y=y\} = \int_1^\infty 1\, y\, e^{-x/y}\, dx = -e^{-x/y} \mid_1^\infty = e^{-1/y}$$

Example 5c The t-distribution

If Z and Y are independent, with Z having a standard normal distribution and Y having a chi-squared distribution with n degrees of freedom, then the random variable T defined by

$$T = ZY/n = n\, ZY$$

is said to have a *t-distribution* with n degrees of freedom. As will be seen in **Section 7.8**, the t-distribution has important applications in statistical inference. At present, we will content ourselves with computing its density function. This will be accomplished by using the conditional density of T given Y to obtain the joint density function of T and Y, from which we will then obtain the marginal density of T. To begin, note that because of the independence of Z and Y, it follows that the conditional distribution of T given that Y=y is the distribution of n/y Z, which is normal with mean 0 and variance n/y. Hence, the conditional density of T given that Y=y is

$$f_{T|Y}(t|y) = \frac{1}{2\pi n/y}\, e^{-t^2 y/2n}, \quad -\infty<t<\infty$$

Using the preceding, along with the following formula for the chi-squared density given in **Example 3b** of this chapter,

$$f_Y(y) = e^{-y/2} y^{n/2-1} 2^{n/2}\, \Gamma(n/2), \quad y>0$$

we obtain that the joint density of T,Y is

$$f_{T,Y}(t,y) = \frac{1}{2\pi n}\, 2^{n/2}\, \Gamma(n/2)\, e^{-t^2 y/2n} e^{-y/2} y^{(n-1)/2} = \frac{1}{\pi n}\, 2^{(n+1)/2}\, \Gamma(n/2) e^{-t^2+n\, 2n}\, y^{(n-1)/2}, \quad y>0, -\infty<t<\infty$$

Letting $c=\frac{t^2+n}{2n}$, and integrating the preceding over all y, gives

$$f_T(t) = \int_0^\infty f_{T,Y}(t,y)\, dy = \frac{1}{\pi n}\, 2^{(n+1)/2}\, \Gamma(n/2) \int_0^\infty e^{-cy}\, y^{(n-1)/2}\, dy = c^{-(n+1)/2} \pi n\, 2^{(n+1)/2}\, \Gamma(n/2) \int_0^\infty e^{-x} x^{(n-1)/2}\, dx \quad \text{(by letting } x=cy) = n^{(n+1)/2}\, \Gamma(\tfrac{n+1}{2})\, \pi n\, (t^2+n)^{(n+1)/2}\, \Gamma(\tfrac{n}{2})$$

$$\left(\text{because } \frac{1}{c} = \frac{2n}{t^2+n}\right) = \frac{\Gamma(\frac{n+1}{2})}{\pi n\, \Gamma(\frac{n}{2})}\, (1+t^2/n)^{-(n+1)/2}, \quad -\infty<t<\infty$$

Example 5d The bivariate normal distribution

One of the most important joint distributions is the bivariate normal distribution. We say that the random variables X,Y have a bivariate normal distribution if, for constants $\mu_x, \mu_y, \sigma_x>0, \sigma_y>0, -1<\rho<1$, their joint density function is given, for all $-\infty<x,y<\infty$, by

$$f(x,y) = \frac{1}{2\pi\, \sigma_x \sigma_y \sqrt{1-\rho^2}}\, \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\, \frac{(x-\mu_x)(y-\mu_y)}{\sigma_x \sigma_y}\right]\right\}$$

where we have made the change of variables y2=1−d−x2. Continuing the string of equalities yields

=3 ∫ 0 1−2d (1−2d− x 1) 2 d x 1 =3 ∫ 0 1−2d y 1 2 d y 1 = (1−2d) 3

Hence, the desired probability that no 2 people are within a distance d of each other when 3 people are uniformly and independently distributed over an interval of size 1 is (1−2d)3 when d≤12. In fact, the same method can be used to prove that when n people are distributed at random over the unit interval, the desired probability is

[1−(n−1)d]n when d≤1n−1

The proof is left as an exercise.

The density function of the jth-order statistic X(j) can be obtained either by integrating the joint density function (**6.1**) or by direct reasoning as follows: In order for X(j) to equal x, it is necessary for j−1 of the n values X1,… ,Xn to be less than x,n−j of them to be greater than x, and 1 of them to equal x. Now, the probability density that any given set of j−1 of the Xi‘s are less than x, another given set of n−j are all greater than x, and the remaining value is equal to x equals

[F(x)]j−1[1−F(x)]n−jf(x)

Hence, since there are

(nj−1,n−j,1)=n!(n−j)!(j−1)!

different partitions of the n random variables X1,… ,Xn into the preceding three groups, it follows that the density function of X(j) is given by

fX(j)(x)=n!(n−j)!(j−1)![F(x)]j−1[1−F(x)]n−jf(x) (6.2)

Example 6b

When a sample of 2n+1 random variables (that is, when 2n+1 independent and identically distributed random variables) is observed, the (n+1) smallest is called the *sample median*. If a sample of size 3 from a uniform distribution over (0, 1) is observed, find the probability that the sample median is between 14 and 34.

Solution

From **Equation (6.2)**, the density of X(2) is given by

fX(2)(x)=3!1!1!x(1−x) 0<x<1

Hence,

P{ 1 4 < X (2) < 3 4 } =6 ∫ 1/4 3/4 x(1−x) dx =6{ x 2 2 − x 3 3 } | x=1/4 x=3/4 = 11 16

The cumulative distribution function of X(j) can be found by integrating **Equation (6.2)**. That is,

F X (j) (y)= n! (n−j)!(j−1)! ∫ −∞ y [F(x)] j−1 [1−F(x)] n−j f(x) dx (6.3)

However, FX(j)(y) could also have been derived directly by noting that the jth order statistic is less than or equal to y if and only if there are j or more of the Xi‘s that are less than or equal to y. Thus, because the number of Xi‘s that are less than or equal to y is a binomial random variable with parameters n,p=F(y), it

follows that

$$F_{X(j)}(y) = P\{X(j) \le y\} = P\{j \text{ or more of the } X_j\text{'s are} \le y\} = \sum_{k=j}^{n} \binom{n}{k} [F(y)]^k [1-F(y)]^{n-k} \tag{6.4}$$

If, in **Equations (6.3)** and **(6.4)**, we take F to be the uniform (0, 1) distribution [that is, f(x)=1,0<x<1], then we obtain the interesting analytical identity

$$\sum_{k=j}^{n} \binom{n}{k} y^k (1-y)^{n-k} = n!\,(n-j)!(j-1)! \int_0^y x^{j-1}(1-x)^{n-j}\,dx \quad 0 \le y \le 1 \tag{6.5}$$

We can employ the same type of argument that we used in establishing **Equation (6.2)** to find the joint density of X(i) and X(j), the ith and jth smallest of the values X1,… ,Xn. For suppose that i<j and xi<xj. Then the event that X(i)=xi,X(j)=xj is equivalent to the event that the n data values can be divided into 5 groups of respective sizes i−1, 1, j−i−1, 1, n−j, that satisfy the condition that all i−1 members of the first group have values less than xi, the one member of the second group has value xi, all j−i−1 members of the third group have values between xi and xj, the one member of the fourth group has value xj, and all n−j members of the last group have values greater than xj. Now, for any specified division of the n values into 5 such groups, the preceding condition will hold with probability (density)

$$F^{i-1}(x_i) f(x_i) (F(x_j)-F(x_i))^{j-i-1} f(x_j) (1-F(x_j))^{n-j}$$

As there are n!(i−1)!1!(j−i−1)!1!(n−j)! such divisions of the n values, and as the condition cannot hold for more than one of these divisions, it follows, for i<j, xi<xj, that

$$f_{X(i),X(j)}(x_i, x_j) = n!\,(i-1)!(j-i-1)!(n-j)!\, F^{i-1}(x_i) f(x_i) [F(x_j)-F(x_i)]^{j-i-1} f(x_j) [1-F(x_j)]^{n-j} \tag{6.6}$$

Example 6c Distribution of the range of a random sample

Suppose that n independent and identically distributed random variables X1,X2,… ,Xn are observed. The random variable R defined by R=X(n)−X(1) is called the *range* of the observed random variables. If the random variables Xi have distribution function F and density function f, then the distribution of R can be obtained from **Equation (6.6)** as follows: For a≥0,

$$P\{R \le a\} = P\{X(n) - X(1) \le a\} = \iint_{x_n - x_1 \le a} f_{X(1),X(n)}(x_1, x_n)\,dx_1\,dx_n = \int_{-\infty}^{\infty} \int_{-x_1}^{x_1+a} n!\,(n-2)! [F(x_n)-F(x_1)]^{n-2} f(x_1) f(x_n)\,dx_n\,dx_1$$

Making the change of variable y=F(xn)−F(x1),dy=f(xn) dxn yields

$$\int_{x_1}^{x_1+a} [F(x_n)-F(x_1)]^{n-2} f(x_n)\,dx_n = \int_0^{F(x_1+a)-F(x_1)} y^{n-2}\,dy = \frac{1}{n-1} [F(x_1+a)-F(x_1)]^{n-1}$$

Thus,

$$P\{R \le a\} = n \int_{-\infty}^{\infty} [F(x_1+a)-F(x_1)]^{n-1} f(x_1)\,dx_1 \tag{6.7}$$

Equation (6.7) can be evaluated explicitly only in a few special cases. One such case is when the Xi's are all uniformly distributed on (0, 1). In this case, we obtain, from **Equation (6.7)**, that for 0<a<1,

$$P\{R<a\} = n \int_0^1 [F(x_1+a)-F(x_1)]^{n-1} f(x_1)\,dx_1 = n \int_0^{1-a} a^{n-1}\,dx_1 + n \int_{1-a}^1 (1-x_1)^{n-1}\,dx_1 = n(1-a)a^{n-1} + a^n$$

Differentiation yields the density function of the range: given in this case by

$$f_R(a) = \begin{cases} n(n-1)a^{n-2}(1-a) & 0 \le a \le 1 \\ 0 & \text{otherwise} \end{cases}$$

That is, the range of n independent uniform (0, 1) random variables is a beta random variable with parameters n−1,2.

6.7 Joint Probability Distribution of Functions of Random Variables

Let X1 and X2 be jointly continuous random variables with joint probability density function fX1,X2. It is sometimes necessary to obtain the joint distribution of the random variables Y1 and Y2, which arise as functions of X1 and X2. Specifically, suppose that Y1=g1(X1,X2) and Y2=g2(X1,X2) for some functions g1 and g2.

Assume that the functions g1 and g2 satisfy the following conditions:

1. The equations y1=g1(x1,x2) and y2=g2(x1,x2) can be uniquely solved for x1 and x2 in terms of y1 and y2, with solutions given by, say, x1=h1(y1,y2),x2=h2(y1, y2).
2. The functions g1 and g2 have continuous partial derivatives at all points (x1,x2) and are such that the 2×2 determinant
   $J(x_1, x_2)=\begin{vmatrix} \partial g_1/\partial x_1 & \partial g_1/\partial x_2 \\ \partial g_2/\partial x_1 & \partial g_2/\partial x_2 \end{vmatrix} \equiv \partial g_1/\partial x_1 \, \partial g_2/\partial x_2 - \partial g_1/\partial x_2 \, \partial g_2/\partial x_1 \neq 0$

   at all points (x1,x2).

Under these two conditions, it can be shown that the random variables Y1 and Y2 are jointly continuous with joint density function given by

$$f_{Y_1Y_2}(y_1,y_2)=f_{X_1,X_2}(x_1,x_2)|J(x_1,x_2)|^{-1} \tag{7.1}$$

where x1=h1(y1, y2), x2=h2(y1,y2).

A proof of **Equation (7.1)** would proceed along the following lines:

$$P\{Y_1 \le y_1, Y_2 \le y_2\}=\iint_{(x_1,x_2):\, g_1(x_1,x_2)\le y_1,\, g_2(x_1,x_2)\le y_2} f_{X_1,X_2}(x_1,x_2)\,dx_1\,dx_2 \tag{7.2}$$

The joint density function can now be obtained by differentiating **Equation (7.2)** with respect to y1 and y2. That the result of this differentiation will be equal to the right-hand side of **Equation (7.1)** is an exercise in advanced calculus whose proof will not be presented in this book.

Example 7a

Let X1 and X2 be jointly continuous random variables with probability density function fX1,X2. Let Y1=X1+X2,Y2=X1−X2. Find the joint density function of Y1 and Y2 in terms of fX1,X2.

Solution

Let g1(x1,x2)=x1+x2 and g2(x1,x2)=x1−x2. Then

J(x1,x2)=|111−1|=−2

Also, since the equations y1=x1+x2 and y2=x1−x2 have x1=(y1+y2)/2, x2=(y1−y2)/2 as their solution, it follows from **Equation (7.1)** that the desired density is

fY1,Y2(y1,y2)=12fX1, X2(y1+y22,y1−y22)

For instance, if X1 and X2 are independent uniform (0, 1) random variables, then

$$f_{Y_1,Y_2}(y_1,y_2)=\begin{cases} \frac{1}{2} & 0\le y_1+y_2\le 2,\ 0\le y_1-y_2\le 2 \\ 0 & \text{otherwise} \end{cases}$$

or if X1 and X2 are independent exponential random variables with respective parameters λ1 and λ2, then

f Y 1 , Y 2 (y 1 , y 2) ={ λ 1 λ 2 2 exp{ − λ 1 (y 1 + y 2 2)− λ 2 (y 1 − y 2 2) } y 1 + y 2 ≥0, y 1 − y 2 ≥0 otherwise

Finally, if X1 and X2 are independent standard normal random variables, then

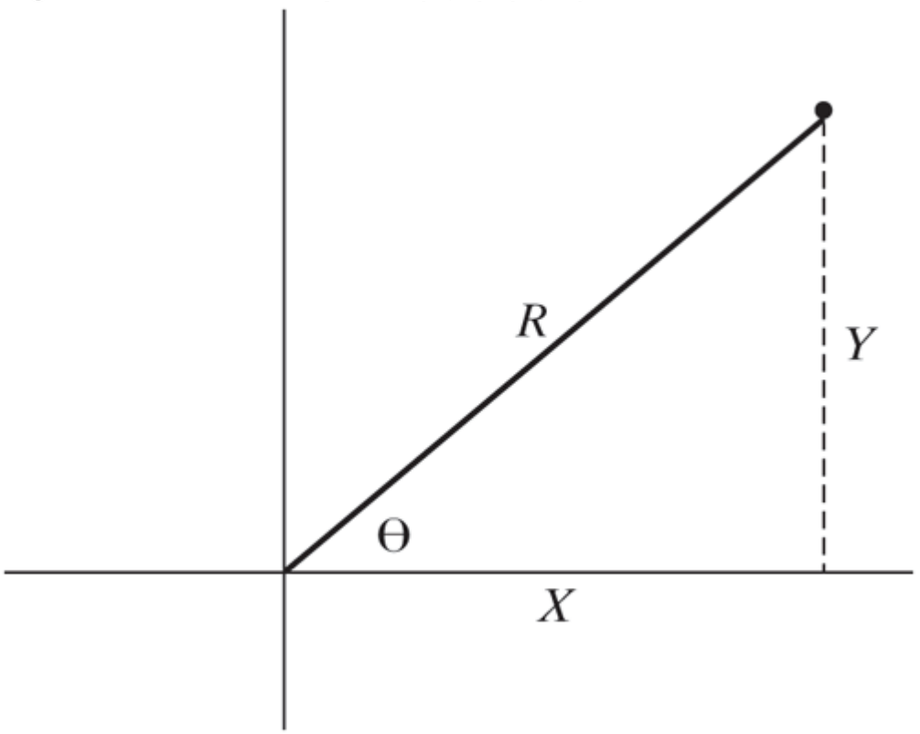
f Y 1 , Y 2 (y 1 , y 2) = 1 4π e −[(y 1 + y 2) 2 /8+ (y 1 − y 2) 2 /8] = 1 4π e −(y 2 1 + y 2 2)/4 = 1 4π e − y 2 1 /4 1 4π e − y 2 2 /4

Thus, not only do we obtain (in agreement with **Proposition 3.2**) that both X1+X2 and X1−X2 are normal with mean 0 and variance 2, but we also conclude that these two random variables are independent. (In fact, it can be shown that if X1 and X2 are independent random variables having a common distribution function F, then X1+X2 will be independent of X1−X2 if and only if F is a normal distribution function.)

Example 7b

Let (X, Y) denote a random point in the plane, and assume that the rectangular coordinates X and Y are independent standard normal random variables. We are interested in the joint distribution of R, Θ, the polar coordinate representation of (x, y). (See **Figure 6.4**.)

Figure 6.4 •= Random point. (X,Y)=(R, Θ).



Suppose first that X and Y are both positive. For x and y positive, letting r=g1(x,y)=x2+y2 and θ=g2(x,y)=tan−1y/x, we see that

∂ g 1 ∂x = x x 2 + y 2 ∂ g 1 ∂y = y x 2 + y 2 ∂ g 2 ∂x = 1 1+ (y/x) 2 (−y x 2)= −y x 2 + y 2 ∂ g 2 ∂y = 1 x[1+ (y/x) 2] = x x 2 + y 2

Hence,

J(x,y)=x2(x2+y2)3/2+y2(x2+y2)3/2=1x2+y2=1r

Because the conditional joint density function of X, Y given that they are both positive is

f(x,y|X>0,Y>0)=f(x,y)P(X>0,Y>0)=2πe−(x2+y2)/2, x>0,y>0

we see that the conditional joint density function of R=X2+Y2 and Θ=tan−1(Y/X), given that X and Y are both positive, is

f(r,θ|X>0,Y>0)= 2 π r e − r 2/2 , 0<θ<π/2, 0<r<∞

Similarly, we can show that

f(r,θ|X<0,Y>0)= 2 π r e − r 2/2 , π/2<θ<π, 0<r<∞ f(r,θ|X<0,Y<0)= 2 π r e − r 2/2 , π<θ<3π/2, 0<r<∞
f(r,θ|X>0,Y<0)= 2 π r e − r 2/2 , 3π/2<θ<2π, 0<r<∞

As the joint density is an equally weighted average of these four conditional joint densities, we obtain that the joint density of R,Θ is given by

f(r,θ)=12πre−r2/2 0<θ<2π, 0<r<∞

Now, this joint density factors into the marginal densities for R and Θ, so R and Θ are independent random variables, with Θ being uniformly distributed over (0,2π) and R having the Rayleigh distribution with density

f(r)=re−r2/2 0<r<∞

(For instance, when one is aiming at a target in the plane, if the horizontal and vertical miss distances are independent standard normals, then the absolute value of the error has the preceding Rayleigh distribution.)

This result is quite interesting, for it certainly is not evident a priori that a random vector whose coordinates are independent standard normal random variables will have an angle of orientation that not only is uniformly distributed, but also is independent of the vector's distance from the origin.

If we wanted the joint distribution of R2 and Θ, then, since the transformation d=g1(x,y)=x2+y2 and θ=g2(x,y)=tan−1y/x has the Jacobian

J=| 2x −y x 2 + y 2 2y x x 2 + y 2 |=2

it follows that

f(d,θ)=12e−d/212π 0<d<∞, 0<θ<2π

Therefore, R2 and Θ are independent, with R2 having an exponential distribution with parameter 12. But because R2=X2+Y2, it follows by definition that R2 has a chi-squared distribution with 2 degrees of freedom. Hence, we have a verification of the result that the exponential distribution with parameter 12 is the same as the chi-squared distribution with 2 degrees of freedom.

The preceding result can be used to simulate (or generate) normal random variables by making a suitable transformation on uniform random variables. Let U1 and U2 be independent random variables, each uniformly distributed over (0, 1). We will transform U1,U2 into two independent standard normal random variables X1 and X2 by first considering the polar coordinate representation (R,Θ) of the random vector (X1,X2). From the preceding, R2 and Θ will be independent, and, in addition,

R2=X21+X22 will have an exponential distribution with parameter λ=12. But −2 log U1 has such a distribution, since, for x>0,

$$P\{-2\log U_1 < x\} = P\{\log U_1 > -x/2\} = P\{U_1 > e^{-x/2}\} = 1 - e^{-x/2}$$

Also, because 2π U2 is a uniform (0,2π) random variable, we can use it to generate Θ. That is, if we let

$$R^2 = -2\log U_1 \quad \Theta = 2\pi U_2$$

then R2 can be taken to be the square of the distance from the origin and θ can be taken to be the angle of orientation of (X1,X2). Now, since X1=R cos Θ,X2=R sin Θ, it follows that

$$X_1 = \sqrt{-2\log U_1}\cos(2\pi U_2) \quad X_2 = \sqrt{-2\log U_1}\sin(2\pi U_2)$$

are independent standard normal random variables.

Example 7c

If X and Y are independent gamma random variables with parameters (α,λ) and (β,λ), respectively, compute the joint density of U=X+Y and V=X/(X+Y).

Solution

The joint density of X and Y is given by

$$f_{X,Y}(x,y) = \lambda e^{-\lambda x}(\lambda x)^{\alpha-1}\Gamma(\alpha)\,\lambda e^{-\lambda y}(\lambda y)^{\beta-1}\Gamma(\beta) = \lambda^{\alpha+\beta}\Gamma(\alpha)\Gamma(\beta)\, e^{-\lambda(x+y)}\, x^{\alpha-1} y^{\beta-1}$$

Now, if g1(x,y)=x+y,g2(x,y)=x/(x+y), then

$$\partial g_1/\partial x = \partial g_1/\partial y = 1 \quad \partial g_2/\partial x = y/(x+y)^2 \quad \partial g_2/\partial y = -x/(x+y)^2$$

so

$$J(x,y) = \begin{vmatrix} 1 & y/(x+y)^2 \\ 1 & -x/(x+y)^2 \end{vmatrix} = -\frac{1}{x+y}$$

Finally, as the equations u=x+y,v=x/(x+y) have as their solutions x=uv,y=u(1−v), we see that

$$f_{U,V}(u,v) = f_{X,Y}[uv, u(1-v)]u = \lambda e^{-\lambda u}(\lambda u)^{\alpha+\beta-1}\Gamma(\alpha+\beta)\, v^{\alpha-1}(1-v)^{\beta-1}\Gamma(\alpha+\beta)\Gamma(\alpha)\Gamma(\beta)$$

Hence, X+Y and X/(X+Y) are independent, with X+Y having a gamma distribution with parameters (α+β,λ) and X/(X+Y) having a beta distribution with parameters (α,β). The preceding reasoning also shows that B(α,β), the normalizing factor in the beta density, is such that

$$B(\alpha,\beta) \equiv \int_0^1 v^{\alpha-1}(1-v)^{\beta-1}\,dv = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$$

This entire result is quite interesting. For suppose there are n+m jobs to be performed, each (independently) taking an exponential amount of time with rate λ to be completed and suppose that we have two workers to perform these jobs. Worker I will do jobs 1,2,… ,n, and worker II will do the remaining m jobs. If we let X and Y denote the total working times of workers I and II, respectively, then (either from the foregoing result or from **Example 3b**) X and Y will be independent gamma random variables having parameters (n,λ) and (m,λ), respectively. It then follows that independently of the working time needed to complete all n+m jobs (that is, of X+Y), the proportion of this work that will be performed by worker I has a beta distribution with parameters (n, m).

When the joint density function of the n random variables X1,X2,… ,Xn is given and we want to compute the joint density function of Y1,Y2,… ,Yn, where

Y1=g1(X1,… ,Xn) Y2=g2(X1,… ,Xn),… Yn=gn(X1,… ,Xn)

the approach is the same—namely, we assume that the functions gi have continuous partial derivatives and that the Jacobian determinant

J(x1,… ,xn)=|∂g1∂x1∂g1∂x2⋯∂g1∂xn∂g2∂x1∂g2∂x2⋯∂g2∂xn∂gn∂x1∂gn∂x2⋯∂gn∂xn|≠0

at all points (x1,… ,xn). Furthermore, we suppose that the equations y1=g1(x1,… ,xn),y2=g2(x1,… ,xn),… ,yn=gn(x1,… ,xn) have a unique solution, say, x1=h1(y1,… ,yn),… ,xn=hn(y1,… ,yn). Under these assumptions, the joint density function of the random variables Yi is given by

fY1,… ,Yn(y1,… ,yn)=fX1,… ,Xn(x1,… ,xn)|J(x1,… ,xn)|−1 (7.3)

where xi=hi(y1,… ,yn),i=1,2,… ,n.

Example 7d

Let X1,X2, and X3 be independent standard normal random variables. If Y1=X1+X2+X3,Y2=X1−X2, and Y3=X1−X3, compute the joint density function of Y1,Y2,Y3.

Solution

Letting Y1=X1+X2+X3,Y2=X1−X2,Y3=X1−X3, the Jacobian of these transformations is given by

J=|1111−1010−1|=3

As the preceding transformations yield that

X1=Y1+Y2+Y33 X2=Y1−2Y2+Y33 X3=Y1+Y2−2Y33

we see from **Equation (7.3)** that

f Y 1 , Y 2 , Y 3 (y 1 , y 2 , y 3) = 1 3 f X 1 , X 2 , X 3 (y 1 + y 2 + y 3 3 , y 1 −2 y 2 + y 3 3 , y 1 + y 2 −2 y 3 3)

Hence, as

f X 1 , X 2 , X 3 (x 1 , x 2 , x 3)= 1 (2π) 3/2 e − ∑ i=1 3 x 2 i /2

we see that

fY1, Y2, Y3(y1, y2, y3)=13(2π)3/2e−Q(y1, y2, y3)/2

where

Q(y 1 , y 2 , y 3) = (y 1 + y 2 + y 3 3) 2 + (y 1 −2 y 2 + y 3 3) 2 + (y 1 + y 2 −2 y 3 3) 2 = y 2 1 3 + 2 3 y 2 2 + 2 3 y 2 3 − 2 3 y 2 y 3

Example 7e

Let X1,X2,… ,Xn be independent and identically distributed exponential random variables with rate λ. Let

Yi=X1+⋯+Xi i=1,… ,n

a. Find the joint density function of Y1,… ,Yn.
b. Use the result of part (a) to find the density of Yn.
c. Find the conditional density of Y1,… ,Yn−1 given that Yn=t.

Solution

a. The Jacobian of the transformations Y1=X1, Y2=X1+X2, … , Yn=X1+⋯+Xn is
J=| 1 0 0 0 ⋯ 0 1 1 0 0 ⋯ 0 1 1 1 0 ⋯ 0 ⋯ ⋯ ⋯ ⋯ 1 1 1 1 ⋯ 1 |

   Since only the first term of the determinant will be nonzero, we have J=1. Now, the joint density function of X1,… ,Xn is given by
fX1,… ,Xn(x1,… ,xn)=∏i=1nλe−λxi 0<xi<∞, i=1,… ,n

   Hence, because the preceding transformations yield
X1=Y1,X2=Y2−Y1,… ,Xi=Yi−Yi−1,… ,Xn=Yn−Yn−1

   it follows from **Equation (7.3)** that the joint density function of Y1,… ,Yn is
f Y 1 ,… , Y n (y 1 ,… , y n)= f X 1 ,… , X n (y 1 , y 2 − y 1 ,… , y n − y n−1) = λ n exp { −λ [y 1 + ∑ i=2 n (y i − y i−1)] } = λ n e −λ y n 0< y 1 ,0< y i − y i−1 ,i=2,… ,n = λ n e −λ y n 0< y 1 < y 2 <⋯< y n

b. To obtain the marginal density of Yn, let us integrate out the other variables one at a time. Doing this gives
f Y 2 ,… , Y n (y 2 ,… , y n) = ∫ 0 y 2 λ n e −λ y n d y 1 = λ n y 2 e −λ y n 0< y 2 < y 3 <⋯< y n

   Continuing, we obtain
f Y 3 ,… , Y n (y 3 ,… , y n) = ∫ 0 y 3 λ n y2 e −λ y n d y 2 = λ n y 3 2 2 e −λ y n 0< y 3 < y 4 <⋯< y n

   The next integration yields
f Y 4 ,… , Y n (y 4 ,… , y n)= λ n y 4 3 3! e −λ y n 0< y 4 <⋯< y n

   Continuing in this fashion gives
f Y n (y n)= λ n y n n−1 (n−1)! e −λ y n 0< y n

   which, in agreement with the result obtained in **Example 3b**, shows that X1+⋯+Xn is a gamma random variable with parameters n and λ.
c. The conditional density of Y1,… ,Yn−1 given that Yn=t is, for 0<y1<… <yn−1<t,
f Y 1 ,… , Y n−1 | Y n (y 1 ,… , y n−1 |t) = f Y 1 ,… , Y n−1 , Y n (y 1 ,… , y n−1 ,t) f Y n (t) = λ n e −λt λ e −λt (λt) n−1 /(n−1)! = (n−1)! t n−1

Because f(y)=1/t, 0<y<t, is the density of a uniform random variable on (0,t), it follows that conditional on Yn=t, Y1,… ,Yn−1 are distributed as the order statistics of n−1 independent uniform (0,t) random variables.

* 6.8 Exchangeable Random Variables

The random variables $X_1, X_2, \ldots, X_n$ are said to be *exchangeable* if, for every permutation $i_1, \ldots, i_n$ of the integers $1, \ldots, n$,

$$P\{X_{i_1} \le x_1, X_{i_2} \le x_2, \ldots, X_{i_n} \le x_n\} = P\{X_1 \le x_1, X_2 \le x_2, \ldots, X_n \le x_n\}$$

for all $x_1, \ldots, x_n$. That is, the n random variables are exchangeable if their joint distribution is the same no matter in which order the variables are observed.

Discrete random variables will be exchangeable if

$$P\{X_{i_1} = x_1, X_{i_2} = x_2, \ldots, X_{i_n} = x_n\} = P\{X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n\}$$

for all permutations $i_1, \ldots, i_n$, and all values $x_1, \ldots, x_n$. This is equivalent to stating that $p(x_1, x_2, \ldots, x_n) = P\{X_1 = x_1, \ldots, X_n = x_n\}$ is a symmetric function of the vector $(x_1, \ldots, x_n)$, which means that its value does not change when the values of the vector are permuted.

Example 8a

Suppose that balls are withdrawn one at a time and without replacement from an urn that initially contains n balls, of which k are considered special, in such a manner that each withdrawal is equally likely to be any of the balls that remains in the urn at the time. Let $X_i = 1$ if the ith ball withdrawn is special and let $X_i = 0$ otherwise. We will show that the random variables $X_1, \ldots, X_n$ are exchangeable. To do so, let $(x_1, \ldots, x_n)$ be a vector consisting of k ones and $n-k$ zeros. However, before considering the joint mass function evaluated at $(x_1, \ldots, x_n)$, let us try to gain some insight by considering a fixed such vector for instance, consider the vector $(1,1,0,1,0,\ldots,0,1)$, which is assumed to have k ones and $n-k$ zeros. Then

$$p(1,1,0,1,0,\ldots,0,1) = \frac{k}{n}\frac{k-1}{n-1}\frac{n-k}{n-2}\frac{k-2}{n-3}\frac{n-k-1}{n-4}\cdots\frac{1}{1}$$

which follows because the probability that the first ball is special is k/n, the conditional probability that the next one is special is $(k-1)/(n-1)$, the conditional probability that the next one is not special is $(n-k)/(n-2)$, and so on. By the same argument, it follows that $p(x_1, \ldots, x_n)$ can be expressed as the product of n fractions. The successive denominator terms of these fractions will go from n down to 1. The numerator term at the location where the vector $(x_1, \ldots, x_n)$ is 1 for the ith time is $k-(i-1)$, and where it is 0 for the ith time it is $n-k-(i-1)$. Hence, since the vector $(x_1, \ldots, x_n)$ consists of k ones and $n-k$ zeros, we obtain

$$p(x_1, \ldots, x_n) = \frac{k!(n-k)!}{n!} \quad x_i = 0,1, \sum_{i=1}^{n} x_i = k$$

Since this is a symmetric function of $(x_1, \ldots, x_n)$, it follows that the random variables are exchangeable.

Remark Another way to obtain the preceding formula for the joint probability mass function is to regard all the n balls as distinguishable from one another. Then, since the outcome of the experiment is an ordering of these balls, it follows that there are $n!$ equally likely outcomes. Finally, because the number of outcomes having special and nonspecial balls in specified places is equal to the number of ways of permuting the special and the nonspecial balls among themselves, namely $k!(n-k)!$, we obtain the preceding mass function.

It is easily seen that if $X_1, X_2, \ldots, X_n$ are exchangeable, then each X_i has the same probability distribution. For instance, if X and Y are exchangeable discrete random variables, then

$$P\{X = x\} = \sum_y P\{X = x, Y = y\} = \sum_y P\{X = y, Y = x\} = P\{Y = x\}$$

For example, it follows from **Example 8a** that the ith ball withdrawn will be special with probability k/n, which is intuitively clear, since each of the n balls is equally likely to be the ith one selected.

Example 8b

In **Example 8a**, let Y1 denote the selection number of the first special ball withdrawn, let Y2 denote the additional number of balls that are then withdrawn until the second special ball appears, and, in general, let Yi denote the additional number of balls withdrawn after the (i−1) special ball is selected until the ith is selected, i=1,… ,k. For instance, if n=4,k=2 and X1=1,X2=0,X3=0,X4=1, then Y1=1,Y2=3. Now, Y1=i1,Y2=i2,… ,Yk=ik⇔Xi1=Xi1+i2=⋯=Xi1+⋯+ik=1,Xj=0, otherwise; thus, from the joint mass function of the Xi, we obtain

$$P\{Y1=i1,Y2=i2,\dots,Yk=ik\}=k!(n-k)!n! \quad i1+\cdots+ik\le n$$

Hence, the random variables Y1,… ,Yk are exchangeable. Note that it follows from this result that the number of cards one must select from a well-shuffled deck until an ace appears has the same distribution as the number of additional cards one must select after the first ace appears until the next one does, and so on.

Example 8c

The following is known as Polya's urn model: Suppose that an urn initially contains n red and m blue balls. At each stage, a ball is randomly chosen, its color is noted, and it is then replaced along with another ball of the same color. Let Xi=1 if the ith ball selected is red and let it equal 0 if the ith ball is blue, i≥1. To obtain a feeling for the joint probabilities of these Xi, note the following special cases:

$$P\{X_1=1, X_2=1, X_3=0, X_4=1, X_5=0\} = \frac{n}{n+m}\,\frac{n+1}{n+m+1}\,\frac{m}{n+m+2}\,\frac{n+2}{n+m+3}\,\frac{m+1}{n+m+4}$$
$$= \frac{n(n+1)(n+2)m(m+1)}{(n+m)(n+m+1)(n+m+2)(n+m+3)(n+m+4)}$$

and

$$P\{X_1=0, X_2=1, X_3=0, X_4=1, X_5=1\} = \frac{m}{n+m}\,\frac{n}{n+m+1}\,\frac{m+1}{n+m+2}\,\frac{n+1}{n+m+3}\,\frac{n+2}{n+m+4}$$
$$= \frac{n(n+1)(n+2)m(m+1)}{(n+m)(n+m+1)(n+m+2)(n+m+3)(n+m+4)}$$

By the same reasoning, for any sequence x1,… ,xk that contains r ones and k−r zeros, we have

$$P\{X_1=x_1,\dots,X_k=x_k\} = \frac{n(n+1)\cdots(n+r-1)m(m+1)\cdots(m+k-r-1)}{(n+m)\cdots(n+m+k-1)}$$

Therefore, for any value of k, the random variables X1,… ,Xk are exchangeable.

An interesting corollary of the exchangeability in this model is that the probability that the ith ball selected is red is the same as the probability that the first ball selected is red, namely, $\frac{n}{n+m}$. (For an intuitive argument for this initially nonintuitive result, imagine that all the n+m balls initially in the urn are of different types. That is, one is a red ball of type 1, one is a red ball of type 2, … , one is a red ball type of n, one is a blue ball of type 1, and so on, down to the blue ball of type m. Suppose that when a ball is selected it is replaced along with another of its type. Then, by symmetry, the ith ball selected is equally likely to be of any of the n+m distinct types. Because n of these n+m types are red, the probability is $\frac{n}{n+m}$.)

Our final example deals with continuous random variables that are exchangeable.

Example 8d

Let X1,X2,… ,Xn be independent uniform (0, 1) random variables, and denote their order statistics by

$f(x,y)=e-(x+y) \qquad 0\le x<\infty, 0\le y<\infty$

Find (a) $P\{X<Y\}$ and (b) $P\{X<a\}$.

6.11. In **Example 1d**, verify that $f(x,y)=2e-xe-2y$, $0<x<\infty$, $0<y<\infty$, is indeed a joint density. function. That is, check that $f(x,y)\ge 0$, and that $\int-\infty\infty\int-\infty\infty f(x,y)dx\,dy=1$.

6.12. The number of people who enter a drugstore in a given hour is a Poisson random variable with parameter $\lambda=10$. Compute the conditional probability that at most 3 men entered the drugstore, given that 10 women entered in that hour. What assumptions have you made?

6.13. A man and a woman agree to meet at a certain location about 12:30 P.M. If the man arrives at a time uniformly distributed between 12:15 and 12:45, and if the woman independently arrives at a time uniformly distributed between 12:00 and 1 P.M., find the probability that the first to arrive waits no longer than 5 minutes. What is the probability that the man arrives first?

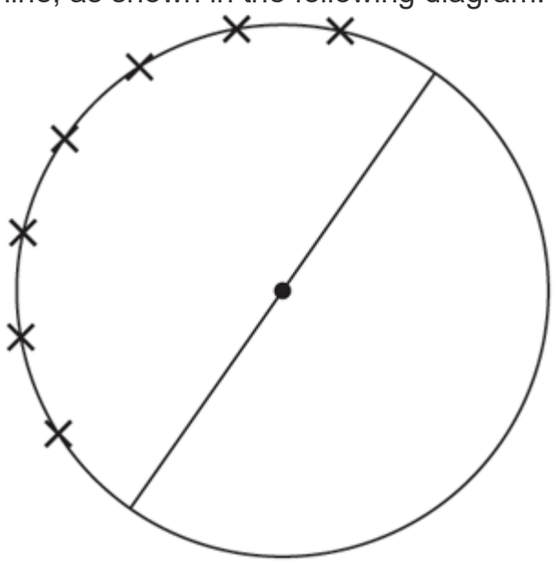
6.14. An ambulance travels back and forth at a constant speed along a road of length L. At a certain moment of time, an accident occurs at a point uniformly distributed on the road. [That is, the distance of the point from one of the fixed ends of the road is uniformly distributed over (0, L).] Assuming that the ambulance's location at the moment of the accident is also uniformly distributed, and assuming independence of the variables, compute the distribution of the distance of the ambulance from the accident.

6.15. The random vector (X, Y) is said to be uniformly distributed over a region R in the plane if, for some constant c, its joint density is

$f(x,y)=\{c \quad if(x, y)\in R0 \quad otherwise$

a. Show that $1/c=$ area of region R.
   Suppose that (X, Y) is uniformly distributed over the square centered at (0, 0) and with sides of length 2.
b. Show that X and Y are independent, with each being distributed uniformly over $(-1,1)$.
c. What is the probability that (X, Y) lies in the circle of radius 1 centered at the origin? That is, find $P\{X2+Y2\le 1\}$.

6.16. Suppose that n points are independently chosen at random on the circumference of a circle, and we want the probability that they all lie in some semicircle. That is, we want the probability that there is a line passing through the center of the circle such that all the points are on one side of that line, as shown in the following diagram:



Let $P1,\ldots,Pn$ denote the n points. Let A denote the event that all the points are contained in some semicircle, and let Ai be the event that all the points lie in the semicircle beginning at the point Pi and going clockwise for $180\circ, i=1,\ldots,n$.

a. Express A in terms of the Ai.

b. Are the Ai mutually exclusive?
c. Find P(A).

6.17. Three points X1,X2,X3 are selected at random on a line L. What is the probability that X2 lies between X1 and X3?
6.18. Let X1 and X2 be independent binomial random variables with Xi having parameters (ni,pi), i=1,2. Find

a. P(X1X2=0);
b. P(X1+X2=1);
c. P(X1+X2=2).

6.19. Show that f(x,y)=1/x, 0<y<x<1, is a joint density function. Assuming that f is the joint density function of X,Y find

a. the marginal density of Y;
b. the marginal density of X;
c. E[X];
d. E[Y].

6.20. The joint density of X and Y is given by
f(x,y)={ x e −(x+y) x>0, y>0 0 otherwise

Are X and Y independent? If, instead, f(x, y) were given by
f(x,y)={ 2 0<x<y, 0<y<1 0 otherwise

would X and Y be independent?
6.21. Let
f(x,y)=24xy 0≤x≤1, 0≤y≤1, 0≤x+y≤1

and let it equal 0 otherwise.

a. Show that f(x y) is a joint probability density function.
b. Find E[X].
c. Find E[Y].

6.22. The joint density function of X and Y is
f(x,y)={ x+y 0<x<1, 0<y<1 0 otherwise

a. Are X and Y independent?
b. Find the density function of X.
c. Find P{X+Y<1}.

6.23. The random variables X and Y have joint density function
f(x,y)=12xy(1−x) 0<x<1, 0<y<1

and equal to 0 otherwise.

a. Are X and Y independent?
b. Find E[X].
c. Find E[Y].
d. Find Var(X).
e. Find Var(Y).

6.24. Consider independent trials, each of which results in outcome i,i=0,1,… ,k with probability pi,∑i=0kpi=1. Let N denote the number of trials needed to obtain an outcome that is not equal to 0,

and let X be that outcome.

a. Find $P\{N=n\}, n \geq 1$.
b. Find $P\{X=j\}, j=1, \ldots, k$.
c. Show that $P\{N=n, X=j\}=P\{N=n\}P\{X=j\}$.
d. Is it intuitive to you that N is independent of X?
e. Is it intuitive to you that X is independent of N?

6.25. Suppose that 106 people arrive at a service station at times that are independent random variables, each of which is uniformly distributed over (0,106). Let N denote the number that arrive in the first hour. Find an approximation for $P\{N=i\}$.

6.26. Suppose that A, B, C, are independent random variables, each being uniformly distributed over (0, 1).

a. What is the joint cumulative distribution function of A, B, C?
b. What is the probability that all of the roots of the equation $Ax^2+Bx+C=0$ are real?

6.27. If X1 and X2 are independent exponential random variables with respective parameters λ_1 and λ_2, find the distribution of $Z=X1/X2$. Also compute $P\{X1<X2\}$.

6.28. The time that it takes to service a car is an exponential random variable with rate 1.

a. If A. J. brings his car in at time 0 and M. J. brings her car in at time t, what is the probability that M. J.'s car is ready before A. J.'s car? (Assume that service times are independent and service begins upon arrival of the car.)
b. If both cars are brought in at time 0, with work starting on M. J.'s car only when A. J.'s car has been completely serviced, what is the probability that M. J.'s car is ready before time 2?

6.29. The gross daily sales at a certain restaurant are a normal random variable with mean $2200 and standard deviation $230. What is the probability that

a. the total gross sales over the next 2 days exceeds $5000;
b. daily sales exceed $2000 in at least 2 of the next 3 days?
   What independence assumptions have you made?

6.30. Jill's bowling scores are approximately normally distributed with mean 170 and standard deviation 20, while Jack's scores are approximately normally distributed with mean 160 and standard deviation 15. If Jack and Jill each bowl one game, then assuming that their scores are independent random variables, approximate the probability that

a. Jack's score is higher;
b. the total of their scores is above 350.

6.31. According to the U.S. National Center for Health Statistics, 25.2 percent of males and 23.6 percent of females never eat breakfast. Suppose that random samples of 200 men and 200 women are chosen. Approximate the probability that

a. at least 110 of these 400 people never eat breakfast;
b. the number of the women who never eat breakfast is at least as large as the number of the men who never eat breakfast.

6.32. Monthly sales are independent normal random variables with mean 100 and standard deviation 5.

a. Find the probability that exactly 3 of the next 6 months have sales greater than 100.
b. Find the probability that the total of the sales in the next 4 months is greater than 420.

6.33. Let X1 and X2 be independent normal random variables, each having mean 10 and variance σ^2. Which probability is larger

a. $P(X1>15)$ or $P(X1+X2>25)$;
b. $P(X1>15)$ or $P(X1+X2>30)$.

6.34. Suppose X and Y are independent normal random variables with mean 10 and variance 4. Find x such that P(X+Y>x)=P(X>15).
6.35. Teams 1,2,3,4 are all scheduled to play each of the other teams 10 times. Whenever team i plays team j, team i is the winner with probability Pi,j, where
P 1,2 =.6, P 1,3 =.7, P 1,4 =.75 P 2,1 =.4, P 2,3 =.6, P 2,4 =.70

a. Approximate the probability that team 1 wins at least 20 games.
   Suppose we want to approximate the probability that team 2 wins at least as many games as does team 1. To do so, let X be the number of games that team 2 wins against team 1, let Y be the total number of games that team 2 wins against teams 3 and 4, and let Z be the total number of games that team 1 wins against teams 3 and 4.
b. Are X,Y,Z independent.
c. Express the event that team 2 wins at least as many games as does team 1 in terms of the random variables X,Y,Z.
d. Approximate the probability that team 2 wins at least as many games as team 1.

Hint: Approximate the distribution of any binomial random variable by a normal with the same mean and variance.
6.36. Let X1,… ,X10 be independent with the same continuous distribution function F, and let m be the median of that distribution. That is, F(m)=.5.

a. If N is the number of the values X1,… ,X10 that are less than m, what type of random variable is N.
b. Let X(1)<X(2)<···<X(10) be the values X1,… ,X10 arranged in increasing order. That is, X(i) is, for i=1, … , 10, the ith smallest of X1,… ,X10. Find P(X(2)<m<X(8)).

6.37. The expected number of typographical errors on a page of a certain magazine is .2. What is the probability that an article of 10 pages contains (a) 0 and (b) 2 or more typographical errors? Explain your reasoning!
6.38. The monthly worldwide average number of airplane crashes of commercial airlines is 2.2. What is the probability that there will be

a. more than 2 such accidents in the next month?
b. more than 4 such accidents in the next 2 months?
c. more than 5 such accidents in the next 3 months?

Explain your reasoning!
6.39. In **Problem 6.4**, calculate the conditional probability mass function of X1 given that

a. X2=1;
b. X2=0.

6.40. In **Problem 6.3**, calculate the conditional probability mass function of Y1 given that

a. Y2=1;
b. Y2=0.

6.41. The discrete integer valued random variables X,Y,Z are independent if for all i,j,k
P(X=i,Y=j,Z=k)=P(X=i)P(Y=j)P(Z=k)

Show that if X,Y,Z are independent then X and Y are independent. That is, show that the preceding implies that
P(X=i,Y=j)=P(X=i)P(Y=j)

6.42. Choose a number X at random from the set of numbers {1,2,3,4,5}. Now choose a number at random from the subset no larger than X, that is, from {1,… ,X}. Call this second number Y.

a. Find the joint mass function of X and Y.
b. Find the conditional mass function of *X* given that Y=i. Do it for i=1,2,3,4,5.
c. Are X and Y independent? Why?

6.43. Two dice are rolled. Let X and Y denote, respectively, the largest and smallest values obtained. Compute the conditional mass function of Y given X=i, for i=1,2,… ,6. Are X and Y independent? Why?

6.44. The joint probability mass function of *X* and Y is given by

p(1,1)= 1 8 p(1,2)= 1 4 p(2,1)= 1 8 p(2,2)= 1 2

a. Compute the conditional mass function of X given Y=i,i=1,2.
b. Are X and Y independent?
c. Compute P{XY≤3},P{X+Y>2},P{X/Y>1}.

6.45. The joint density function of X and Y is given by

f(x,y)=xe−x(y+1) x>0, y>0

a. Find the conditional density of X, given Y=y, and that of Y, given X=x.
b. Find the density function of Z=XY.

6.46. The joint density of X and Y is

f(x,y)=c(x2−y2)e−x 0≤x<∞, −x≤y≤x

Find the conditional distribution of Y, given X=x.

6.47. An insurance company supposes that each person has an accident parameter and that the yearly number of accidents of someone whose accident parameter is λ is Poisson distributed with mean λ. They also suppose that the parameter value of a newly insured person can be assumed to be the value of a gamma random variable with parameters s and α. If a newly insured person has n accidents in her first year, find the conditional density of her accident parameter. Also, determine the expected number of accidents that she will have in the following year.

6.48. If X1,X2,X3 are independent random variables that are uniformly distributed over (0, 1), compute the probability that the largest of the three is greater than the sum of the other two.

6.49. A complex machine is able to operate effectively as long as at least 3 of its 5 motors are functioning. If each motor independently functions for a random amount of time with density function f(x)=xe−x,x>0, compute the density function of the length of time that the machine functions.

6.50. If 3 trucks break down at points randomly distributed on a road of length L, find the probability that no 2 of the trucks are within a distance d of each other when d≤L/2.

6.51. Consider a sample of size 5 from a uniform distribution over (0, 1). Compute the probability that the median is in the interval (14,34).

6.52. If X1,X2,X3,X4,X5 are independent and identically distributed exponential random variables with the parameter λ, compute

a. P{min(X1,… ,X5)≤a};
b. P{max(X1,… ,X5)≤a}.

6.53. Let X(1),X(2),…,X(n) be the order statistics of a set of n independent uniform (0,1) random variables. Find the conditional distribution of X(n) given that X(1)=s1,X(2)=s2,… ,X(n−1)=sn−1.

6.54. Let Z1 and Z2 be independent standard normal random variables. Show that X,Y has a bivariate normal distribution when X=Z1, Y=Z1+Z2.

6.55. Derive the distribution of the range of a sample of size 2 from a distribution having density function f(x)=2x, 0<x<1.

6.56. Let X and Y denote the coordinates of a point uniformly chosen in the circle of radius 1 centered at the origin. That is, their joint density is

f(x,y)=1π x2+y2≤1

Find the joint density function of the polar coordinates R=(X2+Y2)1/2 and Θ=tan−1Y/X.
6.57. If X and Y are independent random variables both uniformly distributed over (0, 1), find the joint density function of R=X2+Y2,Θ=tan−1Y/X.
6.58. If U is uniform on (0, 2π) and Z, independent of U, is exponential with rate 1, show directly (without using the results of **Example 7b**) that X and Y defined by
X= 2Z cos U Y= 2Z sin U

are independent standard normal random variables.
6.59. X and Y have joint density function
f(x,y)=1x2y2 x≥1, y≥1

a. Compute the joint density function of U=XY,V=X/Y.
b. What are the marginal densities?

6.60. If X and Y are independent and identically distributed uniform random variables on (0, 1), compute the joint density of
a. U=X+Y,V=X/Y;
b. U=X,V=X/Y;
c. U=X+Y,V=X/(X+Y).

6.61. Repeat **Problem 6.60** when X and Y are independent exponential random variables, each with parameter λ=1.
6.62. If X1 and X2 are independent exponential random variables, each having parameter λ, find the joint density function of Y1=X1+X2 and Y2=eX1.
6.63. If X, Y, and Z are independent random variables having identical density functions f(x)=e−x,0<x<∞, derive the joint distribution of U=X+Y, V=X+Z,W=Y+ Z.
6.64. In Example 8b, let Yk+1=n+1−∑i=1kYi. Show that Y1,… ,Yk,Yk+1 are exchangeable. Note that Yk+1 is the number of balls one must observe to obtain a special ball if one considers the balls in their reverse order of withdrawal.
6.65. Consider an urn containing n balls numbered 1,… ,n, and suppose that k of them are randomly withdrawn. Let Xi equal 1 if ball number i is removed and let Xi be 0 otherwise. Show that X1,… ,Xn are exchangeable.

Theoretical exercises

6.1. Suppose X, Y have a joint distribution function F(x,y). Show how to obtain the distribution functions FX(x)=P{X≤x} and FY(y)=P{Y≤y}.
6.2. Suppose that X and Y are integer valued random variables and have a joint distribution function F(i,j)=P(X≤i,Y≤j).

a. Give an expression, in terms of the joint distribution function, for P(X=i,Y≤j).
b. Give an expression, in terms of the joint distribution function, for P(X=i,Y=j).

6.3. Suggest a procedure for using Buffon's needle problem to estimate π. Surprisingly enough, this was once a common method of evaluating π.
6.4. Solve Buffon's needle problem when L>D.
ANSWER: 2LπD(1−sin θ)+2θ/π, where cos θ=D/L.
6.5. If X and Y are independent continuous positive random variables, express the density function of (a) Z=X/Y and (b) Z=XY in terms of the density functions of X and Y. Evaluate the density functions in the special case where X and Y are both exponential random variables.
6.6. If X and Y are jointly continuous with joint density function fX,Y(x,y), show that X+Y is continuous with density function
fX+Y(t)=∫−∞∞fX,Y(x,t−x) dx

6.7.

a. If X has a gamma distribution with parameters (t,λ), what is the distribution of cX,c>0?
b. Show that
   12λχ2n2

   has a gamma distribution with parameters n,λ when n is a positive integer and χ22n is a chi-squared random variable with 2n degrees of freedom.

6.8. Let X and Y be independent continuous random variables with respective hazard rate functions λX(t) and λY(t), and set W=min(X,Y).

a. Determine the distribution function of W in terms of those of X and Y.
b. Show that λW(t), the hazard rate function of W, is given by
   λW(t)=λX(t)+λY(t)

6.9. Let X1,… ,Xn be independent exponential random variables having a common parameter λ. Determine the distribution of min(X1,… ,Xn).
6.10. The lifetimes of batteries are independent exponential random variables, each having parameter λ. A flashlight needs 2 batteries to work. If one has a flashlight and a stockpile of n batteries, what is the distribution of time that the flashlight can operate?
6.11. Let X1, X2, X3, X4, X5 be independent continuous random variables having a common distribution function F and density function f, and set
I=P{X1<X2<X3<X4<X5}

a. Show that I does not depend on F.
   Hint: Write I as a five-dimensional integral and make the change of variables ui=F(xi),i=1,… ,5.
b. Evaluate I.
c. Give an intuitive explanation for your answer to (b).

6.12. Show that the jointly continuous (discrete) random variables X1,… ,Xn are independent if and only if their joint probability density (mass) function f(x1,… ,xn) can be written as
f(x1,… ,xn)=∏i=1ngi(xi)

for nonnegative functions $g_i(x), i=1,\ldots,n$.

6.13. In **Example 5e**, we computed the conditional density of a success probability for a sequence of trials when the first $n+m$ trials resulted in n successes. Would the conditional density change if we specified which n of these trials resulted in successes?

6.14. Suppose that X and Y are independent geometric random variables with the same parameter p.

a. Without any computations, what do you think is the value of $P\{X=i|X+Y=n\}$?

   Hint: Imagine that you continually flip a coin having probability p of coming up heads. If the second head occurs on the nth flip, what is the probability mass function of the time of the first head?

b. Verify your conjecture in part (a).

6.15. Consider a sequence of independent trials, with each trial being a success with probability p. Given that the kth success occurs on trial n, show that all possible outcomes of the first $n-1$ trials that consist of $k-1$ successes and $n-k$ failures are equally likely.

6.16. If X and Y are independent binomial random variables with identical parameters n and p, show analytically that the conditional distribution of X given that $X+Y=m$ is the hypergeometric distribution. Also, give a second argument that yields the same result without any computations.

Hint: Suppose that $2n$ coins are flipped. Let X denote the number of heads in the first n flips and Y the number in the second n flips. Argue that given a total of m heads, the number of heads in the first n flips has the same distribution as the number of white balls selected when a sample of size m is chosen from n white and n black balls.

6.17. Suppose that $X_i, i=1,2,3$ are independent Poisson random variables with respective means $\lambda_i, i=1,2,3$. Let $X=X_1+X_2$ and $Y=X_2+X_3$. The random vector X,Y is said to have a bivariate Poisson distribution. Find its joint probability mass function. That is, find $P\{X=n, Y=m\}$.

6.18. Suppose X and Y are both integer-valued random variables. Let

$$p(i|j)=P(X=i|Y=j)$$

and

$$q(j|i)=P(Y=j|X=i)$$

Show that

$$P(X=i,Y=j)= p(i|j) \sum_i p(i|j)\, q(j|i)$$

6.19. Let X_1, X_2, X_3 be independent and identically distributed continuous random variables. Compute

a. $P\{X_1>X_2|X_1>X_3\}$;
b. $P\{X_1>X_2|X_1<X_3\}$;
c. $P\{X_1>X_2|X_2>X_3\}$;
d. $P\{X_1>X_2|X_2<X_3\}$.

6.20. Let U denote a random variable uniformly distributed over $(0, 1)$. Compute the conditional distribution of U given that

a. $U>a$;
b. $U<a$;

where $0<a<1$.

6.21. Suppose that W, the amount of moisture in the air on a given day, is a gamma random variable with parameters (t,β). That is, its density is $f(w)=\beta e^{-\beta w}(\beta w)^{t-1}/\Gamma(t)$, $w>0$. Suppose also that given that $W=w$, the number of accidents during that day—call it N—has a Poisson distribution with mean w. Show that the conditional distribution of W given that $N=n$ is the gamma distribution with parameters $(t+n,\beta+1)$.

6.22. Let W be a gamma random variable with parameters (t,β), and suppose that conditional on

W=w,X1,X2,… ,Xn are independent exponential random variables with rate w. Show that the conditional distribution of W given that X1=x1,X2=x2,… ,Xn=xn is gamma with parameters (t+n,β+∑i=1nxi).

6.23. A rectangular array of *mn* numbers arranged in n rows, each consisting of m columns, is said to contain a *saddlepoint* if there is a number that is both the minimum of its row and the maximum of its column. For instance, in the array

1 3 2 0 −2 6 .5 12 3

the number 1 in the first row, first column is a saddlepoint. The existence of a saddlepoint is of significance in the theory of games. Consider a rectangular array of numbers as described previously and suppose that there are two individuals—A and B—who are playing the following game: A is to choose one of the numbers 1,2,… ,n and B one of the numbers 1,2,… ,m. These choices are announced simultaneously, and if A chose i and B chose j, then A wins from B the amount specified by the number in the ith row, jth column of the array. Now suppose that the array contains a saddlepoint—say the number in the row r and column k call this number xrk. Now if player A chooses row r, then that player can guarantee herself a win of at least xrk (since xrk is the minimum number in the row r). On the other hand, if player B chooses column k, then he can guarantee that he will lose no more than xrk (since xrk is the maximum number in the column k). Hence, as A has a way of playing that guarantees her a win of xrk and as B has a way of playing that guarantees he will lose no more than xrk, it seems reasonable to take these two strategies as being optimal and declare that the value of the game to player A is xrk.

If the *nm* numbers in the rectangular array described are independently chosen from an arbitrary continuous distribution, what is the probability that the resulting array will contain a saddlepoint?

6.24. If X is exponential with rate λ, find P{[X]=n,X−[X]≤x}, where [x] is defined as the largest integer less than or equal to x. Can you conclude that [X] and X−[X] are independent?

6.25. Suppose that F(x) is a cumulative distribution function. Show that (a) Fn(x) and (b) 1−[1−F(x)]n are also cumulative distribution functions when n is a positive integer.

Hint: Let X1,… ,Xn be independent random variables having the common distribution function F. Define random variables Y and Z in terms of the Xi so that P{Y≤x}=Fn(x) and P{Z≤x}=1−[1−F(x)]n.

6.26. Show that if n people are distributed at random along a road L miles long, then the probability that no 2 people are less than a distance D miles apart is when D≤L/(n−1),[1−(n−1)D/L]n. What if D>L/(n−1)?

6.27. Suppose that X1,… ,Xn are independent exponential random variables with rate λ. Find

a. fX1|X1+… +Xn(x|t), the conditional density of X1 given that X1+… +Xn=t;
b. P(X1<x|X1+… +Xn=t).

6.28. Establish **Equation (6.2)** by differentiating **Equation (6.4)**.

6.29. Show that the median of a sample of size 2n+1 from a uniform distribution on (0, 1) has a beta distribution with parameters (n+1, n+1).

6.30. Suppose that X1,… ,Xn are independent and identically distributed continuous random variables. For A={X1<⋯<Xj>Xj+1>⋯>Xn}, find P(A). That is, find the probability that the function X(i)=Xi, i=1,… ,n, is a unimodal function with maximal value X(j). **Hint:** Write

$$A=\{\max(X_1,\ldots,X_j)=\max(X_1,\ldots,X_n),\ X_1<\cdots<X_j,\ X_{j+1}>\cdots>X_n\}$$

6.31. Compute the density of the range of a sample of size n from a continuous distribution having density function f.

6.32. Let X(1)≤X(2)≤⋯≤X(n) be the ordered values of n independent uniform (0, 1) random variables. Prove that for 1≤k≤n+1,

P{X(k)−X(k−1)>t}=(1−t)n

where X(0)≡0,X(n+1)≡1, and 0<t<1.

6.33. Let X1,… ,Xn be a set of independent and identically distributed continuous random variables having distribution function F, and let X(i),i=1,… ,n denote their ordered values. If X, independent of the Xi,i=1,… ,n, also has distribution F, determine

a. P{X>X(n)};
b. P{X>X(1)};
c. P{X(i)<X<X(j)},1≤i<j≤n.

6.34. Let X1,… ,Xn be independent and identically distributed random variables having distribution function F and density f. The quantity M≡[X(1)+X(n)]/2, defined to be the average of the smallest and largest values in X1,… ,Xn, is called the *midrange* of the sequence. Show that its distribution function is
FM(m)=n∫−∞m[F(2m−x)−F(x)]n−1f(x) dx

6.35. Let X1,… ,Xn be independent uniform (0, 1) random variables. Let R=X(n)−X(1) denote the range and M=[X(n)+X(1)]/2 the midrange of X1,… ,Xn. Compute the joint density function of R and M.
6.36. If X and Y are independent standard normal random variables, determine the joint density function of
U=X V=XY

Then use your result to show that X/Y has a Cauchy distribution.
6.37. Suppose that (X,Y) has a bivariate normal distribution with parameters μx,μy,σx,σy,ρ.
a. Show that (X− μ x σx , Y− μ y σy) has a bivariate normal distribution with parameters 0,1,0,1,ρ.
b. What is the joint distribution of (aX+b,cY+d).

6.38. Suppose that X has a beta distribution with parameters (a,b), and that the conditional distribution of N given that X=x is binomial with parameters (n+m,x). Show that the conditional density of X given that N=n is the density of a beta random variable with parameters (n+a,m+b). N is said to be a *beta binomial* random variable.
6.39. Consider an experiment with n possible outcomes, having respective probabilities P1,… ,Pn, ∑i=1nPi=1, and suppose we want to assume a probability distribution on the probability vector (P1,… ,Pn). Because ∑i=1nPi=1, we cannot define a density on P1,… ,Pn, but what we can do is to define one on P1,… ,Pn−1 and then take Pn=1−∑i=1n−1Pi. The *Dirichlet* distribution takes (P1,… ,Pn−1) to be uniformly distributed over the set S={(p1,… ,pn−1):∑i=1n−1pi<1, pi>0, i=1,… ,n−1}. That is, the Dirichlet density is
f P 1 ,… , P n−1 (p 1 ,… , p n−1)=C, p i >0,i=1,… ,n−1, ∑ i=1 n−1 p i <1

a. Determine C. **Hint**: Use results from **Section 6.3.1**.
Let U1,… ,Un be independent uniform (0,1) random variables.
b. Show that the Dirichlet density is the conditional density of U1,… ,Un−1 given that ∑i=1n−1Ui<1.
c. * Show that U(1),U(2)−U(1),… ,U(n)−U(n−1) has a Dirichlet distribution, where U(1),… ,U(n) are the order statistics of U1,… ,Un.

6.40. Let FX1,… ,Xn(x1,… ,xn) and fX1,… ,Xn(x1,… ,xn) be, respectively, the joint distribution function and the joint density function of X1,… ,Xn.
Show that
∂n∂x1⋯∂xnFX1,… ,Xn(x1,… ,xn)=fX1,… ,Xn(x1,… ,xn).

6.41. For given constants ci>0, let Yi=ciXi, i=1,… ,n, and let FY1,… ,Yn(x1,… ,xn) and fY1,… ,Yn(x1,… ,xn) be, respectively the joint distribution function and the joint density function of Y1,… ,Yn
a. Express FY1,… ,Yn(x1,… ,xn) in terms of the joint distribution function of X1,… ,Xn
b. Express fY1,… ,Yn(x1,… ,xn) in terms of the joint density function of X1,… ,Xn.
c. Use **Equation (7.3)** to verify your answer to part (b).

Self-Test Problems and Exercises

6.1. Each throw of an unfair die lands on each of the odd numbers 1, 3, 5 with probability C and on each of the even numbers with probability 2C.

a. Find C.
b. Suppose that the die is tossed. Let X equal 1 if the result is an even number, and let it be 0 otherwise. Also, let Y equal 1 if the result is a number greater than three and let it be 0 otherwise. Find the joint probability mass function of X and Y. Suppose now that 12 independent tosses of the die are made.
c. Find the probability that each of the six outcomes occurs exactly twice.
d. Find the probability that 4 of the outcomes are either one or two, 4 are either three or four, and 4 are either five or six.
e. Find the probability that at least 8 of the tosses land on even numbers.

6.2. The joint probability mass function of the random variables X, Y, Z is
p(1,2,3)=p(2,1,1)=p(2,2,1)=p(2,3,2)=14

Find (a) E[XYZ], and (b) E[XY+XZ+YZ].
6.3. The joint density of X and Y is given by
f(x,y)=C(y−x)e−y −y<x<y, 0<y<∞

a. Find C.
b. Find the density function of X.
c. Find the density function of Y.
d. Find E[X].
e. Find E[Y].

6.4. Let r=r1+… +rk, where all ri are positive integers. Argue that if X1,… ,Xr has a multinomial distribution, then so does Y1,… ,Yk where, with r0=0,
Yi=∑j=ri−1+1ri−1+riXj , i≤k

That is, Y1 is the sum of the first r1 of the X′s, Y2 is the sum of the next r2, and so on.
6.5. Suppose that X, Y, and Z are independent random variables that are each equally likely to be either 1 or 2. Find the probability mass function of (a) XYZ, (b) XY+XZ+YZ, and (c) X2+YZ.
6.6. Let X and Y be continuous random variables with joint density function
f(x,y)={ x 5 +cy 0<x<1, 1<y<5 0 otherwise

where c is a constant.

a. What is the value of c?
b. Are X and Y independent?
c. Find P{X+Y>3}.

6.7. The joint density function of X and Y is
f(x,y)={ xy 0<x<1, 0<y<2 0 otherwise

a. Are X and Y independent?
b. Find the density function of X.
c. Find the density function of Y.
d. Find the joint distribution function.
e. Find E[Y].
f. Find P{X+Y<1}.

6.8. Consider two components and three types of shocks. A type 1 shock causes component 1 to fail, a type 2 shock causes component 2 to fail, and a type 3 shock causes both components 1 and 2 to fail. The times until shocks 1, 2, and 3 occur are independent exponential random variables with respective rates $\lambda 1, \lambda 2$, and $\lambda 3$. Let Xi denote the time at which component i fails, i=1,2. The random variables X1,X2 are said to have a joint bivariate exponential distribution. Find $P\{X1>s, X2>t\}$.

6.9. Consider a directory of classified advertisements that consists of m pages, where m is very large. Suppose that the number of advertisements per page varies and that your only method of finding out how many advertisements there are on a specified page is to count them. In addition, suppose that there are too many pages for it to be feasible to make a complete count of the total number of advertisements and that your objective is to choose a directory advertisement in such a way that each of them has an equal chance of being selected.

a. If you randomly choose a page and then randomly choose an advertisement from that page, would that satisfy your objective? Why or why not?
   Let n(i) denote the number of advertisements on page i,i=1,… ,m, and suppose that whereas these quantities are unknown, we can assume that they are all less than or equal to some specified value n. Consider the following algorithm for choosing an advertisement.
   Step 1Choose a page at random. Suppose it is page X. Determine n(X) by counting the number of advertisements on page X.
   Step 2"Accept" page X with probability n(X)/n. If page X is accepted, go to step 3. Otherwise, return to step 1.
   Step 3Randomly choose one of the advertisements on page X.

   Call each pass of the algorithm through step 1 an iteration. For instance, if the first randomly chosen page is rejected and the second accepted, then we would have needed 2 iterations of the algorithm to obtain an advertisement.
b. What is the probability that a single iteration of the algorithm results in the acceptance of an advertisement on page i?
c. What is the probability that a single iteration of the algorithm results in the acceptance of an advertisement?
d. What is the probability that the algorithm goes through k iterations, accepting the jth advertisement on page i on the final iteration?
e. What is the probability that the jth advertisement on page i is the advertisement obtained from the algorithm?
f. What is the expected number of iterations taken by the algorithm?

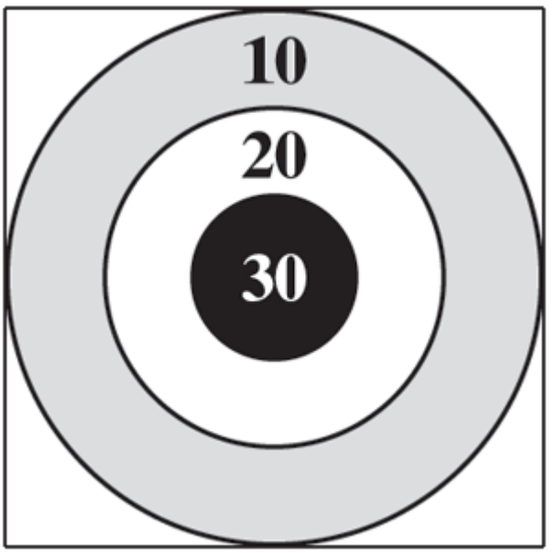
6.10. The "random" parts of the algorithm in Self-Test **Problem 6.9** can be written in terms of the generated values of a sequence of independent uniform (0, 1) random variables, known as random numbers. With [x] defined as the largest integer less than or equal to x, the first step can be written as follows:
Step 1Generate a uniform (0, 1) random variable U. Let X=[mU]+1, and determine the value of n(X).

a. Explain why the above is equivalent to step 1 of **Problem 6.8**.
   ***Hint*:** What is the probability mass function of X?
b. Write the remaining steps of the algorithm in a similar style.

6.11. Let X1,X2,… be a sequence of independent uniform (0, 1) random variables. For a fixed constant c, define the random variable N by
N=min{n:Xn>c}

Is N independent of XN? That is, does knowing the value of the first random variable that is greater than c affect the probability distribution of when this random variable occurs? Give an intuitive explanation for your answer.

6.12. The accompanying dartboard is a square whose sides are of length 6:



The three circles are all centered at the center of the board and are of radii 1, 2, and 3, respectively. Darts landing within the circle of radius 1 score 30 points, those landing outside this circle, but within the circle of radius 2, are worth 20 points, and those landing outside the circle of radius 2, but within the circle of radius 3, are worth 10 points. Darts that do not land within the circle of radius 3 do not score any points. Assuming that each dart that you throw will, independently of what occurred on your previous throws, land on a point uniformly distributed in the square, find the probabilities of the accompanying events:

a. You score 20 on a throw of the dart.
b. You score at least 20 on a throw of the dart.
c. You score 0 on a throw of the dart.
d. The expected value of your score on a throw of the dart.
e. Both of your first two throws score at least 10.
f. Your total score after two throws is 30.

6.13. A model proposed for NBA basketball supposes that when two teams with roughly the same record play each other, the number of points scored in a quarter by the home team minus the number scored by the visiting team is approximately a normal random variable with mean 1.5 and variance 6. In addition, the model supposes that the point differentials for the four quarters are independent. Assume that this model is correct.

a. What is the probability that the home team wins?
b. What is the conditional probability that the home team wins, given that it is behind by 5 points at halftime?
c. What is the conditional probability that the home team wins, given that it is ahead by 5 points at the end of the first quarter?

6.14. Let N be a geometric random variable with parameter p. Suppose that the conditional distribution of X given that N=n is the gamma distribution with parameters n and λ. Find the conditional probability mass function of N given that X=x.

6.15. Let X and Y be independent uniform (0, 1) random variables.

a. Find the joint density of U=X,V=X+Y.
b. Use the result obtained in part (a) to compute the density function of V.

6.16. You and three other people are to place bids for an object, with the high bid winning. If you win, you plan to sell the object immediately for $10,000. How much should you bid to maximize your expected profit if you believe that the bids of the others can be regarded as being independent and uniformly distributed between $7,000 and $10,000 thousand dollars?

6.17. Find the probability that X1,X2,… ,Xn is a permutation of 1,2,… ,n, when X1,X2,… ,Xn are

independent and

a. each is equally likely to be any of the values 1,… , n;
b. each has the probability mass function P{Xi=j}=pj,j=1,… , n.

6.18. Let X1,… ,Xn and Y1,… ,Yn be independent random vectors, with each vector being a random ordering of k ones and n−k zeros. That is, their joint probability mass functions are
P{ X 1 = i 1 ,… , X n = i n } =P{ Y 1 = i 1 ,… , Y n = i n } = 1 (n k) , i j =0,1, ∑ j=1 n i j =k

Let
N=∑i=1n|Xi−Yi|

denote the number of coordinates at which the two vectors have different values. Also, let M denote the number of values of i for which Xi=1,Yi=0.

a. Relate N to M.
b. What is the distribution of M?
c. Find E[N].
d. Find Var(N).

*6.19. Let Z1,Z2,… ,Zn be independent standard normal random variables, and let
Sj=∑i=1jZi

a. What is the conditional distribution of Sn given that Sk=y. Find it for k=1,… ,n−1.
b. Show that, for 1≤k≤n, the conditional distribution of Sk given that Sn=x is normal with mean xk/n and variance k(n−k)/n.

6.20. Let X1,X2,… be a sequence of independent and identically distributed continuous random variables. Find

a. P{X6>X1|X1=max(X1,… ,X5)}
b. P{X6>X2|X1=max(X1,… ,X5)}

6.21. Prove the identity
P{ X≤s,Y≤t }=P{ X≤s }+P{ Y≤t }+P{X>s,Y>t}−1

Hint: Derive an expression for P{X>s,Y>t} by taking the probability of the complementary event.
6.22. In **Example 1c**, find P(Xr=i,Ys=j) when j<i.
6.23. A Pareto random variable X with parameters a>0, >0 has distribution function F(x)=1− a λ x −λ ,x>a. x>a. For x0>a, verify that the conditional distribution of X given that X>x0 is that of a Pareto random variable with parameters (x0,λ) by evaluating P(X>x|X>x0).
6.24. Verify the identity fX(x)=∫−∞∞fX|Y(x|y)fY(y)dy.
6.25. In a contest originating with n players, each player independently advances to the next round, with player i advancing with probability pi. If no players advance to the next round, then the contest ends and all the players in the just concluded round are declared co-winners. If only one player advances, then that player is declared the winner and the contest ends. If two or more players advance, then those players play another round. Let Xi denote the number of rounds that i plays.

a. Find P(Xi≥k). **Hint:** Note that {Xi≥k} will occur if i advances at least k times and at least one of the other players advances at least k−1 times.
b. Find *P*(*i* is either the sole winner or one of the co-winners).
   Hint: It might help to imagine that a player always continues to play rounds until he or she fails to advance. (That is, if there is a sole winner then imagine that that player continues on until a failure occurs.)
c. Find *P*(*i* is the sole winner)

6.26. Let X1,… ,Xn be independent nonnegative integer valued random variables, and let

Suppose that we are presented with a set of n distinct values x1,x2,…,xn and that we desire to put them in increasing order, or as it is commonly stated, to *sort* them. An efficient procedure for accomplishing this task is the quick-sort algorithm, which is defined as follows: When n=2, the algorithm compares the two values and then puts them in the appropriate order. When n>2, one of the elements is randomly chosen say it is xi and then all of the other values are compared with xi. Those smaller than xi are put in a bracket to the left of xi and those larger than xi are put in a bracket to the right of xi. The algorithm then repeats itself on these brackets and continues until all values have been sorted. For instance, suppose that we desire to sort the following 10 distinct values:

5,9,3,10,11,14,8,4,17,6

We start by choosing one of them at random (that is, each value has probability 110 of being chosen). Suppose, for instance, that the value 10 is chosen. We then compare each of the others to this value, putting in a bracket to the left of 10 all those values smaller than 10 and to the right all those larger. This gives

{5,9,3,8,4,6},10,{11,14,17}

We now focus on a bracketed set that contains more than a single value say–the one on the left of the preceding–and randomly choose one of its values–say that 6 is chosen. Comparing each of the values in the bracket with 6 and putting the smaller ones in a new bracket to the left of 6 and the larger ones in a bracket to the right of 6 gives

{5,3,4},6,{9,8},10,{11,14,17}

If we now consider the leftmost bracket, and randomly choose the value 4 for comparison, then the next iteration yields

{3},4,{5},6,{9,8},10,{11,14,17}

This continues until there is no bracketed set that contains more than a single value.

If we let X denote the number of comparisons that it takes the quick-sort algorithm to sort n distinct numbers, then E[X] is a measure of the effectiveness of this algorithm. To compute E[X], we will first express X as a sum of other random variables as follows. To begin, give the following names to the values that are to be sorted: Let 1 stand for the smallest, let 2 stand for the next smallest, and so on. Then, for $1 \le i < j \le n$, let I(i,j) equal 1 if i and j are ever directly compared, and let it equal 0 otherwise. With this definition, it follows that

$$X=\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} I(i,j)$$

implying that

$$E[X]=E\left[\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} I(i,j)\right]=\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} E[I(i,j)]=\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} P\{i \text{ and } j \text{ are ever compared}\}$$

To determine the probability that i and j are ever compared, note that the values i,i+1,…,j−1,j will initially be in the same bracket (since all values are initially in the same bracket) and will remain in the same bracket if the number chosen for the first comparison is not between i and j. For instance, if the comparison number is larger than j, then all the values i,i+1,…,j−1,j will go in a bracket to the left of the comparison number, and if it is smaller than i, then they will all go in a bracket to the right. Thus all the values i,i+1,…,j−1,j will remain in the same bracket until the first time that one of them is chosen as a comparison value. At that point all the other values between i and j will be compared with this comparison value. Now, if this comparison value is neither i nor j, then upon comparison with it, i will go into a left bracket and j into a right bracket, and thus i and j will be in different brackets and so will never be compared. On the other hand, if the comparison value of the set i,i+1,…,j−1,j is either i or j, then there will be a direct comparison between i and j. Now, given that the comparison value is one of the values between i and j, it follows that it is equally likely to be any of these j−i+1 values, and thus the probability that it is either i or j is 2/(j−i+1). Therefore, we can conclude that

$$P\{i \text{ and } j \text{ are ever compared}\}=2j-i+1$$

and

$$E[X]=\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}2j-i+1$$

To obtain a rough approximation of the magnitude of E[X] when n is large, we can approximate the sums by integrals. Now

$$\sum_{j=i+1}^{n}2j-i+1\approx\int_{i+1}^{n}2x-i+1\,dx=2\log(x-i+1)\Big|_{i+1}^{n}=2\log(n-i+1)-2\log(2)\approx 2\log(n-i+1)$$

Thus

$$E[X]\approx\sum_{i=1}^{n-1}2\log(n-i+1)\approx 2\int_{1}^{n-1}\log(n-x+1)\,dx=2\int_{2}^{n}\log(y)\,dy=2(y\log(y)-y)\Big|_{2}^{n}\approx 2n\log(n)$$

Thus we see that when n is large, the quick-sort algorithm requires, on average, approximately 2n log(n) comparisons to sort n distinct values.

Example 2n The probability of a union of events

Let $A_1,\ldots A_n$ denote events, and define the indicator variables $X_i, i=1,\ldots,n$, by

$$X_i=\begin{cases}1 & \text{if } A_i \text{ occurs}\\ 0 & \text{otherwise}\end{cases}$$

Now, note that

$$1-\prod_{i=1}^{n}(1-X_i)=\begin{cases}1 & \text{if } \cup A_i \text{ occurs}\\ 0 & \text{otherwise}\end{cases}$$

Hence,

$$E\left[1-\prod_{i=1}^{n}(1-X_i)\right]=P(\cup_{i=1}^{n}A_i)$$

Expanding the left side of the preceding formula yields

$$P(\cup_{i=1}^{n}A_i)=E\left[\sum_{i=1}^{n}X_i-\sum\sum_{i<j}X_iX_j+\sum\sum\sum_{i<j<k}X_iX_jX_k-\cdots+(-1)^{n+1}X_1\cdots X_n\right] \tag{2.3}$$

However,

$$X_{i_1}X_{i_2}\cdots X_{i_k}=\begin{cases}1 & \text{if } A_{i_1}A_{i_2}\cdots A_{i_k} \text{ occurs}\\ 0 & \text{otherwise}\end{cases}$$

so

$$E[X_{i_1}\cdots X_{i_k}]=P(A_{i_1}\cdots A_{i_k})$$

Thus, **Equation (2.3)** is just a statement of the well-known inclusion-exclusion formula for the union of events:

$$P(\cup A_i)=\sum_i P(A_i)-\sum\sum_{i<j}P(A_iA_j)+\sum\sum\sum_{i<j<k}P(A_iA_jA_k)-\cdots+(-1)^{n+1}P(A_1\cdots A_n)$$

When one is dealing with an infinite collection of random variables $X_i, i\geq 1$, each having a finite expectation, it is not necessarily true that

$$E\left[\sum_{i=1}^{\infty}X_i\right]=\sum_{i=1}^{\infty}E[X_i] \tag{2.4}$$

To determine when (**2.4**) is valid, we note that $\sum_{i=1}^{\infty}X_i=\lim_{n\to\infty}\sum_{i=1}^{n}X_i$. Thus,

$$E\left[\sum_{i=1}^{\infty} X_i\right] = E\left[\lim_{n\to\infty} \sum_{i=1}^{n} X_i\right] = ? \lim_{n\to\infty} E\left[\sum_{i=1}^{n} X_i\right] = \lim_{n\to\infty} \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{\infty} E[X_i] \quad (2.5)$$

Hence, **Equation (2.4)** is valid whenever we are justified in interchanging the expectation and limit operations in **Equation (2.5)**. Although, in general, this interchange is *not* justified, it can be shown to be valid in two important special cases:

1. The Xi are all nonnegative random variables. (That is, P{Xi≥0}=1 for all i.)
2. ∑i=1∞E[|Xi|]<∞.

Example 2o

Consider any nonnegative, integer-valued random variable X. If, for each i≥1, we define

$$X_i = \begin{cases} 1 & \text{if } X \geq i \\ 0 & \text{if } X < i \end{cases}$$

then

$$\sum_{i=1}^{\infty} X_i = \sum_{i=1}^{X} X_i + \sum_{i=X+1}^{\infty} X_i = \sum_{i=1}^{X} 1 + \sum_{i=X+1}^{\infty} 0 = X$$

Hence, since the Xi are all nonnegative, we obtain

$$E[X] = \sum_{i=1}^{\infty} E(X_i) = \sum_{i=1}^{\infty} P\{X \geq i\} \quad (2.6)$$

a useful identity.

Example 2p

Suppose that n elements—call them 1, 2, …, n—must be stored in a computer in the form of an ordered list. Each unit of time, a request will be made for one of these elements —i being requested, independently of the past, with probability P(i), i≥1, ∑iP(i)=1. Assuming that these probabilities are known, what ordering minimizes the average position in the line of the element requested?

Solution

Suppose that the elements are numbered so that P(1)≥P(2)≥···≥P(n). To show that 1, 2, …, n is the optimal ordering, let X denote the position of the requested element. Now, under any ordering say, O=i1,i2,…,in,

$$P_O\{X \geq k\} = \sum_{j=k}^{n} P(i_j) \geq \sum_{j=k}^{n} P(j) = P_{1,2,\ldots,n}\{X \geq k\}$$

Summing over k and using **Equation (2.6)** yields

$$E_O[X] \geq E_{1,2,\ldots,n}[X]$$

thus showing that ordering the elements in decreasing order of the probability that they are requested minimizes the expected position of the element requested.

*7.2.1 Obtaining bounds from expectations via the probabilistic method

The probabilistic method is a technique for analyzing the properties of the elements of a set by introducing probabilities on the set and then studying an element chosen according to those probabilities. The technique was previously seen in Example 4l of **Chapter 3**, where it was used to show that a set contained an element that satisfied a certain property. In this subsection, we show how it can sometimes be used to bound complicated functions.

Let f be a function on the elements of a finite set A, and suppose that we are interested in

$m=\max_{s\in A} f(s)$

A useful lower bound for m can often be obtained by letting S be a random element of A for which the expected value of f(S) is computable and then noting that $m\geq f(S)$ implies that

$m\geq E[f(S)]$

with strict inequality if f(S) is not a constant random variable. That is, E[f(S)] is a lower bound on the maximum value.

Example 2q The maximum number of hamiltonian paths in a tournament

A round-robin tournament of $n>2$ contestants is a tournament in which each of the $\binom{n}{2}$ pairs of contestants play each other exactly once. Suppose that the players are numbered 1,2,3,…,n. The permutation $i_1,i_2,\ldots i_n$ is said to be a *Hamiltonian path* if i_1 beats i_2, i_2 beats i_3,…, and i_{n-1} beats i_n. A problem of some interest is to determine the largest possible number of Hamiltonian paths.

As an illustration, suppose that there are 3 players. On the one hand, one of them wins twice, then there is a single Hamiltonian path. (For instance, if 1 wins twice and 2 beats 3, then the only Hamiltonian path is 1, 2, 3.) On the other hand, if each of the players wins once, then there are 3 Hamiltonian paths. (For instance, if 1 beats 2, 2 beats 3, and 3 beats 1, then 1, 2, 3; 2, 3, 1; and 3, 1, 2, are all Hamiltonians.) Hence, when n=3, there is a maximum of 3 Hamiltonian paths.

We now show that there is an outcome of the tournament that results in more than $n!/2^{n-1}$ Hamiltonian paths. To begin, let the outcome of the tournament specify the result of each of the $\binom{n}{2}$ games played, and let A denote the set of all $2^{\binom{n}{2}}$ possible tournament outcomes. Then, with f(s) defined as the number of Hamiltonian paths that result when the outcome is $s\in A$, we are asked to show that

$\max_s f(s)\geq n!2^{n-1}$

To show this, consider the randomly chosen outcome S that is obtained when the results of the $\binom{n}{2}$ games are independent, with each contestant being equally likely to win each encounter. To determine E[f(S)], the expected number of Hamiltonian paths that result from the outcome S, number the n! permutations, and, for i=1,…,n!, let

$$X_i=\begin{cases}1 & \text{if permutation } i \text{ is a Hamiltonian}\\ 0 & \text{otherwise}\end{cases}$$

Since

$f(S)=\sum_i X_i$

it follows that

$E[f(S)]=\sum_i E[X_i]$

Because, by the assumed independence of the outcomes of the games, the probability that any specified permutation is a Hamiltonian is $(1/2)^{n-1}$, it follows that

$E[X_i]=P\{X_i=1\}=(1/2)^{n-1}$

Therefore,

$E[f(S)]=n!(1/2)^{n-1}$

Since f(S) is not a constant random variable, the preceding equation implies that there is an outcome of the tournament having more than $n!/2^{n-1}$ Hamiltonian paths.

Example 2r

A grove of 52 trees is arranged in a circular fashion. If 15 chipmunks live in these trees, show that there is a group of 7 consecutive trees that together house at least 3 chipmunks.

Solution

Let the neighborhood of a tree consist of that tree along with the next six trees visited by moving in the clockwise direction. We want to show that for any choice of living accommodations of the 15 chipmunks, there is a tree that has at least 3 chipmunks living in its neighborhood. To show this, choose a tree at random and let X denote the number of chipmunks that live in its neighborhood. To determine E[X], arbitrarily number the 15 chipmunks and for i=1,…,15, let

$$X_i=\begin{cases} 1, & \text{if chipmunk } i \text{ lives in the neighborhood of the randomly chosen tree} \\ 0, & \text{otherwise} \end{cases}$$

Because

$$X=\sum_{i=1}^{15}X_i$$

we obtain that

$$E[X]=\sum_{i=1}^{15}E[X_i]$$

However, because Xi will equal 1 if the randomly chosen tree is any of the 7 trees consisting of the tree in which chipmunk i lives along with its 6 neighboring trees when moving in the counterclockwise direction,

$$E[X_i]=P\{X_i=1\}=\frac{7}{52}$$

Consequently,

$$E[X]=\frac{105}{52}>2$$

showing that there exists a tree with more than 2 chipmunks living in its neighborhood.

*7.2.2 The maximum–minimums identity

We start with an identity relating the maximum of a set of numbers to the minimums of the subsets of these numbers.

Proposition 2.2

For arbitrary numbers $x_i, i=1,\ldots,n$,

$$\max_i x_i=\sum_i x_i-\sum_{i<j}\min(x_i,x_j)+\sum_{i<j<k}\min(x_i,x_j,x_k)$$

$$+\ldots+(-1)^{n+1}\min(x_1,\ldots,x_n)$$

Proof We will give a probabilistic proof of the proposition. To begin, assume that all the x_i are in the interval [0, 1]. Let U be a uniform (0, 1) random variable, and define the events $A_i, i=1,\ldots,n$, by $A_i=\{U<x_i\}$. That is, A_i is the event that the uniform random variable is less than x_i. Because at least one of these events A_i will occur if U is less than at least one of the values x_i, we have that

$$\cup_i A_i=\{U<\max_i x_i\}$$

Therefore,

$P(\cup_i A_i) = P\{U < \max_i x_i\} = \max_i x_i$

Also,

$P(A_i) = P\{U < x_i\} = x_i$

In addition, because all of the events $A_{i1}, \ldots, A_{ir}$ will occur if U is less than all the values $x_{i1}, \ldots, x_{ir}$, we see that the intersection of these events is

$A_{i1} \ldots A_{ir} = \{U < \min_{j=1,\ldots r} x_{ij}\}$

implying that

$P(A_{i1} \ldots A_{ir}) = P\{U < \min_{j=1,\ldots r} x_{ij}\} = \min_{j=1,\ldots r} x_{ij}$

Thus, the proposition follows from the inclusion–exclusion formula for the probability of the union of events:

$$P(\cup_i A_i) = \sum_i P(A_i) - \sum_{i<j} P(A_i A_j) + \sum_{i<j<k} P(A_i A_j A_k) + \ldots + (-1)^{n+1} P(A_1 \ldots A_n)$$

When the x_i are nonnegative, but not restricted to the unit interval, let c be such that all the x_i are less than c. Then the identity holds for the values $y_i = x_i/c$, and the desired result follows by multiplying through by c. When the x_i can be negative, let b be such that $x_i + b > 0$ for all i. Therefore, by the preceding,

$$\max_i (x_i + b) = \sum_i (x_i + b) - \sum_{i<j} \min(x_i + b, x_j + b) + \cdots + (-1)^{n+1} \min(x_1 + b, \ldots, x_n + b)$$

Letting

$$M = \sum_i x_i - \sum_{i<j} \min(x_i, x_j) + \cdots + (-1)^{n+1} \min(x_1, \ldots, x_n)$$

we can rewrite the foregoing identity as

$$\max_i x_i + b = M + b\left(n - \binom{n}{2} + \cdots + (-1)^{n+1} \binom{n}{n}\right)$$

But

$$0 = (1-1)^n = 1 - n + \binom{n}{2} + \cdots + (-1)^n \binom{n}{n}$$

The preceding two equations show that

$$\max_i x_i = M$$

and the proposition is proven.

It follows from **Proposition 2.2** that for any random variables $X_1, \ldots, X_n$,

$$\max_i X_i = \sum_i X_i - \sum_{i<j} \min(X_i, X_j) + \cdots + (-1)^{n+1} \min(X_1, \ldots, X_n)$$

Taking expectations of both sides of this equality yields the following relationship between the expected value of the maximum and those of the partial minimums:

$$E[\max_i X_i] = \sum_i E[X_i] - \sum_{i<j} E[\min(X_i, X_j)] + \cdots + (-1)^{n+1} E[\min(X_1, \ldots, X_n)] \quad (2.7)$$

Example 2s Coupon collecting with unequal probabilities

Suppose there are n types of coupons and that each time one collects a coupon, it is, independently of previous coupons collected, a type i coupon with probability p_i, $\sum_{i=1}^n p_i = 1$. Find the expected number of

coupons one needs to collect to obtain a complete set of at least one of each type.

Solution

If we let Xi denote the number of coupons one needs to collect to obtain a type i, then we can express X as

$$X=\max_{i=1,\ldots,n} X_i$$

Because each new coupon obtained is a type i with probability pi, Xi is a geometric random variable with parameter pi. Also, because the minimum of Xi and Xj is the number of coupons needed to obtain either a type i or a type j, it follows that for $i \neq j$, min (Xi,Xj) is a geometric random variable with parameter pi+pj. Similarly, min (Xi,Xj,Xk), the number needed to obtain any of types i, j, or k, is a geometric random variable with parameter pi+pj+pk, and so on. Therefore, the identity (**2.7**) yields

$$E[X]=\sum_{i=1} p_i-\sum_{i<j} p_i+p_j+\sum_{i<j<k} p_i+p_j+p_k+\cdots+(-1)^{n+1} p_1+\cdots+p_n$$

Noting that

$$\int_0^\infty e^{-px}\,dx=\frac{1}{p}$$

and using the identity

$$1-\prod_{i=1}^n (1-e^{-p_i x})=\sum_i e^{-p_i x}-\sum_{i<j} e^{-(p_i+p_j)x}+\cdots+(-1)^{n+1} e^{-(p_1+\cdots+p_n)x}$$

shows, upon integrating the identity, that

$$E[X]=\int_0^\infty \left(1-\prod_{i=1}^n (1-e^{-p_i x})\right)\,dx$$

which is a more useful computational form.

7.3 Moments of the Number of Events that Occur

Many of the examples solved in the previous section were of the following form: For given events A1,…,An, find E[X], where X is the number of these events that occur. The solution then involved defining an indicator variable Ii for event Ai such that

Ii={1, if Ai occurs0, otherwise

Because

X=∑i=1nIi

we obtained the result

E[X]=E[∑i=1nIi]=∑i=1nE[Ii]=∑i=1nP(Ai) (3.1)

Now suppose we are interested in the number of *pairs* of events that occur. Because IiIj will equal 1 if both Ai and Aj occur and will equal 0 otherwise, it follows that the number of pairs is equal to ∑i<jIiIj. But because X is the number of events that occur, it also follows that the number of pairs of events that occur is (X2). Consequently,

(X2)=∑i<jIiIj

where there are (n2) terms in the summation. Taking expectations yields

E[(X2)]=∑i<j E[IiIj]=∑i<jP(AiAj) (3.2)

or

E[X(X−1)2]=∑i<jP(AiAj)

giving that

E[X2]−E[X]=2∑i<jP(AiAj) (3.3)

which yields $E[X^2]$, and thus Var (X)=E[X2]−(E[X])2.

Moreover, by considering the number of distinct subsets of k events that all occur, we see that

(Xk)=∑i1<i2<…<ikIi1Ii2⋯Iik

Taking expectations gives the identity

E[(Xk)]=∑i1<i2<…<ikE[Ii1Ii2⋯Iik]=∑i1<i2<…<ikP(Ai1Ai2⋯Aik) (3.4)

Example 3a Moments of binomial random variables

Consider n independent trials, with each trial being a success with probability p . Let A_i be the event that trial i is a success. When i=j, P(AiAj)=p2. Consequently, **Equation (3.2**) yields

E[(X2)]=∑i<jp2=(n2)p2

or

$$E[X(X-1)]=n(n-1)p2$$

or

$$E[X2]-E[X]=n(n-1)p2$$

Now, $E[X]=\sum_{i=1}^{n}P(Ai)=np$, so, from the preceding equation

$$\text{Var}(X)=E[X2]-(E[X])2=n(n-1)p2+np-(np)2=np(1-p)$$

which is in agreement with the result obtained in **Section 4.6.1**.

In general, because $P(Ai1Ai2\cdots Aik)=pk$, we obtain from **Equation (3.4)** that

$$E[(Xk)]=\sum_{i1<i2<\ldots<ik}pk=(nk)pk$$

or, equivalently,

$$E[X(X-1)\cdots(X-k+1)]=n(n-1)\cdots(n-k+1)pk$$

The successive values $E[Xk]$, $k\geq 3$, can be recursively obtained from this identity. For instance, with $k=3$, it yields

$$E[X(X-1)(X-2)]=n(n-1)(n-2)p3$$

or

$$E[X3-3X2+2X]=n(n-1)(n-2)p3$$

or

$$E[\,X3\,]=3E[\,X2\,]-2E[X]+n(n-1)(n-2)p3=3n(n-1)p2+np+n(n-1)(n-2)p3$$

Example 3b Moments of hypergeometric random variables

Suppose n balls are randomly selected from an urn containing N balls, of which m are white. Let Ai be the event that the ith ball selected is white. Then X, the number of white balls selected, is equal to the number of the events A1,…,An that occur. Because the ith ball selected is equally likely to be any of the N balls, of which m are white, $P(Ai)=m/N$. Consequently, **Equation (3.1)** gives that $E[X]=\sum_{i=1}^{n}P(Ai)=nm/N$. Also, since

$$P(AiAj)=P(Ai)P(Aj|Ai)=mNm-1N-1$$

we obtain, from **Equation (3.2)**, that

$$E[(X2)]=\sum_{i<j}m(m-1)N(N-1)=(n2)m(m-1)N(N-1)$$

or

$$E[X(X-1)]=n(n-1)m(m-1)N(N-1)$$

showing that

$$E[X2]=n(n-1)m(m-1)N(N-1)+E[X]$$

This formula yields the variance of the hypergeometric, namely,

$$Var\ (X)=E[\ X2\ ]-(E[X])2=n(n-1)m(m-1)N(N-1)+nmN-n2m2N2=mnN[\ (n-1)(m-1)N-1+1-mnN\ ]$$

which agrees with the result obtained in **Example 8j** of **Chapter 4**.

Higher moments of X are obtained by using **Equation (3.4)**. Because

$$P(Ai1Ai2\cdots Aik)=m(m-1)\cdots(m-k+1)N(N-1)\cdots(N-k+1)$$

Equation (3.4) yields

$$E[(Xk)]=(nk)m(m-1)\cdots(m-k+1)N(N-1)\cdots(N-k+1)$$

or

$$E[\ X(X-1)\cdots(X-k+1\ )]\qquad =n(n-1)\cdots(n-k+1)m(m-1)\cdots(m-k+1)N(N-1)\cdots(N-k+1)$$

Example 3c Moments in the match problem

For i=1,…,N, let Ai be the event that person i selects his or her own hat in the match problem. Then

$$P(AiAj)=P(Ai)P(Aj|Ai)=1N1N-1$$

which follows because, conditional on person i selecting her own hat, the hat selected by person j is equally likely to be any of the other N−1 hats, of which one is his own. Consequently, with X equal to the number of people who select their own hat, it follows from **Equation (3.2)** that

$$E[(X2)]=\sum i<j1N(N-1)=(N2)1N(N-1)$$

thus showing that

$$E[X(X-1)]=1$$

Therefore, $E[\ X2\ ]=1+E[\ X\ ]$. Because $E[\ X\ ]=\sum i=1NP(Ai) = 1$, we obtain that

$$Var\ (X)=E[X2]-(E[X])2=1.$$

Hence, both the mean and variance of the number of matches is 1. For higher moments, we use **Equation (3.4)**, along with the fact that $P(Ai1Ai2\cdots Aik)=1N(N-1)\cdots(N-k+1)$, to obtain

$$E[\ (X\ k)\ ]=(N\ k)1N(N-1)\cdots(N-k+1)$$

or

$$E[X(X-1)\cdots(X-k+1)]=1$$

Example 3d Another coupon-collecting problem

Suppose that there are N distinct types of coupons and that, independently of past types collected, each new one obtained is type j with probability pj, $\sum j=1Npj=1$. Find the expected value and variance of the

number of different types of coupons that appear among the first n collected.

Solution

We will find it more convenient to work with the number of uncollected types. So, let Y equal the number of types of coupons collected, and let X=N−Y denote the number of uncollected types. With Ai defined as the event that there are no type i coupons in the collection, X is equal to the number of the events A1,…,AN that occur. Because the types of the successive coupons collected are independent, and, with probability 1−pi each new coupon is not type i, we have

$$P(A_i)=(1-p_i)^n$$

Hence, $E[X]=\sum_{i=1}^{N}(1-p_i)^n$, from which it follows that

$$E[Y]=N-E[X]=N-\sum_{i=1}^{N}(1-p_i)^n$$

Similarly, because each of the n coupons collected is neither a type i nor a type j coupon, with probability 1−pi−pj, we have

$$P(A_iA_j)=(1-p_i-p_j)^n, \quad i\neq j$$

Thus,

$$E[X(X-1)]=2\sum_{i<j}P(A_iA_j)=2\sum_{i<j}(1-p_i-p_j)^n$$

or

$$E[X^2]=2\sum_{i<j}(1-p_i-p_j)^n+E[X]$$

Hence, we obtain

$$\text{Var}(Y)=\text{Var}(X)=E[X^2]-(E[X])^2=2\sum_{i<j}(1-p_i-p_j)^n+\sum_{i=1}^{N}(1-p_i)^n-\left(\sum_{i=1}^{N}(1-p_i)^n\right)^2$$

In the special case where $p_i=1/N$, $i=1,\ldots,N$, the preceding formulas give

$$E[Y]=N[1-(1-1N)^n]$$

and

$$\text{Var}(Y)=N(N-1)(1-2N)^n+N(1-1N)^n-N^2(1-1N)^{2n}$$

Example 3e The negative hypergeometric random variables

Suppose an urn contains n+m balls, of which n are special and m are ordinary. These items are removed one at a time, with each new removal being equally likely to be any of the balls that remain in the urn. The random variable Y, equal to the number of balls that need be withdrawn until a total of r special balls have been removed, is said to have a *negative hypergeometric distribution*. The negative hypergeometric distribution bears the same relationship to the hypergeometric distribution as the negative binomial does to the binomial. That is, in both cases, rather than considering a random variable equal to the number of successes in a fixed number of trials (as are the binomial and hypergeometric variables), they refer to the number of trials needed to obtain a fixed number of successes.

To obtain the probability mass function of a negative hypergeometric random variable Y, note that Y will

equal k if both

1. the first k−1 withdrawals consist of r−1 special and k−r ordinary balls and
2. the kth ball withdrawn is special.

Consequently,

$$P\{ Y=k \}=(\ nr-1)(\ mk-r)(n+mk-1)n-r+1n+m-k+1$$

We will not, however, utilize the preceding probability mass function to obtain the mean and variance of Y. Rather, let us number the m ordinary balls as o1,…,om, and then, for each i=1,…,n, let Ai be the event that oi is withdrawn before r special balls have been removed. Then, if X is the number of the events A1,…,Am that occur, it follows that X is the number of ordinary balls that are withdrawn before a total of r special balls have been removed. Consequently,

$$Y=r+X$$

showing that

$$E[Y]=r+E[X]=r+\sum i=1mP(Ai)$$

To determine P(Ai), consider the n+1 balls consisting of oi along with the n special balls. Of these n+1 balls, oi is equally likely to be the first one withdrawn, or the second one withdrawn, …, or the final one withdrawn. Hence, the probability that it is among the first r of these to be selected (and so is removed before a total or r special balls have been withdrawn) is rn+1. Consequently,

$$P(Ai)=rn+1$$

and

$$E[Y]=r+mrn+1=r(n+m+1)n+1$$

Thus, for instance, the expected number of cards of a well-shuffled deck that would need to be turned over until a spade appears is 1+3914=3.786, and the expected number of cards that would need to be turned over until an ace appears is 1+485=10.6.

To determine $Var(Y)=Var(X)$, we use the identity

$$E[X(X-1)]=2\sum i<jP(AiAj)$$

Now, P(AiAj) is the probability that both oi and oj are removed before there have been a total of r special balls removed. So consider the n+2 balls consisting of oi,oj, and the n special balls. Because all withdrawal orderings of these balls are equally likely, the probability that oi and oj are both among the first r+1 of them to be removed (and so are both removed before r special balls have been withdrawn) is

$$P(AiAj)=(22)(\ nr-1)(n+2r+1)=r(r+1)(n+1)(n+2)$$

Consequently,

$$E[X(X-1)]=2(m2)r(r+1)(n+1)(n+2)$$

so

$$E[X2]=m(m-1)r(r+1)(n+1)(n+2)+E[X]$$

Because E[X]=mrn+1, this yields

$$\mathrm{Var}(Y)=\mathrm{Var}(X)=m(m-1)r(r+1)(n+1)(n+2)+mrn+1-(mrn+1)2$$

A little algebra now shows that

$$\mathrm{Var}(Y)=mr(n+1-r)(n+m+1)(n+1)2(n+2)$$

Example 3f Singletons in the coupon collector's problem

Suppose that there are n distinct types of coupons and that, independently of past types collected, each new one obtained is equally likely to be any of the n types. Suppose also that one continues to collect coupons until a complete set of at least one of each type has been obtained. Find the expected value and variance of the number of types for which exactly one coupon of that type is collected.

Solution

Let X equal the number of types for which exactly one of that type is collected. Also, let Ti denote the ith type of coupon to be collected, and let Ai be the event that there is only a single type Ti coupon in the complete set. Because X is equal to the number of the events A1,…,An that occur, we have

$$E[X]=\sum_{i=1}^{n}P(A_i)$$

Now, at the moment when the first type Ti coupon is collected, there remain n−i types that need to be collected to have a complete set. Because, starting at this moment, each of these n−i+1 types (the $n-i$ not yet collected and type T_i) is equally likely to be the last of these types to be collected, it follows that the type T_i will be the last of these types (and so will be a singleton) with probability 1n−i+1. Consequently, P(Ai)=1n−i+1, yielding

$$E[X]=\sum_{i=1}^{n}\frac{1}{n-i+1}=\sum_{i=1}^{n}\frac{1}{i}$$

To determine the variance of the number of singletons, let Si, j, for i<j, be the event that the first type Ti coupon to be collected is still the only one of its type to have been collected at the moment that the first type Tj coupon has been collected. Then

$$P(A_iA_j)=P(A_iA_j|S_{i,j})P(S_{i,j})$$

Now, P(Si,j) is the probability that when a type Ti has just been collected, of the n−i+1 types consisting of type Ti and the n−i as yet uncollected types, a type Ti is not among the first j−i of these types to be collected. Because type Ti is equally likely to be the first, or second, or …,n−i+1 of these types to be collected, we have

$$P(S_{i,j})=\frac{1-j-i}{n-i+1}=\frac{n+1-j}{n+1-i}$$

Now, conditional on the event Si,j, both Ai and Aj will occur if, at the time the first type Tj coupon is collected, of the n−j+2 types consisting of types Ti,Tj, and the n−j as yet uncollected types, Ti and Tj are both collected after the other n−j. But this implies that

$$P(A_iA_j|S_{i,j})=\frac{2}{n-j+2}\frac{1}{n-j+1}$$

Therefore,

$$P(A_iA_j)=\frac{2}{(n+1-i)(n+2-j)},\quad i<j$$

yielding

$$E[X(X-1)]=4\sum_{i<j}^{1}(n+1-i)(n+2-j)$$

Consequently, using the previous result for E[X], we obtain

$$\text{Var}(X)=4\sum_{i<j}^{1}(n+1-i)(n+2-j)+\sum_{i=1}^{n}1i-(\sum_{i=1}^{n}1i)2$$

7.4 Covariance, Variance of Sums, and Correlations

The following proposition shows that the expectation of a product of independent random variables is equal to the product of their expectations.

Proposition 4.1

If X and Y are independent, then, for any functions h and g,

$$E[g(X)h(Y)]=E[g(X)]E[h(Y)]$$

Proof Suppose that X and Y are jointly continuous with joint density f(x,y). Then

$$E[g(x)h(Y)]=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}g(x)\,h(y)\,f(x,y)\,dx\,dy=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}g(x)\,h(y)\,f_X(x)\,f_Y(y)\,dx\,dy=\int_{-\infty}^{\infty}h(y)\,f_Y(y)dy\int_{-\infty}^{\infty}g(x)\,f_X(x)\,dx=E[h(Y)]E[g(X)]$$

The proof in the discrete case is similar.

Just as the expected value and the variance of a single random variable give us information about that random variable, so does the covariance between two random variables give us information about the relationship between the random variables.

> Definition
> The covariance between X and Y, denoted by Cov (X,Y), is defined by
>
> $$Cov(X,Y)=E[(X-E[X])(Y-E[Y])]$$

Upon expanding the right side of the preceding definition, we see that

$$Cov\,(X,Y)=E[XY-E[X]Y-XE[Y]+E[Y]E[X]]=E[XY]-E[X]E[Y]-E[X]E[Y]+E[X]E[Y]=E[XY]-E[X]E[Y]$$

Note that if X and Y are independent, then, by **Proposition 4.1**, Cov (X,Y)=0. However, the converse is not true. A simple example of two dependent random variables X and Y having zero covariance is obtained by letting X be a random variable such that

$$P\{X=0\}=P\{X=1\}=P\{X=-1\}=13$$

and defining

$$Y=\begin{cases}0 & \text{if } X\neq 0\\ 1 & \text{if } X=0\end{cases}$$

Now,XY=0, so E[XY]=0. Also,E[X]=0. Thus,

$$Cov\,(X,Y)=E[XY]-E[X]E[Y]=0$$

However,X and Y are clearly not independent.

The following proposition lists some of the properties of covariance.

Proposition 4.2

i. Cov (X,Y)= Cov (Y,X)
ii. Cov (X,X)=Cov(X)
iii. Cov (aX,Y)=a Cov (X,Y)
iv. $Cov\ (\sum_{i=1}^{n}X_i, \sum_{j=1}^{m}Y_j)=\sum_{i=1}^{n}\sum_{j=1}^{m} Cov\,(X_i,Y_j)$

Proof of Proposition 4.2 Parts (i) and (ii) follow immediately from the definition of covariance, and part (iii) is left as an exercise for the reader. To prove part (iv), which states that the covariance operation is additive (as is the operation of taking expectations), let $\mu_i=E[X_i]$ and $v_j=E[Y_j]$. Then

$$E[\sum_{i=1}^{n}X_i]=\sum_{i=1}^{n}\mu_i,\quad E[\sum_{j=1}^{m}Y_j]=\sum_{j=1}^{m}v_j$$

and

$$Cov\,(\sum_{i=1}^{n}X_i, \sum_{j=1}^{m}Y_j)=E\,[\,(\sum_{i=1}^{n}X_i-\sum_{i=1}^{n}\mu_i)(\sum_{j=1}^{m}Y_j-\sum_{j=1}^{m}v_j)\,]=E\,[\,\sum_{i=1}^{n}(X_i-\mu_i)-\sum_{j=1}^{m}(Y_j-v_j)\,]=E\,[\,\sum_{i=1}^{n}\sum_{j=1}^{m}(X_i-\mu_i)(Y_j-v_j)]=\sum_{i=1}^{n}\sum_{j=1}^{m}E\,[\,(X_i-\mu_i\,)(Y_j-v_j)]$$

where the last equality follows because the expected value of a sum of random variables is equal to the sum of the expected values.

It follows from parts (ii) and (iv) of **Proposition 4.2**, upon taking Yj=Xj,j=1,…,n, that

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right)=\text{Cov}\left(\sum_{i=1}^{n}X_i,\sum_{j=1}^{n}X_j\right)=\sum_{i=1}^{n}\sum_{j=1}^{n}\text{Cov}(X_i,X_j)=\sum_{i=1}^{n}\left(\text{Cov}(X_i,X_i)+\sum_{j:j\neq i}\text{Cov}(X_i,X_j)\right)=\sum_{i=1}^{n}\text{Var}(X_i)+\sum\sum_{i\neq j}\text{Cov}(X_i,X_j)$$

Since each pair of indices i,j,i≠j, appears twice in the double summation, the preceding formula is equivalent to

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right)=\sum_{i=1}^{n}\text{Var}(X_i)+2\sum\sum_{i<j}\text{Cov}(X_i,X_j) \qquad (4.1)$$

If X1,…,Xn are pairwise independent, in that Xi and Xj are independent for i≠j, then **Equation (4.1)** reduces to

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right)=\sum_{i=1}^{n}\text{Var}(X_i)$$

The following examples illustrate the use of **Equation (4.1)**.

Example 4a

Let X1,…,Xn be independent and identically distributed random variables having expected value μ and variance σ2, and as in **Example 2c**, let $\bar{X}=\sum_{i=1}^{n}X_i/n$ be the sample mean. The quantities $X_i-\bar{X}$,i=1,…,n, are called *deviations*, as they equal the differences between the individual data and the sample mean. The random variable

$$S^2=\sum_{i=1}^{n}(X_i-\bar{X})^2/n-1$$

is called the *sample variance*. Find (a) $\text{Var}(\bar{X})$ and (b)E[S2].

Solution

a. $\text{Var}(\bar{X})=(1/n)^2\,\text{Var}\left(\sum_{i=1}^{n}X_i\right)=(1/n)^2\sum_{i=1}^{n}\text{Var}(X_i)$ by independence$=\sigma^2/n$

b. We start with the following algebraic identity:
$$(n-1)S^2=\sum_{i=1}^{n}(X_i-\mu+\mu-\bar{X})^2=\sum_{i=1}^{n}(X_i-\mu)^2+\sum_{i=1}^{n}(\bar{X}-\mu)^2-2(\bar{X}-\mu)\sum_{i=1}^{n}(X_i-\mu)=\sum_{i=1}^{n}(X_i-\mu)^2+n(\bar{X}-\mu)^2-2(\bar{X}-\mu)n(\bar{X}-\mu)$$
$$=\sum_{i=1}^{n}(X_i-\mu)^2-n(\bar{X}-\mu)^2$$

Taking expectations of the preceding yields

$$(n-1)E[S^2]=\sum_{i=1}^{n}E[(X_i-\mu)^2]-nE[(\bar{X}-\mu)^2]=n\sigma^2-n\,\text{Var}(\bar{X})=(n-1)\sigma^2$$

where the final equality made use of part (a) of this example and the one preceding it made use of the result of **Example 2c**, namely, that $E[\bar{X}]=\mu$. Dividing through by n−1 shows that the expected value of the sample variance is the distribution variance σ2.

Our next example presents another method for obtaining the variance of a binomial random variable.

Example 4b Variance of a binomial random variable

Compute the variance of a binomial random variable X with parameters n and p.

Solution

Since such a random variable represents the number of successes in n independent trials when each trial has the common probability p of being a success, we may write

$$X=X_1+\cdots+X_n$$

where the Xi are independent Bernoulli random variables such that

$$X_i=\begin{cases}1 & \text{if the } i\text{th trial is a success}\\ 0 & \text{otherwise}\end{cases}$$

Hence, from **Equation (4.1)**, we obtain

$$\text{Var}(X)=\text{Var}(X_1)+\cdots+\text{Var}(X_n)$$

But

$$\text{Var}(X_i) = E[X_{2i}] - (E[X_i])^2 = E[X_i] - (E[X_i])^2 \quad \text{since } X_{2i} = X_i = p - p^2$$

Thus,

$$\text{Var}(X) = np(1-p)$$

Example 4c Sampling from a finite population

Consider a set of N people, each of whom has an opinion about a certain subject that is measured by a real number v that represents the person's "strength of feeling" about the subject. Let vi represent the strength of feeling of person i,i=1,…N.

Suppose that the quantities vi,i=1,…,N, are unknown and, to gather information, a group of n of the N people is "randomly chosen" in the sense that all of the (N2) subsets of size n are equally likely to be chosen. These n people are then questioned and their feelings determined. If S denotes the sum of the n sampled values, determine its mean and variance.

An important application of the preceding problem is to a forthcoming election in which each person in the population is either for or against a certain candidate or proposition. If we take vi to equal 1 if person i is in favor and 0 if he or she is against, then $\bar{v} = \sum_{i=1}^{N} v_i / N$ represents the proportion of the population that is in favor. To estimate $\bar{v}$, a random sample of n people is chosen, and these people are polled. The proportion of those polled who are in favor – that is, *S/n* – is often used as an estimate of $\bar{v}$.

Solution

For each person i,i=1,…,N, define an indicator variable Ii to indicate whether or not that person is included in the sample. That is,

$$I_i = \begin{cases} 1 & \text{if person } i \text{ is in the random sample} \\ 0 & \text{otherwise} \end{cases}$$

Now,S can be expressed by

$$S = \sum_{i=1}^{N} v_i I_i$$

so

$$E[S] = \sum_{i=1}^{N} v_i E[I_i] \quad \text{Var}(S) = \sum_{i=1}^{N} \text{Var}(v_i I_i) + 2\sum\sum_{i<j} \text{Cov}(v_i I_i, v_j I_j) = \sum_{i=1}^{N} v_i^2 \text{Var}(I_i) + 2\sum\sum_{i<j} v_i v_j \text{Cov}(I_i, I_j)$$

Because

$$E[I_i] = \frac{n}{N} \quad E[I_i I_j] = \frac{n}{N}\frac{n-1}{N-1}$$

it follows that

$$\text{Var}(I_i) = \frac{n}{N}\left(1 - \frac{n}{N}\right) \quad \text{Cov}(I_i, I_j) = \frac{n(n-1)}{N(N-1)} - \left(\frac{n}{N}\right)^2 = -\frac{n(N-n)}{N^2(N-1)}$$

Hence,

$$E[S] = n \sum_{i=1}^{N} v_i / N = n\bar{v} \quad \text{Var}(S) = \frac{n}{N}\left(\frac{N-n}{N}\right) \sum_{i=1}^{N} v_i^2 - 2\frac{n}{N}\left(\frac{N-n}{N}\right)\frac{1}{2(N-1)} \sum\sum_{i<j} v_i v_j$$

The expression for Var (S) can be simplified somewhat by using the identity $(v_1 + \cdots + v_N)^2 = \sum_{i=1}^{N} v_i^2 + 2\sum\sum_{i<j} v_i v_j$. After some simplification, we obtain

$$\text{Var}(S) = \frac{n(N-n)}{N-1}\left(\sum_{i=1}^{N} v_i^2 / N - \bar{v}^2\right)$$

Consider now the special case in which *Np* of the v's are equal to 1 and the remainder equal to 0. Then, in this case,S is a hypergeometric random variable and has mean and variance given, respectively, by

$$E[S] = n\bar{v} = np \quad \text{since } \bar{v} = Np/N = p$$

and

$$\text{Var}(S) = \frac{n(N-n)}{N-1}\left(\frac{Np}{N} - p^2\right) = \frac{n(N-n)}{N-1}p(1-p)$$

The quantity *S/n*, equal to the proportion of those sampled that have values equal to 1, is such that

$$E[S_n]=p \quad \text{Var}(S_n)=\frac{N-n}{n(N-1)}p(1-p)$$

The correlation of two random variables X and Y, denoted by ρ(X,Y), is defined, as long as Var (X)Var (Y) is positive, by

$$\rho(X,Y)=\frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)\,\text{Var}(Y)}}$$

It can be shown that

$$-1\le\rho(X,Y)\le 1 \tag{4.2}$$

To prove **Equation (4.2)**, suppose that and Y have variances given by σ2x and σ2y, respectively. Then, on the one hand,

$$0\le\text{Var}(X\sigma x+Y\sigma y)=\text{Var}(X)\sigma^2x+\text{Var}(Y)\sigma^2y+2\,\text{Cov}(X,Y)\sigma x\sigma y=2[1+\rho(X,Y)]$$

implying that

$$-1\le\rho(X,Y)$$

On the other hand,

$$0\le\text{Var}(X\sigma x-Y\sigma y)=\text{Var}(X)\sigma^2x+\text{Var}(Y)(-\sigma y)^2-2\,\text{Cov}(X,Y)\sigma x\sigma y=2[1+\rho(X,Y)]$$

implying that

$$\rho(X,Y)\le 1$$

which completes the proof of **Equation (4.2)**.

In fact, since Var (Z)=0 implies that Z is constant with probability 1 (this intuitive relationship will be rigorously proven in **Chapter 8**), it follows from the proof of **Equation (4.2)** that ρ(X,Y)=1 implies that Y=a+bX, where b=σy/σx>0 and ρ(X,Y)=−1 implies that Y=a+bX, where b=−σy/σx<0. We leave it as an exercise for the reader to show that the reverse is also true: that if Y=a+bX, then ρ(X,Y) is either +1 or −1, depending on the sign of b.

The correlation coefficient is a measure of the degree of linearity between X and Y. A value of ρ(X,Y) near +1 or −1 indicates a high degree of linearity between X and Y, whereas a value near 0 indicates that such linearity is absent. A positive value of ρ(X,Y) indicates that Y tends to increase when X does, whereas a negative value indicates that Y tends to decrease when X increases. If ρ(X,Y)=0, then X and Y are said to be *uncorrelated*.

Example 4d

Let IA and IB be indicator variables for the events A and B. That is,

$$I_A=\begin{cases}1 & \text{if } A \text{ occurs}\\ 0 & \text{otherwise}\end{cases} \qquad I_B=\begin{cases}1 & \text{if } B \text{ occurs}\\ 0 & \text{otherwise}\end{cases}$$

Then

$$E[I_A]=P(A)\quad E[I_B]=P(B)\quad E[I_AI_B]=P(AB)$$

so

$$\text{Cov}(I_A,I_B)=P(AB)-P(A)P(B)=P(B)[P(A|B)-P(A)]$$

Thus, we obtain the quite intuitive result that the indicator variables for A and B are either positively correlated, uncorrelated, or negatively correlated, depending on whether P(A|B) is, respectively, greater than, equal to, or less than P(A).

Our next example shows that the sample mean and a deviation from the sample mean are uncorrelated.

Example 4e

Let X1,…,Xn be independent and identically distributed random variables having variance σ2. Show that

$$\text{Cov}(X_i-\bar{X},\bar{X})=0$$

Solution

We have

$$\text{Cov}(X_i - \bar{X}, \bar{X}) = \text{Cov}(X_i, \bar{X}) - \text{Cov}(\bar{X}, \bar{X}) = \text{Cov}\left(X_i, \frac{1}{n}\sum_{j=1}^{n} X_j\right) - \text{Var}(\bar{X}) = \frac{1}{n}\sum_{j=1}^{n}\text{Cov}(X_i, X_j) - \frac{\sigma^2}{n} = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0$$

where the next-to-last equality uses the result of **Example 4a** and the final equality follows because

$$\text{Cov}(X_i, X_j) = \begin{cases} 0 & \text{if } j \neq i \text{ by independence} \\ \sigma^2 & \text{if } j = i \text{ since } \text{Var}(X_i) = \sigma^2 \end{cases}$$

Although $\bar{X}$ and the deviation $X_i - \bar{X}$ are uncorrelated, they are not, in general, independent. However, in the special case where the X_i are normal random variables, it turns out that not only is $\bar{X}$ independent of a single deviation, but it is independent of the entire sequence of deviations $X_j - \bar{X}, j = 1, \ldots, n$. This result will be established in **Section 7.8**, where we will also show that, in this case, the sample mean $\bar{X}$ and the sample variance S^2 are independent, with $(n-1)S^2/\sigma^2$ having a chi-squared distribution with $n-1$ degrees of freedom. (See **Example 4a** for the definition of S^2.)

Example 4f

Consider m independent trials, each of which results in any of r possible outcomes with probabilities $p_1, \ldots, p_r, \sum_{i=1}^{r} p_i = 1$. If we let $N_i, i = 1, \ldots, r$, denote the number of the m trials that result in outcome i, then $N_1, N_2, \ldots, N_r$ have the multinomial distribution

$$P\{N_1 = n_1, \ldots, N_r = n_r\} = \frac{m!}{n_1! \ldots n_r!} p_1^{n_1} \cdots p_r^{n_r}, \quad \sum_{i=1}^{r} n_i = m$$

For $i \neq j$, it seems likely that when N_i is large, N_j would tend to be small; hence, it is intuitive that they should be negatively correlated. Let us compute their covariance by using **Proposition 4.2**(iv) and the representation

$$N_i = \sum_{k=1}^{m} I_i(k) \qquad \text{and} \qquad N_j = \sum_{k=1}^{m} I_j(k)$$

where

$$I_i(k) = \begin{cases} 1 & \text{if trial } k \text{ results in outcome } i \\ 0 & \text{otherwise} \end{cases} \qquad I_j(k) = \begin{cases} 1 & \text{if trial } k \text{ results in outcome } j \\ 0 & \text{otherwise} \end{cases}$$

From **Proposition 4.2**(iv), we have

$$\text{Cov}(N_i, N_j) = \sum_{\ell=1}^{m}\sum_{k=1}^{m} \text{Cov}(I_i(k), I_j(\ell))$$

Now, on the one hand, when $k \neq \ell$,

$$\text{Cov}(I_i(k), I_j(\ell)) = 0$$

since the outcome of trial k is independent of the outcome of trial ℓ. On the other hand,

$$\text{Cov}(I_i(\ell), I_j(\ell)) = E[I_i(\ell) I_j(\ell)] - E[I_i(\ell)]E[I_j(\ell)] = 0 - p_i p_j = -p_i p_j$$

where the equation uses the fact that $I_i(\ell) I_j(\ell) = 0$, since trial ℓ cannot result in both outcome i and outcome j. Hence, we obtain

$$\text{Cov}(N_i, N_j) = -m p_i p_j$$

which is in accord with our intuition that N_i and N_j are negatively correlated.

7.5 Conditional Expectation

7.5.1 Definitions

Recall that if X and Y are jointly discrete random variables, then the conditional probability mass function of X, given that Y=y, is defined for all y such that P{Y=y}>0, by

pX|Y(x|y)=P{X=x|Y=y}=p(x,y)pY(y)

It is therefore natural to define, in this case, the conditional expectation of X given that Y=y, for all values of y such that pY(y)>0, by

E[X|Y=y]=∑x xP{X=x|Y=y}=∑x xpX|Y(x|y)

> ***Example 5a***
>
> If X and Y are independent binomial random variables with identical parameters n and p, calculate the conditional expected value of X given that X+Y=m.
>
> Solution
>
> Let us first calculate the conditional probability mass function of X given that X+Y=m. For k≤min(n,m),
>
> P{X=k|X+Y=m}=P{ X=k,X+Y=m }P{ X+Y=m }=P{ X=k,Y=m−k }P{ X+Y=m }=P{ X=k }P{ Y=m−k }P{ X+Y=m }=(nk) pk(1−p)n−k (nm−k) pm−k(1−p)n−m+k(2nm) pm(1−p)2n−m=(nk) (nm−k)(2nm)
>
> where we have used the fact (see **Example 3f** of **Chapter 6**) that X+Y is a binomial random variable with parameters 2n and p. Hence, the conditional distribution of X, given that X+Y=m, is the hypergeometric distribution, and from **Example 2g**, we obtain
>
> E[X|X+Y=m]=m2

Similarly, let us recall that if X and Y are jointly continuous with a joint probability density function f(x,y), then the conditional probability density of X, given that Y=y, is defined for all values of y such that fY(y)>0 by

fX|Y(x|y)=f(x,y)fY(y)

It is natural, in this case, to define the conditional expectation of X, given that Y=y, by

E[X|Y=y]=∫−∞∞xfX|Y(x|y) dx

provided that fY(y)>0.

> ***Example 5b***
>
> Suppose that the joint density of X and Y is given by
>
> f(x,y)=e−x/ye−yy 0<x<∞, 0<y<∞
>
> Compute E[X|Y=y].
>
> Solution
>
> We start by computing the conditional density
>
> fX|Y(x|y)=f(x,y)fY(y)=f(x,y)∫−∞∞f(x,y) dx=(1/y)e−x/ye−y∫0∞(1/y)e−x/ye−y dx=(1/y)e−x/y∫0∞(1/y)e−x/y dx=1ye−x/y
>
> Hence, the conditional distribution of X, given that Y=y, is just the exponential distribution with mean y. Thus,
>
> E[X|Y=y]=∫0∞xye−x/y dx=y

Remark Just as conditional probabilities satisfy all of the properties of ordinary probabilities, so do conditional expectations satisfy the properties of ordinary expectations. For instance, such formulas as

$$E[g(X)|Y=y]=\begin{cases}\sum_x g(x)\, p_{X|Y}(x|y) & \text{in the discrete case}\\ \int_{-\infty}^{\infty} g(x)\, f_{X|Y}(x|y)\, dx & \text{in the continuous case}\end{cases}$$

and

$$E\left[\sum_{i=1}^{n} X_i|Y=y\right]=\sum_{i=1}^{n} E[X_i|Y=y]$$

remain valid. As a matter of fact, conditional expectation given that Y=y can be thought of as being an ordinary expectation on a reduced sample space consisting only of outcomes for which Y=y.

7.5.2 Computing expectations by conditioning

Let us denote by E[X|Y] that function of the random variable Y whose value at Y=y is E[X|Y=y]. Note that E[X|Y] is itself a random variable. An extremely important property of conditional expectations is given by the following proposition.

Proposition 5.1 The conditional expectation formula

$$E[X]=E[E[X|Y]] \tag{5.1}$$

If Y is a discrete random variable, then **Equation (5.1)** states that

$$E[X]=\sum_y E[X|Y=y]P\{Y=y\} \tag{5.1a}$$

whereas if Y is continuous with density $f_Y(y)$, then **Equation (5.1)** states

$$E[X]=\int_{-\infty}^{\infty} E[X|Y=y]\, f_Y(y)\, dy \tag{5.1b}$$

We now give a proof of **Equation (5.1)** in the case where X and Y are both discrete random variables.

Proof of Equation (5.1) When X and Y Are Discrete: We must show that

$$E[X]=\sum_y E[X|Y=y]P\{Y=y\} \tag{5.2}$$

Now, the right-hand side of **Equation (5.2)** can be written as

$$\sum_y E[X|Y=y]P\{Y=y\}=\sum_y\sum_x x P\{X=x|Y=y\}P\{Y=y\}=\sum_y\sum_x x P\{X=x,Y=y\}P\{Y=y\}P\{Y=y\}=\sum_y\sum_x x P\{X=x,Y=y\}=\sum_x x\sum_y P\{X=x,Y=y\}=\sum_x x P\{X=x\}=E[X]$$

and the result is proved.

One way to understand **Equation (5.2)** is to interpret it as follows: To calculate E[X], we may take a weighted average of the conditional expected value of X given that Y=y, each of the terms E[X|Y=y] being weighted by the probability of the event on which it is conditioned. (Of what does this remind you?) This is an extremely useful result that often enables us to compute expectations easily by first conditioning on some appropriate random variable. The following examples illustrate its use.

Example 5c

A miner is trapped in a mine containing 3 doors. The first door leads to a tunnel that will take him to safety after 3 hours of travel. The second door leads to a tunnel that will return him to the mine after 5 hours of travel. The third door leads to a tunnel that will return him to the mine after 7 hours. If we assume that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until he reaches safety?

Solution

Let X denote the amount of time (in hours) until the miner reaches safety, and let Y denote the door he initially chooses. Now,

$$E[X] = E[X|Y=1]P\{Y=1\}+E[X|Y=2]P\{Y=2\}+E[X|Y=3]P\{Y=3\} = \frac{1}{3}(E[X|Y=1]+E[X|Y=2]+E[X|Y=3])$$

However,

$$E[X|Y=1]=3E[X|Y=2]=5+E[X]E[X|Y=3]=7+E[X] \quad (5.3)$$

To understand why **Equation (5.3)** is correct, consider, for instance,E[X|Y=2] and reason as follows: If the miner chooses the second door, he spends 5 hours in the tunnel and then returns to his cell. But once he returns to his cell, the problem is as before; thus, his expected additional time until safety is just E[X]. Hence,E[X|Y=2]=5+E[X]. The argument behind the other equalities in **Equation (5.3)** is similar. Hence,

$$E[X]=13(3+5+E[X]+7+E[X])$$

or

$$E[X]=15$$

Example 5d Expectation of a sum of a random number of random variables

Suppose that the number of people entering a department store on a given day is a random variable with mean 50. Suppose further that the amounts of money spent by these customers are independent random variables having a common mean of $8. Finally, suppose also that the amount of money spent by a customer is also independent of the total number of customers who enter the store. What is the expected amount of money spent in the store on a given day?

Solution

If we let N denote the number of customers who enter the store and Xi the amount spent by the ith such customer, then the total amount of money spent can be expressed as $\sum_{i=1}^{N} X_i$. Now,

$$E\left[\sum_{1}^{N} X_i\right]=E\left[E\left[\sum_{1}^{N} X_i|N\right]\right]$$

But

$$E\left[\sum_{1}^{N} X_i|N=n\right] = E\left[\sum_{1}^{n} X_i|N=n\right] = E\left[\sum_{1}^{n} X_i\right] \text{ by the independence of the } X_i \text{ and } N = nE[X] \text{ where } E[X]=E[X_i]$$

which implies that

$$E\left[\sum_{1}^{N} X_i|N\right]=NE[X]$$

Thus,

$$E\left[\sum_{i=1}^{N} X_i\right]=E[NE[X]]=E[N]E[X]$$

Hence, in our example, the expected amount of money spent in the store is 50×$8, or $400.

Example 5e

The game of craps is begun by rolling an ordinary pair of dice. If the sum of the dice is 2, 3, or 12, the player loses. If it is 7 or 11, the player wins. If it is any other number i, the player continues to roll the dice until the sum is either 7 or i. If it is 7, the player loses; if it is i, the player wins. Let R denote the number of rolls of the dice in a game of craps. Find -5pt

a. E[R];
b. E[R| player wins];
c. E[R| player loses].

Solution

If we let Pi denote the probability that the sum of the dice is i, then

$$P_i=P_{14-i}=i-136, \quad i=2,\ldots,7$$

To compute E[R], we condition on S, the initial sum, giving

$$E[R]=\sum_{i=2}^{12}E[R|S=i]P_i$$

However,

$$E[R|S=i]=\begin{cases}1, & \text{if } i=2,3,7,11,12\\ 1+P_i+P_7, & \text{otherwise}\end{cases}$$

The preceding equation follows because if the sum is a value i that does not end the game, then the dice will continue to be rolled until the sum is either i or 7, and the number of rolls until this occurs is a geometric random variable with parameter P_i+P_7. Therefore,

$$E[R]=1+\sum_{i=4}^{6}\frac{P_i}{P_i+P_7}+\sum_{i=8}^{10}\frac{P_i}{P_i+P_7}=1+2(3/9+4/10+5/11)=3.376$$

To determine E[R| win], let us start by determining p, the probability that the player wins. Conditioning on S yields

$$p=\sum_{i=2}^{12}P\{\text{win}|S=i\}P_i=P_7+P_{11}+\sum_{i=4}^{6}\frac{P_i}{P_i+P_7}P_i+P_7P_i+\sum_{i=8}^{10}\frac{P_i}{P_i+P_7}P_i+P_7P_i=0.493$$

where the preceding uses the fact that the probability of obtaining a sum of i before one of 7 is $P_i/(P_i+P_7)$. Now, let us determine the conditional probability mass function of S, given that the player wins. Letting $Q_i=P\{S=i|\text{ win}\}$, we have

$$Q_2=Q_3=Q_{12}=0,\quad Q_7=P_7/p,\quad Q_{11}=P_{11}/p$$

and, for i=4,5,6,8,9,10,

$$Q_i=P\{S=i,\text{ win}\}P\{\text{win}\}=P_iP\{\text{win}|S=i\}p=\frac{P_i}{2}\frac{p}{(P_i+P_7)}$$

Now, conditioning on the initial sum gives

$$E[R|\text{ win }]=\sum_i E[R|\text{ win },S=i]Q_i$$

However, as was noted in **Example 2j** of **Chapter 6**, given that the initial sum is i, the number of additional rolls needed and the outcome (whether a win or a loss) are independent. (This is easily seen by first noting that conditional on an initial sum of i, the outcome is independent of the number of additional dice rolls needed and then using the symmetry property of independence, which states that if event A is independent of event B, then event B is independent of event A.) Therefore,

$$E[R|\text{win }]=\sum_i E[R|S=i]Q_i=1+\sum_{i=4}^{6}\frac{Q_iP_i}{P_i+P_7}+\sum_{i=8}^{10}\frac{Q_iP_i}{P_i+P_7}=2.938$$

Although we could determine E[R| player loses] exactly as we did E[R| player wins], it is easier to use

$$E[R]=E[R|\text{ win }]p+E[R|\text{ lose }](1-p)$$

implying that

$$E[R|\text{ lose }]=\frac{E[R]-E[R|\text{ win }]p}{1-p}=3.801$$

Example 5f

As defined in **Example 5d** of **Chapter 6**, the bivariate normal joint density function of the random variables X and Y is

$$f(x,y)=\frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}\exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2+\left(\frac{y-\mu_y}{\sigma_y}\right)^2-2\rho\frac{(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y}\right]\right\}$$

We will now show that ρ is the correlation between X and Y. As shown in **Example 5c**, $\mu_x=E[X],\sigma_x^2=\text{Var}(X)$, and $\mu_y=E[Y],\sigma_y^2=\text{Var}(Y)$. Consequently,

$$\text{Corr}(X,Y)=\frac{\text{Cov}(X,Y)}{\sigma_x\sigma_y}=\frac{E[XY]-\mu_x\mu_y}{\sigma_x\sigma_y}$$

To determine E[XY], we condition on Y. That is, we use the identity

$$E[XY]=E[E[XY|Y]]$$

Recalling from **Example 5d** that the conditional distribution of X given that Y=y is normal with mean μx+ρσxσy(y−μy), we see that

$$E[XY|Y=y]=E[Xy|Y=y]=yE[X|Y=y]=y[\mu x+\rho\sigma x\sigma y(y-\mu y)]=y\mu x+\rho\sigma x\sigma y(y2-\mu yy)$$

Consequently,

$$E[XY|Y]=Y\mu x+\rho\sigma x\sigma y(Y2-\mu yY)$$

implying that

$$E[XY]=E[Y\mu x+\rho\sigma x\sigma y(Y2-\mu yY)]=\mu xE[Y]+\rho\sigma xE[Y2-\mu yY]=\mu x\mu y+\rho\sigma x\sigma y(E[Y2]-\mu 2y)=\mu x\mu y+\rho\sigma x\sigma y \text{ Var } (Y)=\mu x\mu y+\rho\sigma x\sigma y$$

Therefore,

$$\text{Corr } (X,Y)=\rho\sigma x\sigma y\sigma x\sigma y=\rho$$

Sometimes E[X] is easy to compute, and we use the conditioning identity to compute a conditional expected value. This approach is illustrated by our next example.

Example 5g

Consider n independent trials, each of which results in one of the outcomes 1,…,k, with respective probabilities p1,…,pk,∑i=1kpi=1. Let Ni denote the number of trials that result in outcome i,i=1,…,k. For i≠j, find

(a) $E[Nj|Ni>0]$ and (b) $E[Nj|Ni>1]$

Solution

To solve (a), let

$$I=\begin{cases}0, & \text{if } Ni=01,\\ 1, & \text{if } Ni>0\end{cases}$$

Then

$$E[Nj]=E[Nj|I=0]P\{I=0\}+E[Nj|I=1]P\{I=1\}$$

or, equivalently,

$$E[Nj]=E[Nj|Ni=0]P\{Ni=0\}+E[Nj|Ni>0]P\{Ni>0\}$$

Now, the unconditional distribution of Nj is binomial with parameters n,pj. Also, given that Ni=r, each of the n−r trials that does not result in outcome i will, independently, result in outcome j with probability P(j| not i)=pj1−pi. Consequently, the conditional distribution of Nj, given that Ni=r, is binomial with parameters n−r,pj1−pi. (For a more detailed argument for this conclusion, see **Example 4c** of **Chapter 6**.) Because P{Ni=0}=(1−pi)n, the preceding equation yields

$$npj=npj1-pi(1-pi)n+E[Nj|Ni>0](1-(1-pi)n)$$

giving the result

$$E[Nj|Ni>0]=npj\ 1-(1-pi)n-11-(1-pi)n$$

We can solve part (b) in a similar manner. Let

$$J=\begin{cases}0, & \text{if } Ni=01,\\ 1, & \text{if } Ni=12,\\ 2, & \text{if } Ni>1\end{cases}$$

Then

$$E[Nj]=E[Nj|J=0]P\{J=0\}+E[Nj|J=1]P\{J=1\}\ +E[Nj|J=2]P\{J=2\}$$

or, equivalently,

$$E[N_j]=E[N_j|N_i=0]P\{N_i=0\}+E[N_j|N_i=1]P\{N_i=1\} \quad +E[N_j|N_i>1]P\{N_i>1\}$$

This equation yields

$$np_j=np_j1-p_i(1-p_i)n+(n-1)p_j1-p_inp_i(1-p_i)n-1 \quad +E[N_j|N_i>1](1-(1-p_i)n-np_i(1-p_i)n-1)$$

giving the result

$$E[N_j|N_i>1]=np_j[1-(1-p_i)n-1-(n-1)p_i(1-p_i)n-2]1-(1-p_i)n-np_i(1-p_i)n-1$$

It is also possible to obtain the variance of a random variable by conditioning. We illustrate this approach by the following example.

Example 5h Variance of the geometric distribution

Independent trials, each resulting in a success with probability p, are successively performed. Let N be the time of the first success. Find Var (N).

Solution

Let Y=1 if the first trial results in a success and Y=0 otherwise. Now,

$$\text{Var}(N)=E[N^2]-(E[N])2$$

To calculate E[N2], we condition on Y as follows:

$$E[N^2]=E[E[N^2|Y]]$$

However,

$$E[N^2|Y=1]=1E[N^2|Y=0]=E[(1+N)2]$$

These two equations follow because, on the one hand, if the first trial results in a success, then, clearly,N=1; thus,N2=1. On the other hand, if the first trial results in a failure, then the total number of trials necessary for the first success will have the same distribution as 1 (the first trial that results in failure) plus the necessary number of additional trials. Since the latter quantity has the same distribution as N, we obtain $E[N^2|Y=0]=E[(1+N)2]$Hence,

$$E[N^2]=E[N^2|Y=1]P\{Y=1\}+E[N^2|Y=0]P\{Y=0\}=p+(1-p)E[(1+N)2]=1+(1-p)E[2N+N^2]$$

However, as was shown in **Example 8b** of **Chapter 4**,E[N]=1/p; therefore,

$$E[N^2]=1+2(1-p)p+(1-p)E[N^2]$$

or

$$E[N^2]=2-pp2$$

Consequently,

$$\text{Var}(N)=E[N^2]-(E[N])2=2-pp2-(1p)2=1-pp2$$

Example 5i

Consider a gambling situation in which there are r players, with player i initially having ni units, ni>0, i=1, … , r. At each stage, two of the players are chosen to play a game, with the winner of the game receiving 1 unit from the loser. Any player whose fortune drops to 0 is eliminated, and this continues until a single player has all $n\equiv\sum i=1rni$ units, with that player designated as the victor. Assuming that the results of successive games are independent and that each game is equally likely to be won by either of its two players, find the average number of stages until one of the players has all n units.

Solution

To find the expected number of stages played, suppose first that there are only 2 players, with players 1 and 2 initially having j and n−j units, respectively. Let Xj denote the number of stages that will be played, and let mj=E[Xj].Then, for j=1,...,n−1,

Xj=1+Aj

where Aj is the additional number of stages needed beyond the first stage. Taking expectations gives

mj=1+E[Aj]

Conditioning on the result of the first stage then yields

mj=1+E[Aj|1 wins first stage]1/2+E[Aj|2 wins first stage]1/2

Now, if player 1 wins at the first stage, then the situation from that point on is exactly the same as in a problem that supposes that player *1* starts with j+1 and player *2* with n−(j+1) units. Consequently,

E[Aj|1 wins first stage]=mj+1

and, analogously,

E[Aj|2 wins first stage]=mj−1

Thus,

mj=1+12 mj+1+12 mj−1

or, equivalently,

$$m_{j+1}=2m_j-m_{j-1}-2, \quad j=1,\ldots,n-1 \tag{5.4}$$

Using that m0=0, the preceding equation yields

m2=2m1−2m3=2m2−m1−2=3m1−6=3(m1−2)m4=2m3−m2−2=4m1−12=4(m1−3)

suggesting that

$$m_i=i(m_1-i+1), \qquad i=1,\ldots,n \tag{5.5}$$

To prove the preceding equality, we use mathematical induction. Since we've already shown the equation to be true for i=1,2, we take as the induction hypothesis that it is true whenever i≤j<n. Now we must prove that it is true for j+1. Using **Equation (5.4)** yields

mj+1=2mj−mj−1−2=2j(m1−j+1)−(j−1)(m1−j+2)−2 (by the induction hypothesis)=(j+1)m1−2j2+2j+j2−3j+2−2=(j+1)m1−j2−j =(j+1)(m1−j)

which completes the induction proof of **(5.5)**. Letting i=n in **(5.5)**, and using that mn=0, now yields that

m1=n−1

which, again using **(5.5)**, gives the result

mi=i(n−i)

Thus, the mean number of games played when there are only *2* players with initial amounts i and n−i is the product of their initial amounts. Because both players play all stages, this is also the mean number of stages involving player 1.

Now let us return to the problem involving r players with initial amounts ni,i=1,...∑i=1r ni=n. Let X denote the number of stages needed to obtain a victor, and let Xi denote the number of stages involving player i. Now, from the point of view of player i, starting with ni, he will continue to play stages, independently being equally likely to win or lose each one, until his fortune is either n or 0. Thus, the number of stages he plays is exactly

the same as when he has a single opponent with an initial fortune of n−ni. Consequently, by the preceding result, it follows that

$$E[X_i]=n_i(n-n_i)$$

so

$$E[\sum_{i=1}^{r}X_i]=\sum_{i=1}^{r}n_i(n-n_i)=n^2-\sum_{i=1}^{r}n_i^2$$

But because each stage involves two players,

$$X=12\sum_{i=1}^{r}X_i$$

Taking expectations now yields

$$E[X]=12(n^2-\sum_{i=1}^{r}n_i^2)$$

It is interesting to note that while our argument shows that the mean number of stages does not depend on the manner in which the teams are selected at each stage, the same is not true for the distribution of the number of stages. To see this, suppose r=3, n1=n2=1, and n3=2. If players *1* and *2* are chosen in the first stage, then it will take at least three stages to determine a winner, whereas if player 3 is in the first stage, then it is possible for there to be only two stages.

In our next example, we use conditioning to verify a result previously noted in **Section 6.3.1**: that the expected number of uniform (0,1) random variables that need to be added for their sum to exceed 1 is equal to e.

Example 5j

Let U1,U2,… be a sequence of independent uniform (0, 1) random variables. Find E[N] when

$$N=\min\{n: \sum_{i=1}^{n}U_i>1\}$$

Solution

We will find E[N] by obtaining a more general result. For $x\in[0,1]$, let

$$N(x)=\min\{n: \sum_{i=1}^{n}U_i>x\}$$

and set

$$m(x)=E[N(X)]$$

That is,N(x) is the number of uniform (0, 1) random variables we must add until their sum exceeds x, and m(x) is its expected value. We will now derive an equation for m(x) by conditioning on U1. This gives, from **Equation (5.1b)**,

$$m(x)=\int_0^1 E[N(X)|U_1=y]\,dy \tag{5.6}$$

Now,

$$E[N(X)|U_1=y]=\begin{cases}1 & \text{if } y>x\\ 1+m(x-y) & \text{if } y\le x\end{cases} \tag{5.7}$$

The preceding formula is obviously true when y>x. It is also true when y≤x, since, if the first uniform value is y, then, at that point, the remaining number of uniform random variables needed is the same as if we were just starting and were going to add uniform random variables until their sum exceeded x−y. Substituting **Equation (5.7)** into **Equation (5.6)** gives

$$m(x)=1+\int_0^x m(x-y)\,dy=1+\int_0^x m(u)\,du \qquad \text{by letting } u=x-y$$

Differentiating the preceding equation yields

$$m'(x)=m(X)$$

or, equivalently,

$$m'(x)m(x)=1$$

Integrating this equation gives

$$\log[m(X)]=x+c$$

or

$$m(x)=kex$$

Since $m(0)=1$, it follows that $k=1$, so we obtain

$$m(x)=ex$$

Therefore,$m(1)$, the expected number of uniform (0, 1) random variables that need to be added until their sum exceeds 1, is equal to e.

7.5.3 Computing probabilities by conditioning

Not only can we obtain expectations by first conditioning on an appropriate random variable, but we can also use this approach to compute probabilities. To see this, let A denote an arbitrary event, and define the indicator random variable X by

$$X=\{\ 1\text{if }\ A\ \text{occurs}0\text{if }\ A\ \text{does not occur}$$

It follows from the definition of X that

$$E[X]=P(A)E[X|Y=y]=P(A|Y=y)\quad \text{for any random variable }\ Y$$

Therefore, from **Equations (5.1a**) and **(5.1b)**, we obtain

$$P(A)=\sum yP(A|Y=y)P(Y=y)\quad \text{if }\ Y\ \text{is discrete}=\int -\infty\infty P(A|Y=y)fY(y)dy\quad \text{if }\ Y\ \text{is continuous} \tag{5.8}$$

Note that if Y is a discrete random variable taking on one of the values $y1,\ldots,yn$, then by defining the events $Bi,i=1,\ldots,n$, by $Bi=\{Y=yi\}$, **Equation (5.8)** reduces to the familiar equation

$$P(A)=\sum i=1nP(A|Bi)P(Bi)$$

where $B1,\ldots,Bn$ are mutually exclusive events whose union is the sample space.

Example 5k The best-prize problem

Suppose that we are to be presented with n distinct prizes, in sequence. After being presented with a prize, we must immediately decide whether to accept it or to reject it and consider the next prize. The only information we are given when deciding whether to accept a prize is the relative rank of that prize compared to ones already seen. That is, for instance, when the fifth prize is presented, we learn how it compares with the four prizes we've already seen. Suppose that once a prize is rejected, it is lost, and that our objective is to maximize the probability of obtaining the best prize. Assuming that all n! orderings of the prizes are equally likely, how well can we do?

Solution

Rather surprisingly, we can do quite well. To see this, fix a value $k,0\le k<n$, and consider the strategy that rejects the first k prizes and then accepts the first one that is better than all of those first k. Let $Pk(\ \text{best}\ )$ denote the probability that the best prize is selected when this strategy is employed. To compute this probability, condition on X, the position of the best prize. This gives

$$Pk(\ \text{best}\ )=\sum i=1nPk(\ \text{best}\ |X=i)P(X=i)$$

$$=1n\sum i=1nPk(\ \text{best}\ |X=i)$$

Now, on the one hand, if the overall best prize is among the first k, then no prize is ever selected under the strategy considered. That is,

$$P_k(\text{ best }|X=i)=0 \quad \text{if } i\leq k$$

On the other hand, if the best prize is in position i, where i>k, then the best prize will be selected if the best of the first i−1 prizes is among the first k(for then none of the prizes in positions k+1,k+2,…,i−1 would be selected). But, conditional on the best prize being in position i, it is easy to verify that all possible orderings of the other prizes remain equally likely, which implies that each of the first i−1 prizes is equally likely to be the best of that batch. Hence, we have

$$P_k(\text{ best }|X=i)=P\{\text{best of first } i-1 \text{ is among the first } k|X=i\}=\frac{k}{i-1} \quad \text{if } i>k$$

From the preceding, we obtain

$$P_k(\text{best })=\frac{k}{n}\sum_{i=k+1}^{n}\frac{1}{i-1}\approx\frac{k}{n}\int_{k+1}^{n}\frac{1}{x-1}\,dx=\frac{k}{n}\log\left(\frac{n-1}{k}\right)\approx\frac{k}{n}\log\left(\frac{n}{k}\right)$$

Now, if we consider the function

$$g(x)=x\log\left(\frac{n}{x}\right)$$

then

$$g'(x)=\log\left(\frac{n}{x}\right)-1$$

so

$$g'(x)=0\Rightarrow\log\left(\frac{n}{x}\right)=1\Rightarrow x=\frac{n}{e}$$

Thus, since $P_k(\text{ best })\approx g(k)$, we see that the best strategy of the type considered is to let the first *n/e* prizes go by and then accept the first one to appear that is better than all of those. In addition, since $g(n/e)=1/e$, the probability that this strategy selects the best prize is approximately $1/e\approx.36788$.

Remark Most people are quite surprised by the size of the probability of obtaining the best prize, thinking that this probability would be close to 0 when n is large. However, even without going through the calculations, a little thought reveals that the probability of obtaining the best prize can be made reasonably large. Consider the strategy of letting half of the prizes go by and then selecting the first one to appear that is better than all of those. The probability that a prize is actually selected is the probability that the overall best is among the second half, and this is $\frac{1}{2}$. In addition, given that a prize is selected, at the time of selection that prize would have been the best of more than n/2 prizes to have appeared and would thus have probability of at least $\frac{1}{2}$ of being the overall best. Hence, the strategy of letting the first half of all prizes go by and then accepting the first one that is better than all of those prizes has a probability greater than $\frac{1}{4}$ of obtaining the best prize.

Example 5l

Let U be a uniform random variable on (0, 1), and suppose that the conditional distribution of X, given that U=p, is binomial with parameters n and p. Find the probability mass function of X.

Solution

Conditioning on the value of U gives

$$P\{X=i\}=\int_0^1 P\{X=i|U=p\}f_U(p)\,dp=\int_0^1 P\{X=i|U=p\}\,dp=\frac{n!}{i!(n-i)!}\int_0^1 p^i(1-p)^{n-i}\,dp$$

Now, it can be shown (a probabilistic proof is given in **Section 6.6)** that

$$\int_0^1 p^i(1-p)^{n-i}dp=\frac{i!(n-i)!}{(n+1)!}$$

Hence, we obtain

$$P\{X=i\}=\frac{1}{n+1} \quad i=0,\ldots,n$$

That is, we obtain the surprising result that if a coin whose probability of coming up heads is uniformly distributed over (0, 1) is flipped n times, then the number of heads occurring is equally likely to be any of the values 0,…,n.

Because the preceding conditional distribution has such a nice form, it is worth trying to find another argument to enhance our intuition as to why such a result is true. To do so, let U,U1,…,Un be n+1 independent uniform (0, 1) random variables, and let X denote the number of the random variables U1,…,Un that are smaller than U.Since all the random variables U,U1,…,Un have the same distribution, it follows that U is equally likely to be the smallest, or second smallest, or largest of them; so X is equally likely to be any of the values 0,1,…,n. However, given that U=p, the number of the Ui that are less than U is a binomial random variable with parameters n and p, thus establishing our previous result.

Example 5m

A random sample of X balls is chosen from an urn that contains n red and m blue balls. If X is equally likely to be any of the values 1,…,n, find the probability that all the balls in the sample are red.

Solution

Conditioning on X yields

$$P(\text{ all balls are red })=\sum_{i=1}^{n} P(\text{ all balls are red } | X=i) P(X=i)$$

Now, given that the sample is of size i, each of the (n+mi) subsets of size i is equally likely to be the chosen set of balls. As (ni) of these subsets have all red balls, it follows that P{ all balls are red|X=i }=(ni)(n+mi) and thus that

$$P(\text{all balls are red})=\frac{1}{n}\sum_{i=1}^{n}\frac{\binom{n}{i}}{\binom{n+m}{i}}$$

However, though not obvious, it turns out that the preceding can be simplified, and indeed yields the surprising result that

$$P(\text{ all balls are red })=\frac{1}{m+1}, \quad \text{for all } n,m$$

To prove the preceding formula, we will not make use of our earlier result, but rather we will use induction on n. When n=1, the urn contains 1 red and m blue balls and so a random sample of size 1 will be red with probability 1m+1. So, assume the result is true whenever the urn contains n−1 red and m blue balls and a random sample whose size is equally likely to be any of 1,…,n−1 is to be chosen. Now consider the case of n red and m blue balls. Start by conditioning not on the value of X but only on whether or not X=1. This yields

$$P(\text{ all balls are red })=P(\text{ all red } | X=1) P(X=1)+P(\text{ all red } | X>1) P(X>1)=\frac{n}{n+m}\frac{1}{n}+P(\text{ all red } | X>1)\frac{n-1}{n}$$

Now, if X>1 then in order for all balls in the sample to be red, the first one chosen must be red, which occurs with probability nn+m, and then all of the X−1 remaining balls in the sample must be red. But given that the first ball chosen is red, the remaining X−1 balls will be randomly selected from an urn containing n−1 red and m blue balls. As X−1, given that X>1, is equally likely to be any of the values 1,…,n−1, it follows by the induction hypothesis that

$$P(\text{ all balls are red } | X>1)=\frac{n}{n+m}\frac{1}{m+1}$$

Thus,

$$P(\text{ all balls are red })=\frac{1}{n+m}+\frac{n}{n+m}\frac{1}{m+1}+\frac{n-1}{n}=\frac{1}{n+m}\left(1+\frac{n-1}{m+1}\right)=\frac{1}{m+1}$$

Example 5n

Suppose that X and Y are independent continuous random variables having densities fX and fY, respectively. Compute P{X<Y}.

Solution

Conditioning on the value of Y yields

$$P\{X<Y\}=\int_{-\infty}^{\infty} P\{X<Y|Y=y\} f_Y(y)\,dy=\int_{-\infty}^{\infty} P\{X<y|Y=y\} f_Y(y)\,dy=\int_{-\infty}^{\infty} P\{X<y\} f_Y(y)\,dy \quad \text{by independence} =\int_{-\infty}^{\infty} F_X(y) f_Y(y)\,dy$$

where

$$F_X(y)=\int_{-\infty}^{y} f_X(x)\,dx$$

Example 5o

Suppose that X and Y are independent continuous random variables. Find the distribution function and the density function of X+Y.

Solution

By conditioning on the value of Y, we obtain

$$P\{X+Y<a\}=\int_{-\infty}^{\infty} P\{X+Y<a|Y=y\}\, f_Y(y)\,dy=\int_{-\infty}^{\infty} P\{X+y<a|Y=y\}\, f_Y(y)\,dy=\int_{-\infty}^{\infty} P\{X<a-y\}\, f_Y(y)\,dy=\int_{-\infty}^{\infty} F_X(a-y)\, f_Y(y)\,dy$$

Differentiation yields the density function of X+Y:

$$f_{X+Y}(a)=\frac{d}{da}\int_{-\infty}^{\infty} F_X(a-y)\, f_Y(y)dy=\int_{-\infty}^{\infty}\frac{d}{da}F_X(a-y)\, f_Y(y)dy=\int_{-\infty}^{\infty} f_X(a-y)\, f_Y(y)dy$$

7.5.4 Conditional variance

Just as we have defined the conditional expectation of X given the value of Y, we can also define the conditional variance of X given that Y=y:

$$\text{Var}(X|Y)\equiv E[(X-E[X|Y])^2|Y]$$

That is, Var (X|Y) is equal to the (conditional) expected square of the difference between X and its (conditional) mean when the value of Y is given. In other words, Var (X|Y) is exactly analogous to the usual definition of variance, but now all expectations are conditional on the fact that Y is known.

There is a very useful relationship between Var (X), the unconditional variance of X, and Var (X|Y), the conditional variance of X given Y, that can often be applied to compute Var (X). To obtain this relationship, note first that by the same reasoning that yields Var (X)=E[X2]−(E[X] stretchy='false')2, we have

$$\text{Var}(X|Y)=E[X^2|Y]-(E[X|Y])^2$$

so

$$E[\text{Var}(X|Y)]=E[E[X^2|Y]]-E[(E[X|Y])^2]=E[X^2]-E[(E[X|Y])^2] \tag{5.9}$$

Also, since E[E[X|Y]]=E[X], we have

$$\text{Var}(E[X|Y])=E[(E[X|Y])^2]-(E[X])^2 \tag{5.10}$$

Hence, by adding **Equations (5.9)** and **(5.10)**, we arrive at the following proposition.

Proposition 5.2 The Conditional Variance Formula

$$\text{Var}(X)=E[\text{Var}(X|Y)]+\text{Var}(E[X|Y])$$

Example 5p

Suppose that by any time t the number of people who have arrived at a train depot is a Poisson random variable with mean λt. If the initial train arrives at the depot at a time (independent of when the passengers arrive) that is uniformly distributed over (0,T), what are the mean and variance of the number of passengers who enter the train?

Solution

For each t≥0, let N(t) denote the number of arrivals by t, and let Y denote the time at which the train arrives. The random variable of interest is then N(Y). Conditioning on Y gives

$E[N(Y)|Y=t]=E[N(t)|Y=t]=E[N(t)]$ by the independence of Y and $N(t)=\lambda t$ since $N(t)$ is Poisson with mean λt

Hence,

$E[N(Y)|Y]=\lambda Y$

so taking expectations gives

$E[N(Y)]=\lambda E[Y]=\lambda T2$

To obtain $Var(N(Y))$, we use the conditional variance formula:

$Var(N(Y)|Y=t)=Var(N(t)|Y=t)=Var[N(t)]$ by independence$=\lambda t$

Thus,

$Var(N(Y)|Y)=\lambda Y E[N(Y)|Y]=\lambda Y$

Hence, from the conditional variance formula,

$Var(N(Y))=E[\lambda Y]+Var(\lambda Y)=\lambda T2+\lambda 2T212$

where we have used the fact that $Var(Y)=T2/12$.

Example 5q Variance of a sum of a random number of random variables

Let $X1,X2,\ldots$ be a sequence of independent and identically distributed random variables, and let N be a nonnegative integer-valued random variable that is independent of the sequence $Xi, i\geq 1$. To compute $Var(\sum i=1NXi)$, we condition on N:

$E[\sum i=1 N X i |N] = NE[X] \; Var(\sum i=1 N X i |N) = N \, Var(X)$

The preceding result follows because, given N, $\sum i=1NXi$ is just the sum of a fixed number of independent random variables, so its expectation and variance are just the sums of the individual means and variances, respectively. Hence, from the conditional variance formula,

$Var(\sum i=1NXi)=E[N] \; Var(X)+(E[X])2 \; Var(N)$

$$E[X]=E[E[X|Y]]$$

In the case of discrete random variables, this equation reduces to the identity

$$E[X]=\sum yE[X|Y=y]P\{Y=y\}$$

and, in the continuous case, to

$$E[X]=\int -\infty\infty E[X|Y=y]fY(y)dy$$

The preceding equations can often be applied to obtain E[X] by first “conditioning” on the value of some other random variable Y. In addition, since, for any event A, P(A)=E[IA], where IA is 1 if A occurs and is 0 otherwise, we can use the same equations to compute probabilities.

The conditional variance of X, given that Y=y, is defined by

$$Var(X|Y=y)=E[(X-E[X|Y=y])2|Y=y]$$

Let Var(X|Y) be that function of Y whose value at Y=y is Var(X|Y=y). The following is known as the *conditional variance formula*:

$$Var(X)=E[\ Var(X|Y)]+\ Var(E[X|Y])$$

Suppose that the random variable X is to be observed and, on the basis of its value, one must then predict the value of the random variable Y. In such a situation, it turns out that among all predictors, E[Y|X] has the smallest expectation of the square of the difference between it and Y.

The *moment generating function* of the random variable X is defined by

$$M(t)=E[etX]$$

The moments of X can be obtained by successively differentiating M(t) and then evaluating the resulting quantity at t=0. Specifically, we have

$$E[Xn]=dndtnM(t)|t=0 \quad n=1,2,\ldots$$

Two useful results concerning moment generating functions are, first, that the moment generating function uniquely determines the distribution function of the random variable and, second, that the moment generating function of the sum of independent random variables is equal to the product of their moment generating functions. These results lead to simple proofs that the sum of independent normal (Poisson, gamma) random variables remains a normal (Poisson, gamma) random variable.

If X1,…,Xm are all linear combinations of a finite set of independent standard normal random variables, then they are said to have a *multivariate normal distribution*. Their joint distribution is specified by the values of E[Xi],Cov(Xi,Xj),i,j=1,…,m.

If X1,…,Xn are independent and identically distributed normal random variables, then their *sample mean*

$$X\bar{}=\sum i=1nXin$$

and their *sample variance*

$$S2=\sum i=1n(Xi-X\bar{})2n-1$$

are independent. The sample mean $\bar{X}$ is a normal random variable with mean μ and variance σ^2/n; the random variable $(n-1)S^2/\sigma^2$ is a chi-squared random variable with $n-1$ degrees of freedom.

Problems

7.1 A player throws a fair die and simultaneously flips a fair coin. If the coin lands heads, then she wins twice, and if tails, then she wins one-half of the value that appears on the die. Determine her expected winnings.

7.2 The game of Clue involves 6 suspects, 6 weapons, and 9 rooms. One of each is randomly chosen and the object of the game is to guess the chosen three.

a. How many solutions are possible?
   In one version of the game, the selection is made and then each of the players is randomly given three of the remaining cards. Let S, W, and R be, respectively, the numbers of suspects, weapons, and rooms in the set of three cards given to a specified player. Also, let X denote the number of solutions that are possible after that player observes his or her three cards.
b. Express X in terms of S, W, and R.
c. Find E[X].

7.3 Gambles are independent, and each one results in the player being equally likely to win or lose 1 unit. Let W denote the net winnings of a gambler whose strategy is to stop gambling immediately after his first win. Find

a. P{W>0}
b. P{W<0}
c. E[W]

7.4. If X and Y have joint density function

$f_{X,Y}(x,y)=\{ 1/y, \text{if } 0<y<1,\ 0<x<y \text{ 0, otherwise}$

find

a. E[XY]
b. E[X]
c. E[Y]

7.5 The county hospital is located at the center of a square whose sides are 3 miles wide. If an accident occurs within this square, then the hospital sends out an ambulance. The road network is rectangular, so the travel distance from the hospital, whose coordinates are (0, 0), to the point (x, y) is |x|+|y|. If an accident occurs at a point that is uniformly distributed in the square, find the expected travel distance of the ambulance.

7.6 A fair die is rolled 10 times. Calculate the expected sum of the 10 rolls.

7.7 Suppose that A and B each randomly and independently choose 3 of 10 objects. Find the expected number of objects

a. chosen by both A and B;
b. not chosen by either A or B;
c. chosen by exactly one of A and B.

7.8. N people arrive separately to a professional dinner. Upon arrival, each person looks to see if he or she has any friends among those present. That person then sits either at the table of a friend or at an unoccupied table if none of those present is a friend. Assuming that each of the (N2) pairs of people is, independently, a pair of friends with probability p, find the expected number of occupied tables.

Hint: Let Xi equal 1 or 0, depending on whether the ith arrival sits at a previously unoccupied table.

7.9. A total of n balls, numbered 1 through n, are put into n urns, also numbered 1 through n in such a way that ball i is equally likely to go into any of the urns 1,2,…,i. Find

a. the expected number of urns that are empty;
b. the probability that none of the urns is empty.

7.10 Consider 3 trials, each having the same probability of success. Let X denote the total number of successes in these trials. If E[X]=1.8, what is

a. the largest possible value of P{X=3}?
b. the smallest possible value of P{X=3}?

In both cases, construct a probability scenario that results in P{X=3} having the stated value.
Hint: For part (b), you might start by letting U be a uniform random variable on (0, 1) and then defining the trials in terms of the value of U.
7.11 Consider n independent flips of a coin having probability p of landing on heads. Say that a changeover occurs whenever an outcome differs from the one preceding it. For instance, if n=5 and the outcome is HHTHT, then there are 3 changeovers. Find the expected number of changeovers.
Hint: Express the number of changeovers as the sum of n−1 Bernoulli random variables.
7.12. A group of n men and n women is lined up at random.

a. Find the expected number of men who have a woman next to them.
b. Repeat part (a), but now assuming that the group is randomly seated at a round table.

7.13. A set of 1000 cards numbered 1 through 1000 is randomly distributed among 1000 people with each receiving one card. Compute the expected number of cards that are given to people whose age matches the number on the card.
7.14. An urn has m black balls. At each stage, a black ball is removed and a new ball that is black with probability p and white with probability 1−p is put in its place. Find the expected number of stages needed until there are no more black balls in the urn.
NOTE: The preceding has possible applications to understanding the AIDS disease. Part of the body's immune system consists of a certain class of cells known as T-cells. There are 2 types of T-cells, called CD4 and CD8. Now, while the total number of T-cells in AIDS sufferers is (at least in the early stages of the disease) the same as that in healthy individuals, it has recently been discovered that the mix of CD4 and CD8 T-cells is different. Roughly 60 percent of the T-cells of a healthy person are of the CD4 type, whereas the percentage of the T-cells that are of CD4 type appears to decrease continually in AIDS sufferers. A recent model proposes that the HIV virus (the virus that causes AIDS) attacks CD4 cells and that the body's mechanism for replacing killed T-cells does not differentiate between whether the killed T-cell was CD4 or CD8. Instead, it just produces a new T-cell that is CD4 with probability .6 and CD8 with probability .4. However, although this would seem to be a very efficient way of replacing killed T-cells when each one killed is equally likely to be any of the body's T-cells (and thus has probability .6 of being CD4), it has dangerous consequences when facing a virus that targets only the CD4 T-cells.
7.15. In **Example 2h**, say that i and j, i≠j, form a matched pair if i chooses the hat belonging to j and j chooses the hat belonging to i. Find the expected number of matched pairs.
7.16. Let Z be a standard normal random variable, and, for a fixed x, set
X={ Zif Z>x0otherwise

Show that E[X]=12πe−x2/2.
7.17. A deck of n cards numbered 1 through n is thoroughly shuffled so that all possible n! orderings can be assumed to be equally likely. Suppose you are to make n guesses sequentially, where the ith one is a guess of the card in position i. Let N denote the number of correct guesses.

a. If you are not given any information about your earlier guesses, show that for any strategy, E[N]=1.
b. Suppose that after each guess you are shown the card that was in the position in question. What do you think is the best strategy? Show that under this strategy, E[N]=1n+1n−1+⋯+1≈∫1n1x dx=log n

c. Suppose that you are told after each guess whether you are right or wrong. In this case, it can be shown that the strategy that maximizes E[N] is one that keeps on guessing the same card until you are told you are correct and then changes to a new card. For this strategy, show that

$$E[N]=1+\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{n!}\approx e-1$$

Hint: For all parts, express N as the sum of indicator (that is, Bernoulli) random variables.

7.18. Cards from an ordinary deck of 52 playing cards are turned face up one at a time. If the 1st card is an ace, or the 2nd a deuce, or the 3rd a three, or …, or the 13th a king, or the 14 an ace, and so on, we say that a match occurs. Note that we do not require that the (13n+1) card be any particular ace for a match to occur but only that it be an ace. Compute the expected number of matches that occur.

7.19. A certain region is inhabited by r distinct types of a certain species of insect. Each insect caught will, independently of the types of the previous catches, be of type i with probability $P_i, i=1,\ldots,r \quad \sum_{1}^{r} P_i=1$

a. Compute the mean number of insects that are caught before the first type 1 catch.
b. Compute the mean number of types of insects that are caught before the first type 1 catch.

7.20 In an urn containing n balls, the ith ball has weight $W(i), i=1,\ldots,n$. The balls are removed without replacement, one at a time, according to the following rule: At each selection, the probability that a given ball in the urn is chosen is equal to its weight divided by the sum of the weights remaining in the urn. For instance, if at some time $i_1,\ldots,i_r$ is the set of balls remaining in the urn, then the next selection will be i_j with probability $W(i_j)/\sum_{k=1}^{r} W(i_k)$, $j=1,\ldots,r$. Compute the expected number of balls that are withdrawn before ball number 1 is removed.

7.21. For a group of 100 people, compute

a. the expected number of days of the year that are birthdays of exactly 3 people;
b. the expected number of distinct birthdays.

7.22. How many times would you expect to roll a fair die before all 6 sides appeared at least once?

7.23. Urn 1 contains 5 white and 6 black balls, while urn 2 contains 8 white and 10 black balls. Two balls are randomly selected from urn 1 and are put into urn 2. If 3 balls are then randomly selected from urn 2, compute the expected number of white balls in the trio.

Hint: Let $X_i=1$ if the ith white ball initially in urn 1 is one of the three selected, and let $X_i=0$ otherwise. Similarly, let $Y_i=1$ if the ith white ball from urn 2 is one of the three selected, and let $Y_i=0$ otherwise. The number of white balls in the trio can now be written as $\sum_{1}^{5} X_i+\sum_{1}^{8} Y_i$.

7.24. A bottle initially contains m large pills and n small pills. Each day, a patient randomly chooses one of the pills. If a small pill is chosen, then that pill is eaten. If a large pill is chosen, then the pill is broken in two; one part is returned to the bottle (and is now considered a small pill) and the other part is then eaten.

a. Let X denote the number of small pills in the bottle after the last large pill has been chosen and its smaller half returned. Find E[X].
   Hint: Define n+m indicator variables, one for each of the small pills initially present and one for each of the m small pills created when a large one is split in two. Now use the argument of **Example 2m**.
b. Let Y denote the day on which the last large pill is chosen. Find E[Y].
   Hint: What is the relationship between X and Y?

7.25. Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed continuous random variables. Let $N \geq 2$ be such that

$$X_1 \geq X_2 \geq \cdots \geq X_{N-1} < X_N$$

That is, N is the point at which the sequence stops decreasing. Show that E[N]=e.

Hint: First find $P\{N \geq n\}$.

7.26. If $X_1, X_2, \ldots, X_n$ are independent and identically distributed random variables having uniform distributions over (0, 1), find

a. $E[\max(X_1, \ldots, X_n)]$;

b. E[min(X1,…,Xn)].

* 7.27. If 101 items are distributed among 10 boxes, then at least one of the boxes must contain more than 10 items. Use the probabilistic method to prove this result.
* 7.28. The k-of-r-out-of-n circular reliability system, $k \le r \le n$, consists of n components that are arranged in a circular fashion. Each component is either functional or failed, and the system functions if there is no block of r consecutive components of which at least k are failed. Show that there is no way to arrange 47 components, 8 of which are failed, to make a functional 3-of-12-out-of-47 circular system.
* 7.29. There are 4 different types of coupons, the first 2 of which comprise one group and the second 2 another group. Each new coupon obtained is type i with probability pi, where $p1=p2=1/8, p3=p4=3/8$. Find the expected number of coupons that one must obtain to have at least one of

a. all 4 types;
b. all the types of the first group;
c. all the types of the second group;
d. all the types of either group.

* 7.30. If X and Y are independent and identically distributed with mean μ and variance σ2, find $E[(X-Y)2]$

7.31. In **Problem 7.6**, calculate the variance of the sum of the rolls.
7.32. In **Problem 7.9**, compute the variance of the number of empty urns.
7.33. If $E[X]=1$ and $Var(X)=5$, find

a. $E[(2+X)2]$;
b. $Var(4+3X)$.

7.34 If 10 married couples are randomly seated at a round table, compute (a) the expected number and (b) the variance of the number of wives who are seated next to their husbands.
7.35 Cards from an ordinary deck are turned face up one at a time. Compute the expected number of cards that need to be turned face up in order to obtain

a. 2 aces;
b. 5 spades;
c. all 13 hearts.

7.36. Let X be the number of 1's and Y the number of 2's that occur in n rolls of a fair die. Compute Cov(X, Y).
7.37. A die is rolled twice. Let X equal the sum of the outcomes, and let Y equal the first outcome minus the second. Compute Cov(X, Y).
7.38. Suppose X and Y have the following joint probability mass function.
$p(1,1)=.10, p(1,2)=.12, p(1,3)=.16 p(2,1)=.08, p(2,2)=.12, p(2,3)=.10 p(3,1)=.06, p(3,2)=.06, p(3,3)=.20$

a. Find E[X] and E[Y].
b. Find Var(X) and Var(Y).
c. Find Cov(X,Y).
d. Find the correlation between X and Y.

7.39 Suppose that 2 balls are randomly removed from an urn containing n red and m blue balls. Let $Xi=1$ if the ith ball removed is red, and let it be 0 otherwise, i=1,2.

a. Do you think that Cov(X1,X2) is negative, zero, or positive.
b. Validate your answer to part (a).
   Suppose the red balls are numbered, and let Yi equal 1 if red ball number i is removed, and let it be 0 if that ball is not removed.

c. Do you think that Cov(Y1,Y2) is negative, zero, or positive.
d. Validate your answer to part (c).

7.40. The random variables X and Y have a joint density function given by
f(x,y)={ 2e−2x/x0≤x<∞,0≤y≤x0otherwise

Compute Cov(X, Y).
7.41. Let X1,… be independent with common mean μ and common variance σ2, and set Yn=Xn+Xn+1+Xn+2. For j≥0, find Cov(Yn,Yn+j).
7.42. The joint density function of X and Y is given by
f(x,y)=1ye−(y+x/y), x>0,y>0

Find E[X], E[Y], and show that Cov(X,Y)=1.
7.43. A pond contains 100 fish, of which 30 are carp. If 20 fish are caught, what are the mean and variance of the number of carp among the 20? What assumptions are you making?
7.44. A group of 20 people consisting of 10 men and 10 women is randomly arranged into 10 pairs of 2 each. Compute the expectation and variance of the number of pairs that consist of a man and a woman. Now suppose the 20 people consist of 10 married couples. Compute the mean and variance of the number of married couples that are paired together.
7.45. Let X1,X2,…,Xn be independent random variables having an unknown continuous distribution function F, and let Y1,Y2,…,Ym be independent random variables having an unknown continuous distribution function G. Now order those n+m variables, and let
Ii={ 1if the ith smallest of the n+mvariables is from the X sample0otherwise

The random variable R=∑i=1n+miIi is the sum of the ranks of the X sample and is the basis of a standard statistical procedure (called theWilcoxon sum-of-ranks test) for testing whether F and G are identical distributions. This test accepts the hypothesis that F=G when R is neither too large nor too small. Assuming that the hypothesis of equality is in fact correct, compute the mean and variance of R.
Hint: Use the results of **Example 3e**.
7.46. Between two distinct methods for manufacturing certain goods, the quality of goods produced by method i is a continuous random variable having distribution Fi,i=1,2. Suppose that n goods are produced by method 1 and m by method 2. Rank the n+m goods according to quality, and let
Xj={ 1if the ith best was produced frommethod 12otherwise

For the vector X1,X2,…,Xn+m, which consists of n 1's and m 2's, let R denote the number of runs of 1. For instance, if n=5,m=2, and X=1,2,1,1,1,1,2, then R=2. If F1=F2(that is, if the two methods produce identically distributed goods), what are the mean and variance of R?
7.47. If X1,X2,X3, and X4 are (pairwise) uncorrelated random variables, each having mean 0 and variance 1, compute the correlations of
A. X1+X2 and X2+X3;
B. X1+X2 and X3+X4.

7.48. Consider the following dice game, as played at a certain gambling casino: Players 1 and 2 roll a pair of dice in turn. The bank then rolls the dice to determine the outcome according to the following rule: Player i,i=1,2, wins if his roll is strictly greater than the bank's. For i=1,2, let
Ii={ 1if i wins0otherwise

and show that I1 and I2 are positively correlated. Explain why this result was to be expected.
7.49 Consider a graph having n vertices labeled 1,2,…,n, and suppose that, between each of the (n2) pairs of distinct vertices, an edge is independently present with probability p. The degree of vertex i, designated as Di, is the number of edges that have vertex i as one of their vertices.
a. What is the distribution of Di?

b. Find ρ(Di,Dj), the correlation between Di and Dj.

7.50. A fair die is successively rolled. Let X and Y denote, respectively, the number of rolls necessary to obtain a 6 and a 5. Find

a. E[X];
b. E[X|Y=1];
c. E[X|Y=5].

7.51. There are two misshapen coins in a box; their probabilities for landing on heads when they are flipped are, respectively, .4 and .7. One of the coins is to be randomly chosen and flipped 10 times. Given that two of the first three flips landed on heads, what is the conditional expected number of heads in the 10 flips?

7.52. The joint density of X and Y is given by

$f(x,y)=e-x/ye-yy, 0<x<\infty, 0<y<\infty$

Compute E[X2|Y=y].

7.53. The joint density of X and Y is given by

$f(x,y)=e-yy, 0<x<y, 0<y<\infty$

Compute E[X3|Y=y].

7.54. A population is made up of r disjoint subgroups. Let pi denote the proportion of the population that is in subgroup i,i=1,…,r. If the average weight of the members of subgroup i is wi,i=1,…,r, what is the average weight of the members of the population?

7.55. A prisoner is trapped in a cell containing 3 doors. The first door leads to a tunnel that returns him to his cell after 2 days' travel. The second leads to a tunnel that returns him to his cell after 4 days' travel. The third door leads to freedom after 1 day of travel. If it is assumed that the prisoner will always select doors 1, 2, and 3 with respective probabilities .5, .3, and .2, what is the expected number of days until the prisoner reaches freedom?

7.56. Consider the following dice game: A pair of dice is rolled. If the sum is 7, then the game ends and you win 0. If the sum is not 7, then you have the option of either stopping the game and receiving an amount equal to that sum or starting over again. For each value of i,i=2,…,12, find your expected return if you employ the strategy of stopping the first time that a value at least as large as i appears. What value of i leads to the largest expected return?

Hint: Let Xi denote the return when you use the critical value i. To compute E[Xi], condition on the initial sum.

7.57. Ten hunters are waiting for ducks to fly by. When a flock of ducks flies overhead, the hunters fire at the same time, but each chooses his target at random, independently of the others. If each hunter independently hits his target with probability .6, compute the expected number of ducks that are hit. Assume that the number of ducks in a flock is a Poisson random variable with mean 6.

7.58. The number of people who enter an elevator on the ground floor is a Poisson random variable with mean 10. If there are N floors above the ground floor, and if each person is equally likely to get off at any one of the N floors, independently of where the others get off, compute the expected number of stops that the elevator will make before discharging all of its passengers.

7.59. Suppose that the expected number of accidents per week at an industrial plant is 5. Suppose also that the numbers of workers injured in each accident are independent random variables with a common mean of 2.5. If the number of workers injured in each accident is independent of the number of accidents that occur, compute the expected number of workers injured in a week.

7.60. A coin having probability p of coming up heads is continually flipped until both heads and tails have appeared. Find

a. the expected number of flips;
b. the probability that the last flip lands on heads.

7.61. A coin that comes up heads with probability p is continually flipped. Let N be the number of flips

until there have been both at least n heads and at least m tails. Derive an expression for E[N] by conditioning on the number of heads in the first n+m flips.

7.62. There are n+1 participants in a game. Each person independently is a winner with probability p. The winners share a total prize of 1 unit. (For instance, if 4 people win, then each of them receives 14, whereas if there are no winners, then none of the participants receives anything.) Let A denote a specified one of the players, and let X denote the amount that is received by A.

a. Compute the expected total prize shared by the players.
b. Argue that E[X]=1−(1−p)n+1n+1.
c. Compute E[X] by conditioning on whether A is a winner, and conclude that

E[(1+B)−1]=1−(1−p)n+1(n+1)p

when B is a binomial random variable with parameters n and p.

7.63. Each of m+2 players pays 1 unit to a kitty in order to play the following game: A fair coin is to be flipped successively n times, where n is an odd number, and the successive outcomes are noted. Before the n flips, each player writes down a prediction of the outcomes. For instance, if n=3, then a player might write down (H,H,T), which means that he or she predicts that the first flip will land on heads, the second on heads, and the third on tails. After the coins are flipped, the players count their total number of correct predictions. Thus, if the actual outcomes are all heads, then the player who wrote (H,H,T), would have 2 correct predictions. The total kitty of m+2 is then evenly split up among those players having the largest number of correct predictions.

Since each of the coin flips is equally likely to land on either heads or tails, m of the players have decided to make their predictions in a totally random fashion. Specifically, they will each flip one of their own fair coins n times and then use the result as their prediction. However, the final 2 of the players have formed a syndicate and will use the following strategy: One of them will make predictions in the same random fashion as the other m players, but the other one will then predict exactly the opposite of the first. That is, when the randomizing member of the syndicate predicts an H, the other member predicts a T. For instance, if the randomizing member of the syndicate predicts (H,H,T), then the other one predicts (T,T,H).

a. Argue that exactly one of the syndicate members will have more than n/2 correct predictions. (Remember, n is odd.)
b. Let X denote the number of the m nonsyndicate players who have more than n/2 correct predictions. What is the distribution of X?
c. With X as defined in part (b), argue that
E[pay off to the syndicate]=(m+2)× E[1X+1]

d. Use part (c) of **Problem 7.62** to conclude that
E[payoff to the syndicate]=2(m+2)m+1×[1−(12)m+1]

and explicitly compute this number when m=1,2, and 3. Because it can be shown that
2(m+2)m+1[1−(12)m+1]>2

it follows that the syndicate's strategy always gives it a positive expected profit.

7.64. The number of goals that J scores in soccer games that her team wins is Poisson distributed with mean 2, while the number she scores in games that her team loses is Poisson distributed with mean 1. Assume that, independent of earlier results, J's team wins each new game it plays with probability p.

a. Find the expected number of goals that J scores in her team's next game.
b. Find the probability that J scores 6 goals in her next 4 games.

Hint: Would it be useful to know how many of those games were won by J's team.

Suppose J's team has just entered a tournament in which it will continue to play games until it loses. Let X denote the total number of goals scored by J in the tournament. Also, let N be the number of

games her team plays in the tournament.

a. Find E[X].
b. Find P(X=0).
c. Find P(N=3|X=5).

7.65. If the level of infection of a tree is x, then each treatment will independently be successful with probability 1−x. Consider a tree whose infection level is assumed to be the value of a uniform (0,1) random variable.

a. Find the probability that a single treatment will result in a cure.
b. Find the probability that the first two treatments are unsuccessful.
c. Find the probability it will take n treatments for the tree to be cured.

7.66. Let X1,… be independent random variables with the common distribution function F, and suppose they are independent of N, a geometric random variable with parameter p. Let M=max(X1,…,XN).

a. Find P{M≤x} by conditioning on N.
b. Find P{M≤x|N=1}.
c. Find P{M≤x|N>1}.
d. Use (b) and (c) to rederive the probability you found in (a).

7.67. Let U1,U2,… be a sequence of independent uniform (0, 1) random variables. In **Example 5i**, we showed that for 0≤x≤1,E[N(x)]=ex, where
N(x)=min{n:∑i=1nUi>x}

This problem gives another approach to establishing that result.

a. Show by induction on n that for 0<x≤1 and all n≥0,

P{N(x)≥n+1}=xnn!

Hint: First condition on U1 and then use the induction hypothesis.
Use part (a) to conclude that
E[N(x)]=ex

7.68. An urn contains 30 balls, of which 10 are red and 8 are blue. From this urn, 12 balls are randomly withdrawn. Let X denote the number of red and Y the number of blue balls that are withdrawn. Find Cov(X,Y)

a. by defining appropriate indicator (that is, Bernoulli) random variables Xi,Yj such that X=∑i=110Xi,Y=∑j=18Yj

b. by conditioning (on either X or Y) to determine E[XY].

7.69. Type i light bulbs function for a random amount of time having mean μi and standard deviation σi,i=1,2. A light bulb randomly chosen from a bin of bulbs is a type 1 bulb with probability p and a type 2 bulb with probability 1−p. Let X denote the lifetime of this bulb. Find

a. E[X];
b. Var(X).

7.70. The number of winter storms in a good year is a Poisson random variable with mean 3, whereas the number in a bad year is a Poisson random variable with mean 5. If next year will be a good year with probability .4 or a bad year with probability .6, find the expected value and variance of the number of storms that will occur.

7.71. In **Example 5c**, compute the variance of the length of time until the miner reaches safety.

7.72. Consider a gambler who, at each gamble, either wins or loses her bet with respective probabilities p and 1−p. A popular gambling system known as the Kelley strategy is to always bet the fraction 2p−1 of your current fortune when p>12. Compute the expected fortune after n gambles of a gambler who starts with x units and employs the Kelley strategy.
7.73. The number of accidents that a person has in a given year is a Poisson random variable with mean λ. However, suppose that the value of λ changes from person to person, being equal to 2 for 60 percent of the population and 3 for the other 40 percent. If a person is chosen at random, what is the probability that he will have (a) 0 accidents and (b) exactly 3 accidents in a certain year? What is the conditional probability that he will have 3 accidents in a given year, given that he had no accidents the preceding year?
7.74. Repeat **Problem 7.73** when the proportion of the population having a value of λ less than x is equal to 1−e−x.
7.75 Consider an urn containing a large number of coins, and suppose that each of the coins has some probability p of turning up heads when it is flipped. However, this value of p varies from coin to coin. Suppose that the composition of the urn is such that if a coin is selected at random from it, then the p-value of the coin can be regarded as being the value of a random variable that is uniformly distributed over [0, 1]. If a coin is selected at random from the urn and flipped twice, compute the probability that

a. the first flip results in a head;
b. both flips result in heads.

7.76. In **Problem 7.75**, suppose that the coin is tossed n times. Let X denote the number of heads that occur. Show that
$P\{X=i\}=1n+1 \quad i=0,1,\ldots,n$

Hint: Make use of the fact that
$\int 01xa-1(1-x)b-1\,dx=(a-1)!(b-1)!(a+b-1)!$

when a and b are positive integers.
7.77. Suppose that in **Problem 7.75**, we continue to flip the coin until a head appears. Let N denote the number of flips needed. Find

a. $P\{N\geq i\},i\geq 1$;
b. $P\{N=i\}$;
c. $E[N]$.

7.78. In **Example 6b**, let S denote the signal sent and R the signal received.

a. Compute E[R].
b. Compute Var(R).
c. Is R normally distributed?
d. Compute Cov(R,S).

7.79. In **Example 6c**, suppose that X is uniformly distributed over (0, 1). If the discretized regions are determined by a0=0,a1=12, and a2=1, calculate the optimal quantizer Y and compute $E[(X-Y)2]$.
7.80 The moment generating function of X is given by $MX(t)=\exp\{2et-2\}$ and that of Y by $MY(t)=(34et+14)10$. If X and Y are independent, what are

a. $P\{X+Y=2\}$?
b. $P\{XY=0\}$?
c. $E[XY]$?

7.81 . Let X be the value of the first die and Y the sum of the values when two dice are rolled. Compute the joint moment generating function of X and Y.
7.82 . The joint density of X and Y is given by
$f(x,y)=12\pi e-ye-(x-y)2/2 \quad 0<y<\infty,-\infty<x<\infty$

a. Compute the joint moment generating function of X and Y.
b. Compute the individual moment generating functions.

7.83. Two envelopes, each containing a check, are placed in front of you. You are to choose one of the envelopes, open it, and see the amount of the check. At this point, either you can accept that amount or you can exchange it for the check in the unopened envelope. What should you do? Is it possible to devise a strategy that does better than just accepting the first envelope?
Let A and B,A<B, denote the (unknown) amounts of the checks, and note that the strategy that randomly selects an envelope and always accepts its check has an expected return of (A+B)/2. Consider the following strategy: Let $F(\cdot)$ be any strictly increasing (that is, continuous) distribution function. Choose an envelope randomly and open it. If the discovered check has the value x, then accept it with probability F(x) and exchange it with probability $1-F(x)$.

a. Show that if you employ the latter strategy, then your expected return is greater than (A+B)/2
b. **Hint:** Condition on whether the first envelope has the value A or B.
c. Now consider the strategy that fixes a value x and then accepts the first check if its value is greater than x and exchanges it otherwise.
    a. Show that for any x, the expected return under the x-strategy is always at least (A+B)/2 and that it is strictly larger than (A+B)/2 if x lies between A and B.
    b. Let X be a continuous random variable on the whole line, and consider the following strategy: Generate the value of X, and if X=x, then employ the x-strategy of part (b). Show that the expected return under this strategy is greater than (A+B)/2.

7.84. Successive weekly sales, in units of $1,000, have a bivariate normal distribution with common mean 40, common standard deviation 6, and correlation .6.

a. Find the probability that the total of the next 2 weeks' sales exceeds 90.
b. If the correlation were .2 rather than .6, do you think that this would increase or decrease the answer to (a)? Explain your reasoning.
c. Repeat (a) when the correlation is .2.

Theoretical exercises

7.1. Show that $E[(X-a)2]$ is minimized at $a=E[X]$.
7.2. Suppose that X is a continuous random variable with density function f. Show that $E[|X-a|]$ is minimized when a is equal to the median of F.
Hint: Write
$E[\ |\ X-a\ |\ ]=\int|\ x-a\ |f(x)\ dx$

Now break up the integral into the regions where $x<a$ and where $x>a$, and differentiate.
7.3 Prove **Proposition 2.1** when

a. X and Y have a joint probability mass function;
b. X and Y have a joint probability density function and $g(x,y)\geq 0$ for all x,y.

7.4 Let X be a random variable having finite expectation μ and variance $\sigma 2$, and let $g(\cdot)$ be a twice differentiable function. Show that
$E[\ g(X)\ ]\approx g(\mu)+g''(\mu)2\sigma 2$

Hint: Expand $g(\cdot)$ in a Taylor series about μ. Use the first three terms and ignore the remainder.
7.5 If $X\geq 0$ and g is a differentiable function such that $g(0)=0$, show that
$E[\ g(X)\ ]=\int 0\infty P(X>t)g'(t)\ dt$

Hint: Define random variables $I(t),t\geq 0$ so that
$g(X)=\int 0Xg'(t)\ dt=\int 0\infty I(t)g'\ (t)\ dt$

7.6. Let $A1,A2,\ldots,An$ be arbitrary events, and define $Ck=\{$ at least k of the Ai occur $\}$. Show that
$\sum k=1nP(Ck)=\sum k=1nP(Ak)$

Hint: Let X denote the number of the Ai that occur. Show that both sides of the preceding equation are equal to E[X].
7.7. In the text, we noted that
$E[\sum i=1\infty Xi]=\sum i=1\infty E[Xi]$

when the Xi are all nonnegative random variables. Since an integral is a limit of sums, one might expect that
$E[\int 0\infty X(t)dt]=\int 0\infty E[X(t)]\ dt$

whenever $X(t),0\leq t<\infty$, are all nonnegative random variables; this result is indeed true. Use it to give another proof of the result that for a nonnegative random variable X,
$E[X]=\int 0\infty P\{X>t\}\ dt$

Hint: Define, for each nonnegative t, the random variable X(t) by
$X(t)=\{\ 1\text{if } t<X0\text{if } t\geq X$

Now relate $\int 0\infty X(t)dt$ to X.
7.8. We say that X is *stochastically larger* than Y, written $X\geq stY$, if, for all t,
$P\{X>t\}\geq P\{Y>t\}$

Show that if $X\geq stY$, then $E[X]\geq E[Y]$ when

a. X and Y are nonnegative random variables;
b. X and Y are arbitrary random variables.

Hint: Write X as
$X=X^+-X^-$

where
$X^+=\{ X \text{if } X\geq 0 \text{if } X<0 , \quad X^-=\{ 0 \text{if } X\geq 0 - X \text{if } X>0$

Similarly, represent Y as Y^+-Y^-. Then make use of part (a).
7.9. Show that X is stochastically larger than Y if and only if
$E[f(X)]\geq E[f(Y)]$

for all increasing functions f.
Hint: Show that $X\geq_{st} Y$, then $E[f(X)]\geq E[f(Y)]$ by showing that $f(X)\geq_{st} f(Y)$ and then using **Theoretical Exercise 7.8**. To show that if $E[f(X)]\geq E[f(Y)]$ for all increasing functions f, then $P\{X>t\}\geq P\{Y>t\}$, define an appropriate increasing function f.
7.10. A coin having probability p of landing on heads is flipped n times. Compute the expected number of runs of heads of size 1, of size 2, and of size k, $1\leq k\leq n$.
7.11. Let $X_1,X_2,\ldots,X_n$ be independent and identically distributed positive random variables. For $k\leq n$, find
$E[\sum_{i=1}^{k} X_i / \sum_{i=1}^{n} X_i]$

7.12. Consider n independent trials, each resulting in any one of r possible outcomes with probabilities $P_1,P_2,\ldots,P_r$. Let X denote the number of outcomes that never occur in any of the trials. Find E[X] and show that among all probability vectors $P_1,\ldots,P_r$, E[X] is minimized when $P_i=1/r, i=1,\ldots,r$.
7.13. Let $X_1,X_2,\ldots$ be a sequence of independent random variables having the probability mass function
$P\{X_n=0\}=P\{X_n=2\}=1/2 , \quad n\geq 1$

The random variable $X=\sum_{n=1}^{\infty} X_n/3^n$ is said to have the *Cantor distribution*. Find E[X] and Var(X).
7.14. Let $X_1,\ldots,X_n$ be independent and identically distributed continuous random variables. We say that a record value occurs at time $j, j\leq n$, if $X_j\geq X_i$ for all $1\leq i\leq j$. Show that

a. E[number of record values]$=\sum_{j=1}^{n} 1/j$;
b. Var(number of record values)$=\sum_{j=1}^{n}(j-1)/j^2$.

7.15 For **Example 2i**, show that the variance of the number of coupons needed to amass a full set is equal to
$\sum_{i=1}^{N-1} iN/(N-i)^2$

When N is large, this can be shown to be approximately equal (in the sense that their ratio approaches 1 as $N\to\infty$) to $N^2\pi^2/6$.
7.16 Consider n independent trials, the ith of which results in a success with probability P_i

a. Compute the expected number of successes in the n trials—call it μ.
b. For a fixed value of μ, what choice of $P_1,\ldots,P_n$ maximizes the variance of the number of successes?
c. What choice minimizes the variance?

* 7.17. Suppose that each of the elements of $S=\{1,2,\ldots,n\}$ is to be colored either red or blue. Show that if $A_1,\ldots,A_r$ are subsets of S, there is a way of doing the coloring so that at most $\sum_{i=1}^{r}(1/2)^{|A_i|-1}$ of these subsets have all their elements the same color (where $|A|$ denotes the number of elements in the set A).
7.18. Suppose that X_1 and X_2 are independent random variables having a common mean μ. Suppose also that $Var(X_1)=\sigma_1^2$ and $Var(X_2)=\sigma_2^2$ The value of μ is unknown, and it is proposed that μ be estimated by a weighted average of X_1 and X_2. That is, $\lambda X_1+(1-\lambda)X_2$ will be used as an

estimate of μ for some appropriate value of λ. Which value of λ yields the estimate having the lowest possible variance? Explain why it is desirable to use this value of λ.

7.19. In **Example 4f**, we showed that the covariance of the multinomial random variables Ni and Nj is equal to −mPiPj by expressing Ni and Nj as the sum of indicator variables. We could also have obtained that result by using the formula

$$\mathrm{Var}(N_i+N_j)= \mathrm{Var}(N_i)+ \mathrm{Var}(N_j)+ 2\,\mathrm{Cov}(N_i,N_j)$$

a. What is the distribution of Ni+Nj?
b. Use the preceding identity to show that Cov(Ni,Nj)=−mPiPj.

7.20. Show that X and Y are identically distributed and not necessarily independent, then

$$\mathrm{Cov}(X+Y,X-Y)=0$$

7.21. *The Conditional Covariance Formula*. The conditional covariance of X and Y, given Z, is defined by

$$\mathrm{Cov}(X,Y|Z)\equiv E[(X-E[X|Z])(Y-E[Y|Z])|Z]$$

a. Show that
$$\mathrm{Cov}(X, Y| Z)=E[XY|Z]-E[X|Z]E[Y|Z]$$

b. Prove the conditional covariance formula
$$\mathrm{Cov}(X,Y)=E[\mathrm{Cov}(X,Y|Z)]+\mathrm{Cov}(E[X| Z] ,E [Y |Z])$$

c. Set X=Y in part (b) and obtain the conditional variance formula.

7.22. Let X(i),i=1,…,n, denote the order statistics from a set of n uniform (0, 1) random variables, and note that the density function of X(i) is given by

$$f(x)=n!(i-1)!(n-i)!x^{i-1}(1-x)^{n-i}\quad 0<x<1$$

a. Compute Var(X(i)),i=1,…,n.
b. Which value of i minimizes, and which value maximizes, Var(X(i))?

7.23. Show that Y=a+bX, then

$$\rho(X,Y)=\begin{cases}+1 & \text{if } b>0\\ -1 & \text{if } b<0\end{cases}$$

7.24. Show that Z is a standard normal random variable and if Y is defined by Y=a+bZ+cZ2, then

$$\rho(Y,Z)=\frac{b}{\sqrt{b^2+2c^2}}$$

7.25. Prove the Cauchy–Schwarz inequality, namely,

$$(E[XY])^2\le E[X^2]E[Y^2]$$

Hint: Unless Y=−tX for some constant, in which case the inequality holds with equality, it follows that for all t,

$$0<E[(tX+Y)^2]=E[X^2]t^2+2E[XY]t+E[Y^2]$$

Hence, the roots of the quadratic equation

$$E[X^2]t^2+2E[XY]t+E[Y^2]=0$$

must be imaginary, which implies that the discriminant of this quadratic equation must be negative.

7.26. Show that if X and Y are independent, then

$$E[X|Y=y]=E[X]\text{ for all } y$$

a. in the discrete case;
b. in the continuous case.

7.27. Prove that E[g(X)Y|X]=g(X)E[Y|X].
7.28. Prove that if E[Y|X=x]=E[Y] for all x, then X and Y are uncorrelated; give a counterexample to show that the converse is not true.
Hint: Prove and use the fact that E[XY]=E[XE[Y|X]].
7.29. Show that Cov(X,E[Y|X])= Cov(X,Y).
7.30. Let X1,…,Xn be independent and identically distributed random variables. Find E[X1|X1+⋯+Xn=x]

7.31. Consider **Example 4f**, which is concerned with the multinomial distribution. Use conditional expectation to compute E[NiNj], and then use this to verify the formula for Cov(Ni,Nj) given in **Example 4f**.
7.32. An urn initially contains b black and w white balls. At each stage, we add r black balls and then withdraw, at random, r balls from the b+w+r balls in the urn. Show that
E[number of white balls after stage t] =(b+wb+w+r)tw

7.33. For an event A, let IA equal 1 if A occurs and let it equal 0 if A does not occur. For a random variable X, show that
E[X|A]=E[XIA]P(A)

7.34. A coin that lands on heads with probability p is continually flipped. Compute the expected number of flips that are made until a string of r heads in a row is obtained.
Hint: Condition on the time of the first occurrence of tails to obtain the equation
E[X]=(1−p)∑i=1rpi−1(i+E[X])+(1−p)∑i=r+1∞pi−1r

Simplify and solve for E[X].
7.35. For another approach to **Theoretical Exercise 7.34**, let Tr denote the number of flips required to obtain a run of r consecutive heads.
a. Determine E[Tr|Tr−1].
b. Determine E[Tr] in terms of E[Tr−1].
c. What is E[T1]?
d. What is E[Tr]?

7.36. The probability generating function of the discrete nonnegative integer valued random variable X having probability mass function pj, j≥0, is defined by
ϕ(s)=E[sX]=∑j=0∞pjsj

Let Y be a geometric random variable with parameter p=1−s, where 0<s<1. Suppose that Y is independent of X, and show that
ϕ(s)=P{X<Y}

7.37. One ball at a time is randomly selected from an urn containing a white and b black balls until all of the remaining balls are of the same color. Let Ma,b denote the expected number of balls left in the urn when the experiment ends. Compute a recursive formula for Ma,b and solve when a=3 and b=5.
7.38. An urn contains a white and b black balls. After a ball is drawn, it is returned to the urn if it is white; but if it is black, it is replaced by a white ball from another urn. Let Mn denote the expected number of white balls in the urn after the foregoing operation has been repeated n times.
a. Derive the recursive equation
Mn+1=(1−1a+b)Mn+1

b.

Use part (a) to prove that
$M_n = a + b - b(1 - 1a + b)^n$

c. What is the probability that the (n+1) ball drawn is white?

7.39. The best linear predictor of Y with respect to X1 and X2 is equal to a+bX1+cX2, where a,b, and c are chosen to minimize
$E[(Y - (a + bX1 + cX2))2]$

Determine a,b, and c.
7.40. The best quadratic predictor of Y with respect to X is a+bX+cX2, where a,b, and c are chosen to minimize $E[(Y - (a + bX + cX2))2]$. Determine a,b, and c.
7.41. Use the conditional variance formula to determine the variance of a geometric random variable X having parameter p.
7.42. Let X be a normal random variable with parameters $\mu = 0$ and $\sigma 2 = 1$, and let I, independent of X, be such that $P\{I = 1\} = 12 = P\{I = 0\}$. Now define Y by
$Y = \{ X \text{if } I = 1 - X \text{if } I = 0$

In words, Y is equally likely to equal either X or −X.

a. Are X and Y independent?
b. Are I and Y independent?
c. Show that Y is normal with mean 0 and variance 1.
d. Show that Cov(X,Y)=0.

7.43. It follows from **Proposition 6.1** and the fact that the best linear predictor of Y with respect to X is $\mu y + \rho\sigma y\sigma x(X - \mu x)$ that if
$E[Y|X] = a + bX$

then
$a = \mu y - \rho\sigma y\sigma x\mu x \; b = \rho\sigma y\sigma x$

(Why?) Verify this directly.
7.44. Show that for random variables X and Z,
$E[(X - Y)2] = E[X2] - E[Y2]$

where
$Y = E[X|Z]$

7.45. Consider a population consisting of individuals able to produce offspring of the same kind. Suppose that by the end of its lifetime, each individual will have produced j new offspring with probability Pj, $j \geq 0$, independently of the number produced by any other individual. The number of individuals initially present, denoted by X0, is called the size of the zeroth generation. All offspring of the zeroth generation constitute the first generation, and their number is denoted by X1. In general, let Xn denote the size of the nth generation. Let $\mu = \sum j=0 \infty jPj$ and $\sigma 2 = \sum j=0 \infty (j - \mu)2Pj$ denote, respectively, the mean and the variance of the number of offspring produced by a single individual. Suppose that X0=1 that is, initially there is a single individual in the population.

a. Show that
$E[Xn] = \mu E[Xn - 1]$

b. Use part (a) to conclude that
$E[Xn] = \mu n$

c. Show that
$\mathrm{Var}(X_n)=\sigma^2\mu^{n-1}+\mu^2\,\mathrm{Var}(X_{n-1})$

d. Use part (c) to conclude that
$\mathrm{Var}(X_n)=\begin{cases}\sigma^2\mu^{n-1}(\mu^{n-1}-1)/(\mu-1) & \text{if } \mu\neq 1\\ n\sigma^2 & \text{if } \mu=1\end{cases}$

e. The model just described is known as a *branching process*, and an important question for a population that evolves along such lines is the probability that the population will eventually die out. Let π denote this probability when the population starts with a single individual. That is,
$\pi=P\{\text{population eventually dies out} \mid X_0=1\}$

f. Argue that π satisfies
$\pi=\sum_{j=0}^{\infty}P_j\pi^j$

Hint: Condition on the number of offspring of the initial member of the population.

7.46. Verify the formula for the moment generating function of a uniform random variable that is given in **Table 7.2**. Also, differentiate to verify the formulas for the mean and variance.

7.47. For a standard normal random variable Z, let $\mu_n=E[Z^n]$. Show that
$\mu_n=\begin{cases}0 & \text{when } n \text{ is odd}\\ \dfrac{(2j)!}{2^j j!} & \text{when } n=2j\end{cases}$

Hint: Start by expanding the moment generating function of Z into a Taylor series about 0 to obtain
$E[e^{tZ}]=e^{t^2/2}=\sum_{j=0}^{\infty}(t^2/2)^j/j!$

7.48. Let X be a normal random variable with mean μ and variance σ^2. Use the results of Theoretical **Exercise 7.47** to show that
$E[X^n]=\sum_{j=1}^{[n/2]}\binom{n}{2j}\mu^{n-2j}\sigma^{2j}(2j)!/2^j j!$

In the preceding equation, $[n/2]$ is the largest integer less than or equal to $n/2$. Check your answer by letting $n=1$ and $n=2$.

7.49. If $Y=aX+b$, where a and b are constants, express the moment generating function of Y in terms of the moment generating function of X.

7.50. The positive random variable X is said to be a *lognormal* random variable with parameters μ and σ^2 if $\log(X)$ is a normal random variable with mean μ and variance σ^2. Use the normal moment generating function to find the mean and variance of a lognormal random variable.

7.51. Let X have moment generating function $M(t)$, and define $\Psi(t)=\log M(t)$. Show that
$\Psi''(t)|_{t=0}=\mathrm{Var}(X)$

7.52. Use **Table 7.2** to determine the distribution of $\sum_{i=1}^{n}X_i$ when $X_1,\ldots,X_n$ are independent and identically distributed exponential random variables, each having mean $1/\lambda$.

7.53. Show how to compute $\mathrm{Cov}(X,Y)$ from the joint moment generating function of X and Y.

7.54. Suppose that $X_1,\ldots,X_n$ have a multivariate normal distribution. Show that $X_1,\ldots,X_n$ are independent random variables if and only if
$\mathrm{Cov}(X_i,X_j)=0 \quad \text{when } i\neq j$

7.55. If Z is a standard normal random variable, what is $\mathrm{Cov}(Z,Z^2)$?

7.56. Suppose that Y is a normal random variable with mean μ and variance σ^2, and suppose also that the conditional distribution of X, given that $Y=y$, is normal with mean y and variance 1.

a. Argue that the joint distribution of X,Y is the same as that of $Y+Z,Y$ when Z is a standard normal random variable that is independent of Y.
b. Use the result of part (a) to argue that X,Y has a bivariate normal distribution.
c. Find $E[X]$, $\mathrm{Var}(X)$, and $\mathrm{Corr}(X,Y)$.
d. Find $E[Y|X=x]$.
e. What is the conditional distribution of Y given that $X=x$?

Self-Test Problems and Exercises

7.1. Consider a list of m names, where the same name may appear more than once on the list. Let n(i), i=1,…,m, denote the number of times that the name in position i appears on the list, and let d denote the number of distinct names on the list.

a. Express d in terms of the variables m,n(i),i=1,…,m. Let U be a uniform (0, 1) random variable, and let X=[mU]+1.
b. What is the probability mass function of X?
c. Argue that E[m/n(X)]=d.

7.2 An urn has n white and m black balls that are removed one at a time in a randomly chosen order. Find the expected number of instances in which a white ball is immediately followed by a black one.

7.3 Twenty individuals consisting of 10 married couples are to be seated at 5 different tables, with 4 people at each table.

a. If the seating is done “at random,” what is the expected number of married couples that are seated at the same table?
b. If 2 men and 2 women are randomly chosen to be seated at each table, what is the expected number of married couples that are seated at the same table?

7.4 If a die is to be rolled until all sides have appeared at least once, find the expected number of times that outcome 1 appears.

7.5. A deck of 2n cards consists of n red and n black cards. The cards are shuffled and then turned over one at a time. Suppose that each time a red card is turned over, we win 1 unit if more red cards than black cards have been turned over by that time. (For instance, if n=2 and the result is r b r b, then we would win a total of 2 units.) Find the expected amount that we win.

7.6. Let A1,A2,…,An be events, and let N denote the number of them that occur. Also, let I=1 if all of these events occur, and let it be 0 otherwise. Prove Bonferroni’s inequality, namely, $P(A_1 \cdots A_n) \geq \sum_{i=1}^{n} P(A_i) - (n-1)$

Hint: Argue first that $N \leq n-1+I$.

7.7. Let X be the smallest value obtained when k numbers are randomly chosen from the set 1,…,n. Find E[X] by interpreting X as a negative hypergeometric random variable.

7.8. An arriving plane carries r families. A total of nj of these families have checked in a total of j pieces of luggage, $\sum_j nj = r$. Suppose that when the plane lands, the $N = \sum_j jnj$ pieces of luggage come out of the plane in a random order. As soon as a family collects all of its luggage, it immediately departs the airport. If the Sanchez family checked in j pieces of luggage, find the expected number of families that depart after they do.

* 7.9. Nineteen items on the rim of a circle of radius 1 are to be chosen. Show that for any choice of these points, there will be an arc of (arc) length 1 that contains at least 4 of them.

7.10. Let X be a Poisson random variable with mean λ. Show that if λ is not too small, then $Var(X) \approx .25$

Hint: Use the result of **Theoretical Exercise 7.4** to approximate E[X].

7.11. Suppose in Self-Test **Problem 7.3** that the 20 people are to be seated at seven tables, three of which have 4 seats and four of which have 2 seats. If the people are randomly seated, find the expected value of the number of married couples that are seated at the same table.

7.12. Individuals 1 through n,n>1, are to be recruited into a firm in the following manner: Individual 1 starts the firm and recruits individual 2. Individuals 1 and 2 will then compete to recruit individual 3. Once individual 3 is recruited, individuals 1, 2, and 3 will compete to recruit individual 4, and so on. Suppose that when individuals 1,2,…,i compete to recruit individual i+1, each of them is equally likely to be the successful recruiter.

a. Find the expected number of the individuals 1,…,n who did not recruit anyone else.
b.

Derive an expression for the variance of the number of individuals who did not recruit anyone else, and evaluate it for n=5.

7.13 The nine players on a basketball team consist of 2 centers, 3 forwards, and 4 backcourt players. If the players are paired up at random into three groups of size 3 each, find (a) the expected value and (b) the variance of the number of triplets consisting of one of each type of player.

7.14 A deck of 52 cards is shuffled and a bridge hand of 13 cards is dealt out. Let X and Y denote, respectively, the number of aces and the number of spades in the hand.

a. Show that X and Y are uncorrelated.
b. Are they independent?

7.15 Each coin in a bin has a value attached to it. Each time that a coin with value p is flipped, it lands on heads with probability p. When a coin is randomly chosen from the bin, its value is uniformly distributed on (0, 1). Suppose that after the coin is chosen but before it is flipped, you must predict whether it will land on heads or on tails. You will win 1 if you are correct and will lose 1 otherwise.

a. What is your expected gain if you are not told the value of the coin?
b. Suppose now that you are allowed to inspect the coin before it is flipped, with the result of your inspection being that you learn the value of the coin. As a function of p, the value of the coin, what prediction should you make?
c. Under the conditions of part (b), what is your expected gain?

7.16. In Self-Test **Problem 7.1**, we showed how to use the value of a uniform (0, 1) random variable (commonly called a *random number*) to obtain the value of a random variable whose mean is equal to the expected number of distinct names on a list. However, its use required that one choose a random position and then determine the number of times that the name in that position appears on the list. Another approach, which can be more efficient when there is a large amount of replication of names, is as follows: As before, start by choosing the random variable X as in **Problem 7.1**. Now identify the name in position X, and then go through the list, starting at the beginning, until that name appears. Let I equal 0 if you encounter that name before getting to position X, and let I equal 1 if your first encounter with the name is at position X. Show that E[mI]=d.

Hint: Compute E[I] by using conditional expectation.

7.17. A total of m items are to be sequentially distributed among n cells, with each item independently being put in cell j with probability pj,j=1,…,n. Find the expected number of collisions that occur, where a collision occurs whenever an item is put into a nonempty cell.

7.18. Let X be the length of the initial run in a random ordering of n ones and m zeros. That is, if the first k values are the same (either all ones or all zeros), then X≥k. Find E[X].

7.19. There are n items in a box labeled H and m in a box labeled T. A coin that comes up heads with probability p and tails with probability 1−p is flipped. Each time it comes up heads, an item is removed from the H box, and each time it comes up tails, an item is removed from the T box. (If a box is empty and its outcome occurs, then no items are removed.) Find the expected number of coin flips needed for both boxes to become empty.

Hint: Condition on the number of heads in the first n+m flips.

7.20. Let X be a nonnegative random variable having distribution function F. Show that if $\bar{F}(x)=1-F(x)$, then

$$E[X^n]=\int_0^\infty x^{n-1}\bar{F}(x)\,dx$$

Hint: Start with the identity

$$X^n=n\int_0^X x^{n-1}\,dx=n\int_0^\infty x^{n-1}I_X(x)\,dx$$

where

$$I_x(x)=\begin{cases}1, & \text{if } x<X\\ 0, & \text{otherwise}\end{cases}$$

* 7.21. Let a1,…,an, not all equal to 0, be such that $\sum_{i=1}^n a_i=0$. Show that there is a permutation

i1,…,in such that ∑j=1naijaij+1<0.
Hint: Use the probabilistic method. (It is interesting that there need not be a permutation whose sum of products of successive pairs is positive. For instance, if n=3, a1=a2=−1, and a3=2, there is no such permutation.)
7.22. Suppose that Xi, i=1,2,3, are independent Poisson random variables with respective means λi, i=1,2,3. Let X=X1+X2 and Y=X2+X3. The random vector X,Y is said to have a bivariate Poisson distribution.

a. Find E[X] and E[Y].
b. Find Cov(X,Y).
c. Find the joint probability mass function P{X=i,Y=j}.

7.23. Let (Xi,Yi), i=1,…, be a sequence of independent and identically distributed random vectors. That is, X1,Y1 is independent of, and has the same distribution as, X2,Y2, and so on. Although Xi and Yi can be dependent, Xi and Yj are independent when i≠j. Let
μx=E[Xi], μy=E[Yi], σx2= Var(Xi),σy2= Var(Yi), ρ=Corr (Xi,Yi)

Find Corr (∑i=1nXi,∑j=1nYj).
7.24. Three cards are randomly chosen without replacement from an ordinary deck of 52 cards. Let X denote the number of aces chosen.

a. Find E[X|the ace of spades is chosen].
b. Find E[X|at least one ace is chosen].

7.25. Let Φ be the standard normal distribution function, and let X be a normal random variable with mean μ and variance 1. We want to find E[Φ(X)]. To do so, let Z be a standard normal random variable that is independent of X, and let
I={ 1,if Z<X0,if Z≥X

a. Show that E[I|X=x]=Φ(x).
b. Show that E[Φ(X)]=P{Z<X}.
c. Show that E[Φ(X)]=Φ(μ2).

Hint: What is the distribution of X−Z?
The preceding comes up in statistics. Suppose you are about to observe the value of a random variable X that is normally distributed with an unknown mean μ and variance 1, and suppose that you want to test the hypothesis that the mean μ is greater than or equal to 0. Clearly you would want to reject this hypothesis if X is sufficiently small. If it results that X=x, then the p-*value* of the hypothesis that the mean is greater than or equal to 0 is defined to be the probability that X would be as small as x if μ were equal to 0 (its smallest possible value if the hypothesis were true). (A small p-value is taken as an indication that the hypothesis is probably false.) Because X has a standard normal distribution when μ=0, the p-value that results when X=x is Φ(x).Therefore, the preceding shows that the expected p-value that results when the true mean is μ is Φ(μ2).
7.26. A coin that comes up heads with probability p is flipped until either a total of n heads or of m tails is amassed. Find the expected number of flips.
Hint: Imagine that one continues to flip even after the goal is attained. Let X denote the number of flips needed to obtain n heads, and let Y denote the number of flips needed to obtain m tails. Note that max(X,Y)+min(X,Y)=X+Y. Compute E[max(X,Y)] by conditioning on the number of heads in the first n+m−1 flips.
7.27. A deck of n cards numbered *1* through n, initially in any arbitrary order, is shuffled in the following manner: At each stage, we randomly choose one of the cards and move it to the front of the deck, leaving the relative positions of the other cards unchanged. This procedure is continued until all but one of the cards has been chosen. At this point, it follows by symmetry that all n! possible orderings are equally likely. Find the expected number of stages that are required.
7.28. Suppose that a sequence of independent trials in which each trial is a success with probability p

is performed until either a success occurs or a total of n trials has been reached. Find the mean number of trials that are performed.
Hint: The computations are simplified if you use the identity that for a nonnegative integer valued random variable X,
$E[X]=\sum_{i=1}^{\infty}P\{X\geq i\}$

7.29. Suppose that X and Y are both Bernoulli random variables. Show that X and Y are independent if and only if $Cov(X,Y)=0$.
7.30. In the generalized match problem, there are n individuals of whom ni wear hat size $i,\sum_{i=1}^{r}n_i=n$. There are also n hats, of which hi are of size $i,\sum_{i=1}^{r}h_i=n$.If each individual randomly chooses a hat (without replacement), find the expected number who choose a hat that is their size.
7.31. For random variables X and Y, show that
$Var(X+Y)\leq Var(X)+Var(Y)$

That is, show that the standard deviation of a sum is always less than or equal to the sum of the standard deviations.
7.32. Let $R_1,\ldots,R_{n+m}$ be a random permutation of $1,\ldots,n+m$. (That is, $R_1,\ldots,R_{n+m}$ is equally likely to be any of the $(n+m)!$ permutations of $1,\ldots,n+m$.) For a given $i\leq n$, let X be the the ith smallest of the values $R_1,\ldots,R_n$. Show that $E[X]=i+\frac{m}{n+1}$.
Hint: Note that if we let I_{n+k} equal 1 if $R_{n+k}<X$ and let it equal 0 otherwise, that
$X=i+\sum_{k=1}^{m}I_{n+k}$

7.33. Suppose that Y is uniformly distributed over (0,1), and that the conditional distribution of X, given that Y=y, is uniform over (0,y).

a. Find E[X].
b. Find Cov(X,Y).
c. Find Var(X).
d. Find $P\{X\leq x\}$.
e. Find the probability density function of X.